

**Physiological and molecular characterization of drought tolerant
biocontrol agents for management of dry root rot of Chickpea
(*Cicer arietinum* L.) caused by *Rhizoctonia bataticola***



THESIS

Submitted in partial fulfilment of the requirements for the degree of

**Doctor of Philosophy
in
Mycology and Plant Pathology**

Supervisor

Prof. H.B. Singh

Submitted by

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Enrolment No. 312631

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biocontrol agents for management of dry root rot of Chickpea (*Cicer
arietinum* L.) caused by *Rhizoctonia bataticola***



THESIS SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE DEGREE OF

**DOCTOR OF PHILOSOPHY (AGRICULTURE)
in
MYCOLOGY AND PLANT PATHOLOGY**

**Institute of Agricultural Sciences
Banaras Hindu University
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by
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ABBREVIATIONS

<	Less than
%	Per cent
°C	Degree Celsius
>	Greater than
±	Plus Minus
≥	Greater than equal to
µl	Micro litre
µm	Micro metre
a.i.	Active Ingredient
ACC	1 aminocyclo-propane carboxylic acid
BOD	Biological Oxygen Demand
CFU	Colony forming unit
cm	Centimetre
CTAB	Cetyl trimethyl ammonium bromide
dapi	Days After Pathogen Inoculation
DAS	Days After Sowing
DMRT	Duncan multiple Range test
EDTA	Ethylene diamine tetraacetic acid
<i>et al.</i>	And co workers
etc.	(et cetera) and the others
Fig.	Figure
FW	Fresh weight
g	Gram
h	Hour
ha	Hectare
HCl	Hydrochloric acid
HgCl ₂	Mercuric chloride
HPLC	High Performance liquid Chromatography
hrs	Hours
i.e	(ed est) that is

KBM	King's B medium
l/ha	Litre per Hectare
M	Molarity
m ²	Square metre
mg	Milligram
min ⁻¹	Per minute
ml	Millilitre
ml l ⁻¹	Millilitre per Litre
ml ⁻¹	Per millilitres
mm	Millimetre
mM	Millimole
mt	Million Tonnes
N	Normality
NaCl	Sodium Chloride
NaOH	Sodium Hydroxide
NB	Nutrient Broth
NBT	Nitroblue tetrazolium chloride
No.	Number
O.D	Optical Density
PAL	Phenylalanine ammonia lyase
PO	Peroxidase
PPO	Polyphenol oxidase
PDA	Potato Dextrose Agar
PDB	Potato Dextrose Broth
PEG	Polyethylene glycol
ppm	Parts Per Million
PSA	Potato Sucrose Agar
p.s.i.	Pound per square inch
R	Replication
RH	Relative humidity
ROS	Reactive oxygen species
SOD	Superoxide dismutase
Sr. No	Serial number
T	Treatment
TPC	Total Phenol content
UV	Ultra violet

viz.	(videlicet) namely
WP	Wettable powder

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PREFACE

Since this thesis is written as the final document for Doctorate degree in Mycology and Plant Pathology and primarily aimed to isolate drought tolerant microorganisms having the potential to mitigate drought stress and dry root rot in chickpea plants. The work will be of great intent for development of bioformulations which have the ability to mitigate drought stress to a certain extent. Also, the study will be helpful to minimize the crop losses and in turn provide food safety and food security in India.

The work described in this thesis was carried out between 2012-16 at the Department of Mycology and Plant Pathology, Institute of Agricultural Sciences, B.H.U., Varanasi, under the supervision of Dr. H.B. Singh, Professor & Head, Department of Mycology and Plant Pathology.

Chapter I - The introduction part which provides much of the general background. This chapter justifies the reason for choosing this research topic, importance of study in reference & lastly objectives of the research work.

Chapter II - The review of literature deals with the findings related to role of beneficial micro-organisms in mitigating drought stress and also dry root rot of chickpea caused by *R. bataticola*. Attempts made by various scientists in managing dry root rot have also been described.

Chapter III - The materials and methods deals with the methodology employed for carrying the research work and methods for analysis.

Chapter IV - The results and discussion which explains the results obtained with appropriate reasons and support.

Chapter V - The summary and conclusion gives a brief description of the results of the investigation and outcomes of the research work concluded.

In the last, references which have been consulted during the course of investigation have been cited.

CHAPTER I

INTRODUCTION

INTRODUCTION

Legumes have remained an integral part of human diet since time immemorial. Moreover, legumes with their unusual flower structure, podded fruit, and the ability of to form nodules with rhizobia (de Faria *et al.*, 1989), they lie second only to the Graminae in their importance to humans. The various genera (670-750) and species (18,000-19,000) of legumes constitute several grain, pasture and agroforestry species (Polhill *et al.*, 1981) among which bean (*Phaseolus vulgaris*), chickpea (*Cicer arietinum*), pea (*Pisum sativum*), pigeon pea (*Cajanus cajan*), broad bean (*Vicia faba*), cowpea (*Vigna unguiculata*), and lentil constitute the primary dietary legumes (National Academy of Science, 1994). Legumes (predominantly soybean [*Glycine max*] and peanut [*Arachis hypogaeae*]) provide more than 35% of the world's processed vegetable oil and are also rich sources of dietary protein for the chicken and pork industries.

Economic importance of legumes

The importance of grain legumes in the world is high due to their significance in human and animal nutrition. Right behind cereals, legumes are the second most important source of human food and animal forage. Legumes are a rich source of starch, protein, vitamins, minerals as well as phenolics, oligosaccharides and inositol phosphates. Legume seeds can also become an industrial basic material with a wide range of non-food applications. However, in spite of increasing interest regarding cultivation of pulses, the growth area and production of seeds, their application is relatively small (Kozłowska *et al.*, 1998). In agricultural practice, legumes are used as and reduce crop nutrient deficiency. Mankind has known for millenniums of the importance of legumes environmentally.

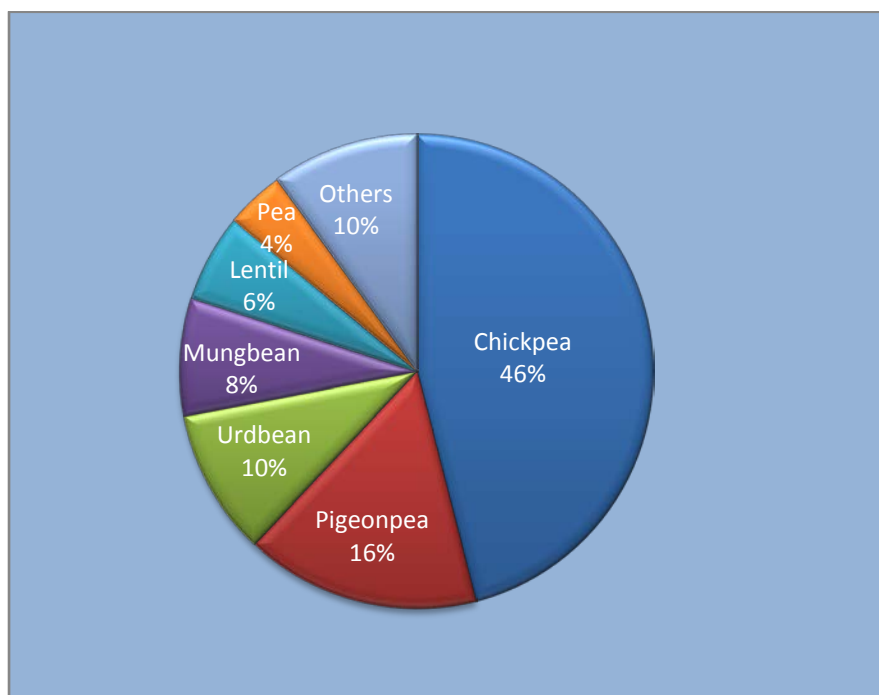


Figure 1.1: Share of different legumes in Indian legume production (ICRISAT 2015)

Plant stress

Plants are subjected to several harsh environmental stresses that adversely affect their growth, metabolism, and yield. Drought, salinity, low and high temperatures, flood, pollutants, and radiation are the important abiotic stress factors limiting the productivity of crops (Lawlor, 2002). Several biotic (insects, bacteria, fungi, and viruses) and abiotic (light, temperature, water availability, nutrients, and soil structure) factors affect the growth in higher plants (Lichtenthaler, 1996, 1998). Among these, drought is a major abiotic factor limiting agricultural crop production. Plants experience drought stress either when the water supply to roots becomes difficult or when the transpiration rate becomes very high. These two conditions often coincide under arid and semi-arid climates. Water stress tolerance is seen in almost all plant species but its extent varies from species to species.

Although the general effects of drought on plant growth are fairly well known, the primary effects of water deficit at the biochemical and the molecular levels are not well understood (Zhu, 2002; Chaitanya *et al.*, 2003; Chaves *et al.*, 2003). In order to improve the agricultural productivity within the limited land resources, it is imperative to ensure higher crop yields against unfavorable environmental stresses. Understanding plant

responses to the external environment is of greater importance and also a fundamental part for making the crops stress tolerant. Drought stress is considered to be a moderate loss of water, which leads to stomatal closure and limitation of gas exchange. Desiccation is much more extensive loss of water, which can potentially lead to gross disruption of metabolism and cell structure and eventually to the cessation of enzyme catalyzed reactions (Smirnoff, 1993; Jaleel *et al.*, 2007d). Drought stress is characterized by reduction of water content, diminished leaf water potential and turgor loss, closure of stomata and decrease in cell enlargement and growth. Severe water stress may result in the arrest of photosynthesis, disturbance of metabolism and finally the death of plant (Jaleel *et al.*, 2008c).

Abiotic stress conditions cause extensive losses to agricultural production worldwide (Boyers, 1982; Bray *et al.*, 2000). Individually, stress conditions such as drought, salinity or heat have been the subject of intense research (Bray *et al.*, 2000; Cushman and Bohnert, 2000). However, in the field, crops and other plants are routinely subjected to a combination of different abiotic stresses (Moffat, 2002; Heyne and Brunson, 1940; Craufurd and Peacock, 1993; Jiang and Huang, 2001). In drought stricken areas, for example, many crops encounter a combination of drought and other stresses, such as heat or salinity (Moffat, 2002; Heyne and Brunson, 1940). Recent studies have revealed that the molecular and metabolic response of plants to a combination of drought and heat is unique and cannot be directly extrapolated from the response of plants to each of these different stresses applied individually (Pnueli *et al.*, 2002; Rizhsky *et al.*, 2002; Rizhsky *et al.*, 2004; Suzuki *et al.*, 2005).

Chickpea

Chickpea (*Cicer arietinum* L.) is the largest produced food legume in South Asia and the third largest produced food legume globally after dry beans and pea. (Anonymous, 2006; Janzen *et al.*, 2006). It is known by different names such as horse gram, chana, Bengal gram etc. At present, chickpea is grown in more than 50 countries and is considered to be the second most important pulse crop in the world. It covers 15% of the cultivated area and nearly contributes to 14% (7.9 million tonnes) of the world's pulse harvest of about 58 million tonnes (Singh, 1997). Besides being an important source of human and animal food, chickpea also plays an important role in the maintenance of soil fertility, particularly in the dry, rainfed areas (Katerji *et al.*, 2001).

Asia accounts 89.7% of the area in chickpea production, followed by 4.3% in Africa, 2.6% in Oceania, 2.9% in Americas and 0.4% in Europe (Gaur, 2010). In India chickpea accounts for 45% of the total pulse produced in the country.

Globally, India is the highest producer and consumer of chickpea. With approximately 75% of total chickpea yield being contributed by India (Reddy and Mishra, 2010). Production of chickpea is mainly concentrated in central and southern parts of India. Madhya Pradesh, Rajasthan, Uttar Pradesh, Maharashtra, Karnataka and Andhra Pradesh are the major producers of chickpea contributing 91% of the produce and 90% of the area.

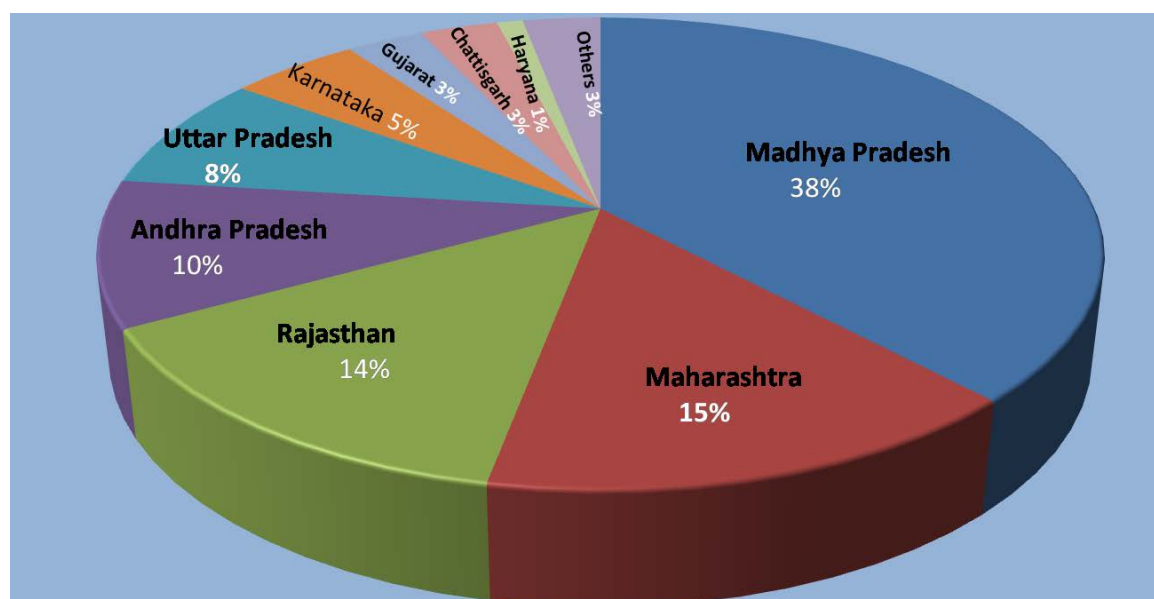


Figure 1.2: Major chickpea producing states of India (AICRP, 2017)

Constraints in Chickpea production

Chickpea is known to be affected by a number of biotic and abiotic factors which limits its growth and yield. The crop is susceptible to many diseases such as Fusarium wilt, Ascochyta blight, wet root rot, dry root rot, wilt complex etc (Nene *et al.*, 1984). Dry root rot (DRR) caused by *Rhizoctonia bataticola* [synonym-*Macrophomina phaseolina*(Tassi) Goid] is one of the major constraint in chickpea production causing an annual loss of 10 to 20% (Dhar and Chaudhary, 2001). The incidence of disease

aggravates in deficit soil moisture conditions. The drastic reduction of cultivable chickpea area in northern India and its quantum increase in hot, drier rainfed central and southern India, dry root rot is emerging as the most destructive constraint to chickpea productivity and production. The disease usually appears during reproductive phase of the crop *i.e* around flowering and podding stage in the form of scattered dried patches of plants in the field. The disease may also appear at seedling stage; however, the old plants are more susceptible towards infection.

Rhizoctonia bataticola

Rhizoctonia bataticola (Taub.) is a very important soil-inhabiting, fungus posing as a serious threat to a wide range of crops. The pathogen is known to incite different types of diseases viz., seedling blight, root rot, stem blight, fruit rot, charcoal rot, wilt, stalk rot, seedling decay and leaf blight in different crop plants (Dhingra and Sinclair, 1978). The genus *Rhizoctonia* is regarded as a heterogeneous group of filamentous fungal taxon that do not produces asexual spores and share a number of similar features in their anamorphic states. The members of this genus are generally soil borne fungi, mostly associated with roots, although there are few reports of their saprophytic and symbiotic ability in few taxa. The Fungi belonging to the genera are distributed worldwide in both agricultural and forest soils and comprising some of the most economically important plant pathogens, which causes foliar and root rot diseases of major crops. *R. bataticola* is important member in the genus that causes root rots and seedling blight in many legumes when they are weakened by other stress factors (Hawang *et al.*, 2003). In contrast to the many pathogens favoured by change to moisture conditions (Garrett *et al.*, 2006), *R. bataticola* becomes more problematic in agricultural areas where climate change results in longer drought periods and higher temperatures.

Drought stress

Among the abiotic stresses, water stress is a significant yield-limiting factor in Chickpea (*Cicer arietinum* L.) production as 90% of the major chickpea growing areas lie under arid and semi-arid zones (Kumar and Abbo, 2001). However, Serraj *et al.* (2004) clarified that yield losses may be due to intermittent drought like conditions during the vegetative phase, reproductive phase or at the end of the crop cycle. Drought impacts

include growth, yield, membrane integrity, pigment content, osmotic adjustment water relations, and photosynthetic activity (Benjamin and Nielsen, 2006; Praba *et al.*, 2009). Drought stress is affected by climatic, edaphic and agronomic factors. The susceptibility of plants to drought stress varies in dependence of stress degree, different accompanying stress factors, plant species, and their developmental stages (Demirevska *et al.*, 2009). Acclimation of plants to water deficit is the result of different events, which lead to adaptive changes in plant growth and physio-biochemical processes, such as changes in plant structure, growth rate, tissue osmotic potential and antioxidant defenses (Duan *et al.*, 2007). It has become imperative to elucidate the responses and adaptation of crops to water deficit, and take actions to improve the drought resistance ability of crop plants and to ensure higher crop yields against unfavorable environmental stresses. Drought stress severely hampered the gas exchange parameters of crop plants and this could be due to decrease in leaf expansion, impaired photosynthetic machinery, premature leaf senescence, oxidation of chloroplast lipids and changes in structure of pigments and proteins (Menconi *et al.*, 1995). Plants grown under drought condition have a lower stomatal conductance which reduces the rate of CO₂ fixation and photosynthesis thereby resulting in lower assimilate production for growth and yield of plants (Kawamitsu *et al.*, 2000). Stomatal closure occurs during mild or moderate drought conditions thereby causing a reduction in leaf internal CO₂ concentration (C_i) and an ultimate decline in rate of photosynthesis (Chaves, 1991; Cornic, 2000; Flexas *et al.*, 2004). Besides, resistance offered by stomata to CO₂ entry is probably the main factor limiting photosynthesis under drought (Boyer, 1970). Under extreme water stress conditions, photosynthesis gets completely inhibited owing to alterations in chlorophyll content and damage of photosynthetic apparatus.

Biological control of dry root rot

The biological management of dry root rot has been attempted by many scientists in the past. According to the overall research done till date, it was found that the disease can be managed to a significant extent by use of fungicides, host plant resistance, biological means combined with other management strategies (Majumdar *et al.*, 1996; Cardoso *et al.*, 1997; Mandhare and Suryawanshi, 2009).

Earlier reports reveal that biocontrol agents like *Pseudomonas fluorescens* effectively colonize and reduce the germination of sclerotia of *M. phaseolina* and considered to be a potent tool agent against root rot fungus (Srivastava *et al.*, 2001). Other bioagents such as *T. viride*, *T. harzianum*, *B. subtilis* and certain botanicals have also been found to have a profound effect in management of the pathogen in chickpea or other crops (Singh *et al.*, 1998; Ahamad and Srivastava, 2000; Ray and Mukerjee, 2002; Naik *et al.*, 2009).

Singh *et al.* (1980) made another attempt in controlling dry root rot by coating seeds with *Bacillus* and *Streptomyces* spp. It was reported that dry root rot disease can be reduced significantly by coating chickpea seeds with certain isolates of *Bacillus* and *Streptomyces* spp. The results revealed that roots were infected only when *R. bataticola* was inoculated in soil and its management can be attributed to biological control. Singh *et al.* (2003) reported the efficacy of various biocontrol agents such as *Trichoderma harzianum*, *T. viride*, *T. hamatum*, *Gliocladium virens*, *Pseudomonas fluorescens* and *Bacillus subtilis* in controlling *R. bataticola* both *in vivo* and *in vitro*. Amongst all the bioagents tested, *T. harzianum* recorded highest control of the pathogen both in lab and field conditions.

Singh and Mehrotra (1980) investigated biological control of *R. bataticola* on chickpea and reported 4 bacteria and 6 actinomycetes proved antagonistic in culture with the exception of *Streptomyces* isolates, all these reduced disease symptoms and increased plant growth and dry matter when tested further by coating chickpea seeds sown in field soil inoculated with the pathogen. Haram *et al.* (1996) evaluated that *T. harzianum* is an efficient biological agent that is commercially produced to prevent development of several soil seed pathogenic fungi like *R. bataticola*, *Fusarium solani* and *F. oxysporum*. Different mechanisms have been suggested as being responsible for their biocontrol activity, which include competition for space and nutrients, secretion of chitinolytic enzymes, mycoparasitism and production of inhibitory compounds.

Major objectives of the study

Keeping in view the above discussed points along with the gaps still remaining to be resolved, the present research work was carried out with following objectives:

1. Collection, identification and purification of the pathogen causing dry root rot of chickpea from different districts of Eastern Uttar Pradesh.
2. Isolation and characterization of agriculturally important micro-organisms (AIMs) from Eastern Uttar Pradesh and Bundelkhand region.
3. To study the impact of individual and combined application of AIMs on chickpea growth and to elucidate their mode of action against *Rhizoctonia bataticola*.
4. To study the resistance response of chickpea under the pathogen challenged condition at physiological and biochemical level.
5. Adaptation of proteomic and genomic approach to understand the resistance response induced by AIMs in chickpea against *Rhizoctonia bataticola*.



CHAPTER II

REVIEW OF LITERATURE

REVIEW OF LITERATURE

2.1 Host crop- Chickpea

Chickpea (*Cicer arietinum* L.) is an annual grain legume crop that is used extensively for human consumption. It is known by various names such as horse gram, garbanzo bean, Bengal gram etc. In India it is popularly known as *chana* or gram. Chickpea is mainly grown owing to its high nutritional value. It is rich in dietary proteins and good for human consumption; moreover, its ability to form nitrogen-fixing nodules via interaction with rhizobia adds to its uniqueness (Ferguson *et al.*, 2010). It is a good source of carbohydrates and protein, together constituting about 80% of the total dry seed mass in comparison to other pulses (Chibbar *et al.*, 2010; Geervani, 1991). In addition, to it cholesterol free and is a good source of dietary fiber, vitamins and minerals (Wood and Grusak, 2007). In semi-arid tropics chickpea is an important component of the diets of those individuals who cannot afford animal proteins and are solely dependent on plant proteins.

2.2 History

Chickpea has been known to mankind for a long period of time. It is considered to be one of the oldest pulses known and cultivated from ancient times both in Asia and in Europe. The probable place of its origin lies in south western Asia such as Afghanistan and Persia. According to Aykroid and Daughy (1964), the centre of origin of chickpea is Eastern Mediterranean. However, De Candolle (1964) suggested that the crop has been under cultivation in India from ancient times longer than any country.

2.3 Botany

Chickpea is an annual cool-season plant with height ranging from one to three feet. It has an indeterminate and branched growth habit, erect or spreading, with hairy leaves, stems, and seed pods that secrete highly acidic exudates. Chickpea is classified as “kabuli-type” or “desi-type,” based primarily on seed size, shape, and color.

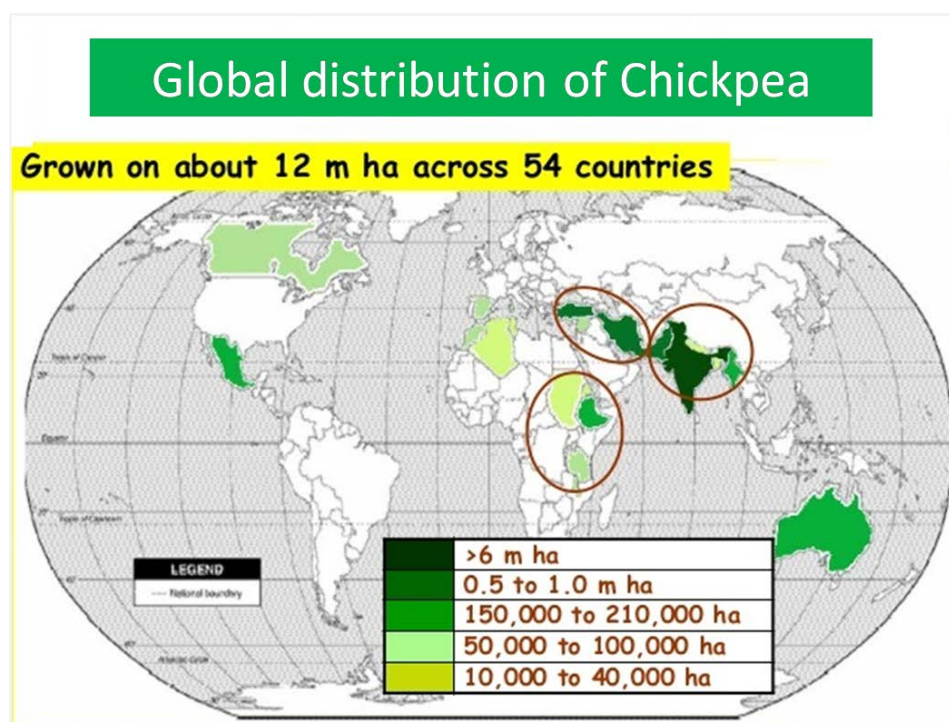
Collectively, both types of chickpea have been referred to as “garbanzo bean.” However several literatures mention “garbanzo bean” specifically for the *Kabuli* type varieties. The *Kabuli*-type chickpea seeds are about twice the size of field pea seed and averages about 1000 seeds per pound, with large variation between varieties and environments. They have a thin seed coat ranging in color from white to pale cream. However, Desi-type chickpea seeds are typically less than one-half the size (2,300 seeds per pound) of the *kabuli*-type chickpea and have a thicker, irregularly shaped seed coat ranging in color from light tan to black. It is believed that *Kabuli*-type varieties have evolved through natural mutation and selections from desi-type chickpea. Pollination in chickpea occurs through self pollination with flowers borne singly at the tip of auxiliary branches and varying in color from white, purple to faded blue. The seed pods usually develop on the top portion of the plant; about six to eight inches above the soil surface, and are relatively shatter resistant. The pods are short and inflated, with commercial types typically having one seed per pod. Each seed is characterized by a median groove around two-thirds of the seed and a “beak” that is formed by the protruding root tip of the exposed embryo.

2.4 Climatic conditions

Chickpea grows well under good moisture conditions with ideal temperature ranging from 24 °C to 30 °C. Chickpea is cultivated under both irrigated and rainfed conditions. The water use efficiency of chickpea plant depends upon various factors such as climatic conditions, soil type, and length of growing season. The plant performs well under annual rainfall between 400 and 600 mm; however its productivity may be enhanced through genotype selection and manipulation of planting density. Owing to its deep taproot, chickpea is moderately drought tolerant as it is able to extract moisture from deep layers of soil profile, but its productivity is reduced by the recurrence of the terminal droughts. The plant requires fertile, sandy-loam soil with good internal drainage and they do not tolerate water-logged conditions. Chickpeas grows well in soils having a pH range of 6.0- 7.0. It prefers soil with good residual soil moisture content or storage. The time of maturity depends on type and variety and ranges from 95 to 110 days.

2.5 Global distribution of chickpea

At present chickpea is grown in more than 50 countries across the Indian subcontinent, North Africa, Middle East, Southern Europe, America and Australia. Globally, chickpea is the third most important pulse crop in production, next to dry beans and field pea (FAOSTAT, 2011). South Asia is the largest producer of chickpea (68%) and India is the largest chickpea producer and consumer with an annual production of 7.70 million tonnes from 8.32 million hectares (FAOSTAT 2012). Asia accounts 89.7% of the area in chickpea production, followed by 4.3% in Africa, 2.6% in Oceania, 2.9% in Americas and 0.4% in Europe (Gaur, 2010). The other major chickpea producing countries include Pakistan, Turkey, Australia, Myanmar, Ethiopia, Iran, Mexico, Canada and USA.



Source: www.slideshare.net

Figure 2.1 Chickpea growing region throughout the world

In India chickpea is mainly grown in central and southern regions. Six states mainly Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Andhra Pradesh and

Karnataka are the leading producers which altogether contribute 91% of the produce and 90% of the area.

2.6 Dry root rot of Chickpea.

Dry root rot (DRR) of chickpea caused by *Rhizoctonia bataticola*, is one of serious disease of chickpea emerging as a potential threat to chickpea production (Pande *et al.*, 2010; Sharma *et al.*, 2010; Ghosh *et al.*, 2013). Savary *et al.* (2011) described DRR as an acute-emerging disease that occurs irregularly, both temporally and spatially, and may cause enormous disruptions in system performances with expanding ranges to new areas. The disease usually appears around flowering and podding time of the crop in the form of scattered dried plants. The disease may also appear at seedling stage; however, the old plants are more susceptible towards infection. The incidence of disease is usually seen when day temperature exceeds 30°C. The plants dry suddenly and affected plants are scattered in the field. The plants are usually straw coloured, however, in some cases the lower leaves and stems show a characteristic brown colour. Drooping of petioles and leaflets is confined to those at the top of the plant. Sometimes when the rest of the plant is dry, the topmost leaves are chlorotic. When the plants are uprooted, the lower portion of the tap root usually remains in the soil. The tap root becomes black, rotted with a low percentage the lateral and fine roots. The dead root is quite brittle and shows shredding of the bark. Dark minute sclerotial bodies seem to be exposed or inside the wood or bark (Nene *et al.*, 1991). When the dry stem of the collar region is split vertically, sparse mycelium or minute sclerotia can be seen in the pith. Sclerotial survivability is greatly reduced in wet soils as compared to dry soils at low moisture levels.

Microsclerotia serve as the primary source of inoculum and have the ability to persist in the soil for around 15 years (Short *et al.*, 1980). They are generally found in clusters on the soil surface and have been reported up to the depth of 0-20 cm in soil (Alabouvette, 1990; Campbell and Van der Gaag, 1993) and are well adapted to survive under adverse environmental conditions (Short *et al.*, 1980). The optimum temperature range for germination of microsclerotia is around 28-35°C (Mihail, 1989). The germ tube leads to formation of appressorium which penetrates

through the host epidermis by secreting cell wall degrading enzyme, although indirect penetration of the fungi through natural openings or wounds has also been reported (Bowers *et al.*, 1999; Mayek-Perez *et al.*, 2002). The mycelium colonizes the vascular tissue by growing through the cortex and finally enters the xylem vessel (Abawi and Pastor-Corrales, 1990). Once inside vascular tissues, the fungus spreads through the tap root and plugs the vessels resulting in wilting of the plant (Wyllie, 1988). The ability of *R. bataticola* to produce pycnidia depends on the host and the specific nature of the fungal isolate, which will determine the epidemiological role of conidia in the disease cycle (Ahmed and Ahmed, 1969).

The incidence of disease can be lowered to a considerable extent by targeting the microsclerotia or pycnidia in the soil which serve as primary inoculums. The survival of *R. bataticola* was reported from 2–15 years and was largely influenced by environmental conditions irrespective of association with the host tissues (Short, 1980; Baird *et al.*, 2003). There are reports of survival of the fungus in sorghum and corn residues, cucurbit roots under dry soil conditions for up to 10, 16 and 18 months, respectively (Ghaffar and Akhtar, 1968; Cook *et al.*, 1973). Repeated freezing and thawing of soil, low C: N ratio and soil moisture content are critical factors affecting microsclerotial survival (Dhingra and Sinclair, 1975). Low soil moisture condition enhances microsclerotia germination whereas high soil moisture has proved detrimental and reduced the survival of *R. bataticola* sclerotia in soil (Dhingra and Sinclair, 1977; Olaya and Abawi, 1996)

Very little is known about the epidemiology of the disease in relation to predisposition of chickpea plants by climate change variables (temperature and soil moisture). Limited literature on the effect of climatic factors indicates that hot, dry weather promotes infection and development of dry root rot (Bhatti and Kraft, 1997).

2.7 Variability among isolates of *R. bataticola*

The variation in the isolates of *R. bataticola* has been studied on the basis of several factors such as morphological, sclerotia formation, pigmentation, pathogenic and genetic characters etc. In a study conducted on pulse crop, eleven isolates of *R. bataticola* from different parts of the pulse crops have shown

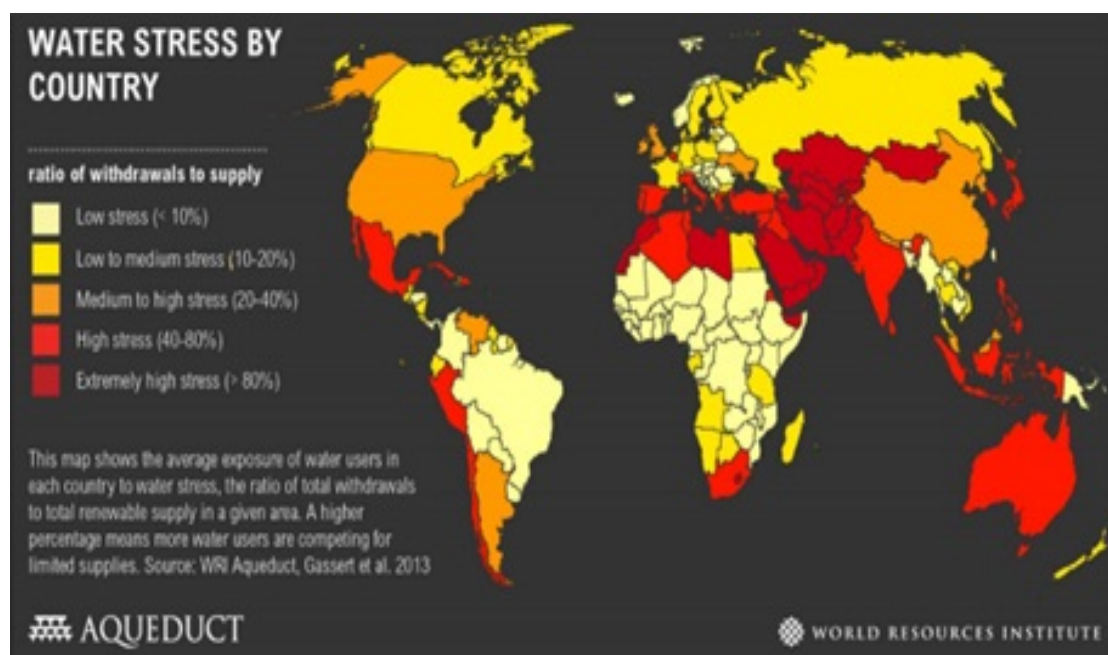
morphological variability, including sclerotial character and pigment productions (Sundravadana *et al.*, 2011). On the morphological characters the isolates were categorized into three groups such as linear, fluffy and linear growth with fluffy mycelial growth at centre. Mycelial growth rate was also classified as fast, moderate and slow. Similarly, Sobti and Sharma (1992), have categorized seven isolates of *R. bataticola* into various groups viz, black in centre and white in periphery, fluffy whitish with black periphery, charcoal black, grayish and submerged. with a black centre.

The degree of production of sclerotia is directly correlated with sclerotial production and pathogenicity of isolates. Based on sclerotial formation, the different isolates were divided as abundant, moderate and less sclerotia forming isolates. However, Jain *et al.* (1973), reported that the size of the sclerotia was found to be maximum in soil and the isolates recovered from stem. Byadgi and Hegde 1985, reported that Gliricidia isolate of *R. bataticola* produced the biggest size of sclerotia with a mean diameter of 101.51µm while cowpea isolate produced the smallest sclerotia (66.88 µm). RAPD-PCR has been successfully used to identify strains and races in phytopathogenic fungi (Williams *et al.*, 1990). It has also been used for studying inter- and intraspecific variability among populations from different as well as from the same geographic regions. The RAPD pattern analysis showed variations at the DNA level, and is thus suitable for differentiation of *M. phaseolina* isolates below species level (Franco *et al.*, 2006). Using PCR, closely related strains of a pathogen can be distinguished without prior knowledge of the nature of polymorphic regions by the use of RAPD. PCR-based DNA fingerprinting, particularly with short oligonucleotide primers, has been used by various researchers for the analysis of genetic variation in plant pathogens (Purkayastha *et al.*, 2006). The results presented here show that, in dendrogram analysis, two clusters were formed (A and B). Cluster A consists of blackgram root, shoot, seed isolates, soybean, root and shoot isolates, cowpea root and shoot isolates, green gram root and shoot isolates and redgram shoot isolate. Red gram root isolate alone, was grouped under cluster B.

Several workers have researched the genetic variability of *M. phaseolina* to work out the pathogenic and genetic patterns of diversity and genetic specialization (Jain *et al.*, 1973; Almeida *et al.*, 2003; Janar *et al.*, 2003). Further, fine tuned characterization of root rot pathogen, *R. bataticola*, and more improvement in pulse root rot resistance and varietal development strategies needs to be done. Finally, the experimental research such as morphological, pathological and genetic variability in *R. bataticola* isolates from pulse crops, significantly determines the effective, future management practices.

2.8 Effect of moisture stress on plants

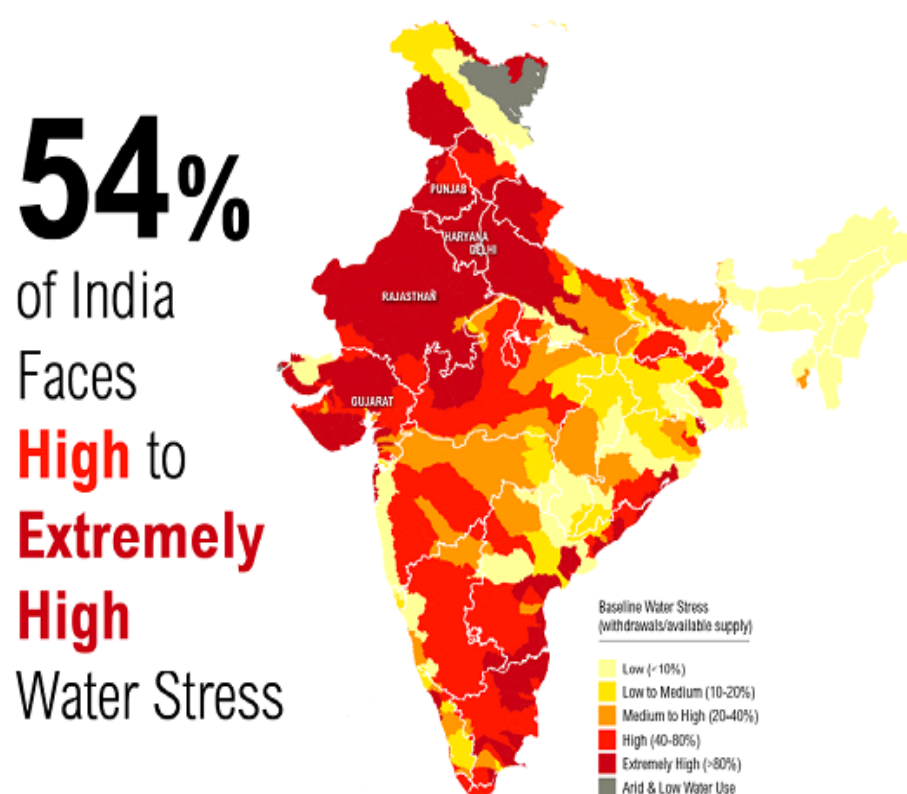
Water plays a key role in growth and development of an organism. It is required during all the growth stages but a few stages in every crop are crucially dependent on it and if water is not provided during those stages then the plant may succumb. Since our country is a highly populated one, a large thrust has been impeded on agriculture to meet the demands of population from limited land resource. Many agricultural crops worldwide are exposed to several abiotic stresses such as extremely high or low temperature, salinity, drought, acidic soils, and metal toxicity. Depending on the type of crop, such abiotic stresses result in yield losses between 50 and 82 % (Kang *et al.*, 2014). Abiotic stress conditions cause extensive losses to agricultural production worldwide (Boyers, 1982; Bray *et al.*, 2000). Individually, stress conditions such as drought, salinity or heat have been the subject of intense research (Bray *et al.*, 2000; Cushman and Bohnert, 2000). However, in the field, crops and other plants are routinely subjected to a combination of different abiotic stresses (Moffat, 2002; Heyne and Brunson, 1940; Craufurd and Peacock, 1993; Jiang and Huang, 2001). Among these various stresses, drought is a major abiotic factor limiting agricultural crop production.



Source: World Resource Institute

Figure 2.2 Regions in world affected by various degrees of water stress

India is highly prone to drought as 54% of the total area faces moderate to high water stress annually. Plants experience drought stress either when the water supply to roots becomes difficult or when the transpiration rate becomes very high. These two conditions often coincide under arid and semi-arid climates. Water stress tolerance is seen in almost all plant species but its extent varies from species to species. Although the general effects of drought on plant growth are fairly well known, the primary effects of water deficit at the biochemical and the molecular levels are not well understood (Zhu, 2002; Chaitanya *et al.*, 2003; Chaves *et al.*, 2003). In order to improve the agricultural productivity within the limited land resources, it is imperative to ensure higher crop yields against unfavorable environmental stresses. Understanding plant responses to the external environment is of greater importance and also a fundamental part for making the crops stress tolerant.



Source: www.induiawatertool.in

Figure 2.3 Regions in India affected by various degrees of water stress

2.9 Microbial management of abiotic stress in plants

Abiotic stresses affect plants in different ways and are causes of decline in crop productivity. In order to enhance crop productivity it becomes necessary to evolve efficient low-cost technologies for abiotic stress management. Microorganisms, the most natural inhabitants of diverse environments exhibit enormous metabolic capabilities to mitigate abiotic stresses. While microbial interactions with plants are an integral part of the living ecosystem, they are

considered as natural partners that modulate local and systemic mechanisms in plants to offer defense under adverse external conditions. Plant– microbe interactions comprise complex mechanisms within the plant cellular system. Biochemical, molecular and physiological studies are paving the way in understanding the complex but integrated cellular processes. Soil microorganisms, surviving in the soil under extreme conditions, have shown great potential which, if exploited can serve agriculture for increasing and maintaining crop productivity. It is well established that beneficial soil microorganisms can promote growth and increase productivity through mechanisms such as nutrient mobilization, hormone secretion and disease suppression; it is also becoming increasingly clear that their effects may be more far-reaching. Several studies have reported that soil microorganisms may have mechanisms for alleviation of abiotic stresses in plants such as water and temperature stress, salinity, heavy metals etc. Some of these include tolerance to salinity, drought (*Azospirillum* sp., *Pseudomonas syringae*, *P. fluorescens*, *Bacillus* sp.) and nutrient deficiency (*Bacillus polymyxa*, *Pseudomonas alcaligenes*). Other than bacteria, salinity and drought-tolerant isolates of *Trichoderma harzianum* and the effect of other strains of *Trichoderma* in amelioration of such abiotic stresses have also been reported. Arbuscular mycorrhizal fungi (*Glomus mosseae*, *G. etunicatum*, *G. intraradices*, *G. fasciculatum*, *G. macrocarpum*, *G. coronatum* etc.) help in alleviating abiotic stresses in different crops by enhancing nutrient uptake (phosphorus, nitrogen, magnesium and calcium), biochemical (accumulation of proline, betaines, polyamines, carbohydrates and antioxidants), and physiological, molecular and ultra-structural changes. The efficient applications of secondary metabolites of *Trichoderma* sp. have been described by Keswani *et al.* (2014). Microbial management is more preferred over other management strategies owing to its ecofriendly nature. Use of these microorganisms can alleviate stresses in agriculture thus opening a new and emerging application of microorganisms.

Table 2.1: Microorganisms conferring abiotic stress tolerance in crop plants

Organism	Crop	Type of Stress	Mechanism	Reference
<i>P. polymyxa</i> and <i>Rhizobium tropici</i>	Common bean	Drought	Change in hormone balance and stomatal conductance	Figueiredo <i>et al.</i> , 2008
<i>Pseudomonas</i> sp.	Pea	Drought	Decreased ethylene production	Arshad <i>et al.</i> , 2008
<i>Pseudomonas mendocina</i> and <i>Glomus intraradices</i>	Lettuce	Drought	Improved antioxidant status	Kohler <i>et al.</i> , 2008
<i>Pseudomonas putida</i> P45	Sunflower	Drought	Improved soil aggregation due to EPS production	Sandhya <i>et al.</i> , 2009a; 2009b
<i>Bacillus megaterium</i> and <i>Glomus</i> sp.	Trifolium	Drought	IAA and proline production	Marulanda <i>et al.</i> , 2007
<i>Pantoea agglomerans</i>	Wheat	Drought	Rhizosphere soil aggregation through EPS	Amellal <i>et al.</i> , 1998
<i>Paenibacillus polymyxa</i>	Arabidopsis	Drought	Induction of stress resistant gene ERD 15	Timmusk and Wagner, 1999
<i>Rhizobium</i> spp.	Sunflower	Drought	Soil aggregation through EPS	Alami <i>et al.</i> , 2000

Review of Literature

<i>Azospirillum</i> spp.	Wheat	Drought	Improved water relations	Creus <i>et al.</i> , 2004
<i>Variovorax paradoxus</i>	Pea	Drought	Synthesis of ACC deaminase	Dodd <i>et al.</i> , 2005
AM fungi	Sorghum	Drought	Improved water relations	Cho <i>et al.</i> , 2006
<i>Paraphaeosphaeria quadriseptata</i>	Arabidopsis	Drought	Induction of heat shock proteins	McLellan <i>et al.</i> , 2007
Brome mosaic virus	Rice	Drought	Unknown	Marquez <i>et al.</i> , 2007
<i>Burkholderia phytofirmans</i> <i>Enterobacter</i> sp. FD17	Maize	Drought	Increased photosynthesis, root and shoot biomass under drought conditions	Naveed <i>et al.</i> , 2014b
<i>Bacillus thuringiensis</i> AZP2	Wheat	Drought	Production of volatile organic compounds	Timmusk <i>et al.</i> , 2014
<i>Pseudomonas chlororaphis</i> O6	<i>Arabidopsis thaliana</i>	Drought	Production of 2R,3R butanediol-a volatile compound	Cho <i>et al.</i> , 2008
<i>Bacillus licheformis</i> strain K11	<i>Capsicum</i> sp.	Drought	Stress related genes and proteins	Lim and Kim, 2013
<i>Bacillus cereus</i>	Cucumber	Drought	Production of	Wang <i>et al.</i> ,

AR156, <i>B. subtilis</i> SM21 and <i>Serratia</i> sp. XY21			monodehydro ascorbate, proline, and antioxidant enzyme, expression of genes	2012
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2.10 Studies conducted on microbial management to abiotic stress

Microbial interactions with crop plants are key to the adaptation and survival of both the partners in any abiotic environment. Induced Systemic Tolerance (IST) is the term being used for microbe-mediated induction of abiotic stress responses. The role of microorganisms to alleviate abiotic stresses in plants has been the area of great concern in past few decades (de Zelicourt *et al.*, 2013; Nadeem *et al.*, 2014; Souza *et al.*, 2015). Microbes with their potential intrinsic metabolic and genetic capabilities, contribute to alleviate abiotic stresses in the plants (Gopalakrishnan *et al.*, 2015). The role of several rhizospheric occupants belonging to the genera *Pseudomonas* (Grichko and Glick, 2001; Ali *et al.*, 2009; Sorty *et al.*, 2016), *Azotobacter* (Sahoo *et al.*, 2014a,b), *Azospirillum* (Creus *et al.*, 2004; Omar *et al.*, 2009), *Rhizobium* (Alami *et al.*, 2000; Remans *et al.*, 2008; Sorty *et al.*, 2016), *Pantoea* (Amellal *et al.*, 1998; Egamberdiyeva and Höflich, 2003; Sorty *et al.*, 2016), *Bacillus* (Ashraf *et al.*, 2004; Marulanda *et al.*, 2007; Tiwari *et al.*, 2011; Vardharajula *et al.*, 2011; Sorty *et al.*, 2016), *Enterobacter* (Grichko and Glick, 2001; Nadeem *et al.*, 2007; Sorty *et al.*, 2016), *Bradyrhizobium* (Fugyeuredi *et al.*, 1999; Swaine *et al.*, 2007; Panlada *et al.*, 2013), *Methylobacterium* (Madhaiyan *et al.*, 2007; Meena *et al.*, 2012), *Burkholderia* (Barka *et al.*, 2006; Oliveira *et al.*, 2009), *Trichoderma* (Ahmad *et al.*, 2015) and cyanobacteria (Singh *et al.*, 2011) in plant growth promotion and mitigation of multiple kinds of abiotic stresses has been documented.

Drought stress is among the most destructive abiotic stresses that increased in intensity over the past decades affecting world's food security. Drought stress may range from moderate and short to extremely severe and prolonged duration, restricting

the crop yields (Austin, 1989; Pereira and Chaves, 1993, 1995; Bottner *et al.*, 1995). Drought is expected to cause serious plant growth problems for more than 50% of the arable lands by 2050 (Ashraf, 1994; Vinocur and Altman, 2005; Kasim *et al.*, 2013). The mechanism of plant drought tolerance induced by PGPR include: (1) production of phyto hormones like abscisic acid (ABA), gibberellic acid, cytokinins, and indole-3-acetic acid (IAA); (2) ACC deaminase to reduce the level of ethylene in the roots; (3) induced systemic tolerance by bacterial compounds; (4) bacterial exopolysaccharides (Yang *et al.*, 2009; Dimkpa *et al.*, 2009; Timmusk and Nevo, 2011; Kim *et al.*, 2013; Timmusk *et al.*, 2014). The very first report on plant drought tolerance enhancement by plant growth promoting rhizobacteria (PGPR) was published in Uppsala, Sweden (Timmusk and Wagner, 1999). It was reported that *Arabidopsis thaliana* inoculated with PGPR *Paenibacillus polymyxa* B2 could survive drought stress remarkably longer compared to the untreated control plant.

Plant-growth-promoting rhizobacteria (PGPR) colonize the rhizosphere of many plant species and confer beneficial effects, such as increased plant growth and reduced susceptibility to diseases caused by plant pathogenic fungi, bacteria, viruses and nematodes (Kloepper *et al.*, 2004). Some PGPR also elicit physical or chemical changes related to plant defense, a process referred to as ‘induced systemic resistance’ (ISR) (Van Loon *et al.*, 2004). ISR elicited by PGPR has suppressed plant diseases caused by a range of pathogens in both the greenhouse and field (Kloepper *et al.*, 2004; Van Loon *et al.*, 2004). However, fewer reports have been published on PGPR as elicitors of tolerance to abiotic stresses, such as drought, salt and nutrient deficiency or excess. Drought stress limits the growth and productivity of crops, particularly in arid and semi-arid areas (Kramer and Boyer, 1995). Early studies on IST to drought (Timusk and Wagner, 1999) reported that inoculation with the PGPR *Paenibacillus polymyxa* enhanced the drought tolerance of *Arabidopsis thaliana*. RNA differential display on parallel RNA preparations from *P. polymyxa* treated and untreated plants revealed that mRNA transcriptions of a drought-response gene, EARLY RESPONSIVE TO DEHYDRATION 15 (ERD15), were also augmented. Another PGPR strain, *Achromobacter piechaudii* ARV8 which produces 1 aminocyclopropane-1-carboxylate (ACC) deaminase, conferred IST to drought stress in pepper (*Capsicum annuum* L.) and tomato (*Solanum lycopersicum* L.) plants

(Mayak *et al.*, 2004). Under stress conditions, including drought, the plant hormone ethylene endogenously regulates plant homeostasis and results in reduced root and shoot growth (Glick *et al.*, 2007). However, degradation of the ethylene precursor ACC by bacterial ACC deaminase releases plant stress and rescues normal plant growth (Glick *et al.*, 2007). Under drought stress, co-inoculation of bean (*Phaseolus vulgaris* L.) with *Rhizobium tropici* and two strains of *P. polymyxa* resulted in augmented plant height; shoot dry weight and nodule number (Figueiredo *et al.*, 2008). Interestingly, the effect on IST and increased nodule number was greater when a mix of two strains of *P. polymyxa* was applied than when one strain was applied, suggesting some synergistic effects from the use of strain mixtures. Co-inoculation of lettuce (*Lactuca sativa* L.) with PGPR *Pseudomonas mendocina* and arbuscular mycorrhizal fungi (*Glomus intraradices* or *G. mosseae*) augmented an antioxidant catalase under severe drought conditions, suggesting that they can be used in inoculants to alleviate the oxidative damage elicited by drought (Kohler *et al.*, 2008).

Bacteria occurring on root surfaces containing ACC deaminase have been shown to modify the sensitivity of root and leaf growth to soil drying, apparently by influencing ethylene signalling. The ACC deaminase activity of *Achromobacter piechaudi* was shown to confer tolerance to water deficit in tomato and pepper, resulting in significant increases in fresh and dry weights. Ethylene production was reduced in inoculated plants compared to non-inoculated controls, with improved recovery from water deficiency, although inoculation did not influence relative water contents (Mayak *et al.*, 2004b).

Under water deficiency, maize seedlings inoculated with *Azospirillum brasilense*, however, displayed improved relative and absolute water contents, in comparison to non inoculated treatments. Bacterial inoculation has also been shown to prevent a significant drop in water potential, in parallel with a concomitant increase in root growth, total aerial biomass and foliar area, as well as proline accumulation in leaves and roots. These effects were more significant at 75% reduction in water supply, compared to 50% reduction (Casanovas *et al.*, 2002). Creus *et al.* (2004) reported on reduced grain yield losses and higher Mg, K and Ca contents in grains of *Azospirillum*-inoculated wheat exposed to water deficit. Increases in water content, relative water content, water potential and apoplastic water fraction were observed.

Furthermore, lower volumetric cell wall moduli of elasticity values were measured. Therefore, in addition to a better water status, an ‘elastic adjustment’ is hypothesized to be crucial in increased drought tolerance.

The alteration in the morphology of host plant upon *Azospirillum* treatment is mainly attributed to the plant’s enhanced tolerance to drought. It is believed that the bacteria produce hormone-like substances having ability to stimulate endogenous hormone levels (Dobbelaere *et al.*, 1999; Cassan *et al.*, 2001). However, another experiment revealed that, under aerobic conditions, *A. brasilense* produces significant amounts of the small diffusible gas, nitric oxide, which has been shown to act as a signalling molecule in an IAA induced pathway involved in adventitious root development (Creus *et al.*, 2005; Molina-Favero *et al.*, 2008). Nitric oxide is involved in the enhancement of lateral root and root hair development in *Azospirillum*-inoculated tomato plants. As with *Azospirillum*, recent studies on droughted pea inoculated with ACC deaminase activity-containing *Variovorax paradoxus*, as against an ACC deaminase mutant strain, showed hormone signalling-mediated plant growth improvement, yield and water-use efficiency (Belimov *et al.*, 2009). At the transcriptional level, the bacterium *P. polymyxa* engendered the induction of a drought-responsive gene, *ERD15*, isolated from drought-stressed *A. thaliana* (Timmusk and Wagner, 1999). Similarly, Rocha *et al.* (2007) reported the differential expression by drought treatment, of as many as 93 sugarcane genes, including well-known drought-responsive genes such as *MRB* and *WRKY* transcription factors; however, treatment of the same plant with beneficial endophytic bacteria (*Herbaspirillum* spp. and *Gluconacetobacter diazotrophicus*) resulted in the induction of resistance (R) and salicylic acid biosynthesis genes.

Plant species inoculated with IAA producing bacteria increased root growth and helps in formation of lateral roots and roots hairs (Dimkpa *et al.*, 2009) thus facilitate more water and nutrient uptake (Mantelin and Touraine, 2004). Bacterial hormone production and their ability to stimulate endogenous hormones play the key role in enhancing tolerance (Dobbelaere *et al.*, 1999; Cassan *et al.*, 2001). *Azospirillum brasilense* Sp245-inoculated seeds of *Triticum aestivum* showed higher osmotic adjustment, increased water content, relative water content, water potential, apoplastic water fraction, and lower cell wall modulus of elasticity values under

drought conditions. Grain yield loss to drought was 26.5% and 14.1% in noninoculated and *Azospirillum* inoculated plants, respectively (Creus *et al.*, 2005; Molina-Favero *et al.*, 2008). Similarly, inoculation with *A. brasilense* Cd increased root projection area, specific root length and specific root area, as compared with non-inoculated controls of common bean (*Phaseolus vulgaris* L.) under drought stress (German *et al.*, 2000). *A. brasilense* inoculation in maize seedlings improved relative and absolute water contents compared to non-inoculated plants under drought stress. Inoculation of *A. brasilense* Sp245 in wheat (*Triticum aestivum*) under drought stress resulted in better grain yield and higher mineral quality (Mg, K and Ca), with improved relative and absolute water content, water potential, apoplastic water fraction and lower volumetric cell wall elasticity suggesting that, in addition to better water status, an 'elastic adjustment' is crucial in increased drought tolerance in plants (Creus *et al.*, 2004). Similarly, *Azospirillum*-wheat association facilitate increase in leaf water content, which was attributed to the production of plant hormones such as IAA by the bacteria that enhanced root growth and formation of lateral roots their by increasing uptake of water and nutrients under drought stress (Arzanesh *et al.*, 2011). *B. thuringiensis* promoted growth of *Lavandula dentate* through production of IAA by the bacterium which aid in improving nutrition, physiology, and metabolic activities of plant (Armada *et al.*, 2014). Sang-Mo *et al.* (2014) reported that Soybean plants inoculated with *P. putida* strain H-2-3 secretes gibberellins which felicitate improved plant growth under drought conditions. *Azospirillum lipoferum* alleviate drought stress in maize plants through production of ABA and gibberellins (Cohen *et al.*, 2009). During water deficit condition, biosynthesis of ABA takes place, which plays a key role in regulating water loss by controlling stomatal closure and stress signal transduction pathways (Yamaguchi-Shinozaki and Shinozaki, 1994; Kaushal and Wani, 2015). *Phyllobacterium brassicacearum* strain STM196 enhance osmotic stress tolerance in Arabidopsis plants by raising ABA content, leading to decreased leaf transpiration (Bresson *et al.*, 2013). Similarly, *Platycladus orientalis* seedlings on treatment with *Bacillus subtilis* elevated the levels of ABA in shoots and increased the stomatal conductance conferring drought stress resistance.

When plants are subjected to multiple stress stress conditions, induction of volatile compounds takes place (Loreto and Schnitzler, 2010; Holopainen and

Gershenzon, 2010). These stress-induced volatile compounds serve as signals for developing priming and systemic responses within the same and in neighboring plants (Heil and Silva Bueno, 2007; Choudhary *et al.*, 2008; Niinemets, 2010). Treatment with *Bacillus thuringiensis* AZP2 in wheat seedlings under drought stress resulted in increased biomass and higher survival due to significant reduction of volatile emissions and higher photosynthetic rate (Timmusk *et al.*, 2014).

Ethylene is an important component regulating plant activities and its biosynthesis is regulated by biotic and abiotic stresses (Hardoim *et al.*, 2008). Under stress conditions, the plant hormone ethylene endogenously regulates plant homeostasis resulting in reduced root and shoot growth. Rhizosphere bacteria *Variovorax paradoxus* 5C-2 containing the enzyme ACC deaminase was tested to overcome the growth inhibition of Pea. ACC deaminase has a significant role in decreasing plant levels of the ethylene precursor ACC (1 aminocyclo-propane carboxylic acid). Upon treatment it was found that the bacterium stimulated root biomass by 20-25% irrespective of soil moisture regime, and whole plant biomass was stimulated also by 25% in plants grown in drying soil (Dodd *et al.*, 2005). Plant ACC is sequestered and degraded by ACC deaminase producing bacteria to supply nitrogen and energy. Removal of ACC helps in reducing the deleterious effect of ethylene, ameliorating plant stress and promoting plant growth (Glick, 2005). Similarly, the inoculation with *Pseudomonas* spp. containing ACC-deaminase partially eliminates the effects of drought Stress on growth, yield, and ripening of pea. The results revealed that inoculation of peas with *Pseudomonas* sp. containing ACC-deaminase significantly minimized the drought effects on the growth and yield of peas. Exposure of plants to drought stress at vegetative growth stage significantly decreased shoot growth by 41% in the case of uninoculated plants, whereas, by only 18% in the case of inoculated plants. The inoculated plants also achieved higher grain yield *i.e* 62% in comparison with uninoculated control. Ripening of pods was also delayed in plants inoculated with PGPR, which signify less ethylene production in inoculated plants. The seedlings treated with *A. piechaudii* ARV8, were found to be significantly better in growth compared to *P. putida* GR12-2 (Lifshitz *et al.*, 1986).

Coinoculation of ACC deaminase producing *Bacillus* isolate 23-B and *Pseudomonas* 6-P with *Mesorhizobium ciceris* for mitigation of drought stress and

plant growth promotion under axenic conditions on *Cicer arietinum* varieties (Kabuli L-552 and Desi GPF-2) significantly improved germination, root and shoot length and fresh weight of chickpea seedlings compared to uninoculated control. Proline content was higher in bacteria treated varieties under water stress. Similarly, pepper inoculated with *Bacillus licheniformis* K11 alleviated drought stress and lead to expression of six stress related protein (Hui and Kim, 2013).

Increased proline content due to up regulation of proline biosynthesis pathway keeps proline in high levels to help in maintaining cell water status, protects membranes and proteins from stress (Sandhya *et al.*, 2010). Similarly, tomato (*Lycopersicon esculentum* Mill) variety Anakha treated with *Bacillus polymyxa* found to produce excess proline in order to cope up with the drought condition (Shintu and Jayaram, 2015). Higher proline accumulation in inoculated plants is an indicator of enhance plant tolerance to water stress (Gusain *et al.*, 2015). Choline plays a critical role in providing resistance to plant stress, by enhancing glycine betaine (GB) synthesis and accumulation (Zeisel, 2006; Zhang *et al.*, 2010). Reports on induced role of *B. subtilis* GB03 in Arabidopsis (Zhang *et al.*, 2010) and *Klebsiella variicola* F2, *P. fluorescens* YX2 and *Raoultella planticola* YL2 in maize showed elevated levels of choline biosynthesis which aid in higher leaf relative water content (RWC) and dry mater weight (Glick *et al.*, 2007; Zhang *et al.*, 2010; Gou *et al.*, 2015).

Plants treated with exopolysaccharide (EPS) producing bacteria display increased resistance to water stress due to improved soil structure (Sandhya *et al.*, 2009b). Sunflower seedlings upon treatment with *Pseudomonas putida* strain P45 helps to mitigate drought stress significantly. Inoculation of *Pseudomonas* sp. strain increased the survival, plant biomass, and root adhering soil/root tissue ratio of sunflower seedlings subjected to drought stress. EPS can also bind to cations including Na thus making it unavailable to plants under saline conditions. The prime mechanism behind alleviation of drought stress is advocated to production of EPS, which protects microorganisms from hydric stress and fluctuations in water potential by enhancing water retention and regulating the diffusion of carbon sources in microbial environment. Similarly, the role of EPS in mitigating drought stress was proved in another experiment conducted by Alami *et al.*, (2000). It was reported that Sunflower seedlings upon inoculation with *Rhizobium* sp. strain YAS34, helps to

mitigate drought stress to a considerable extent through production of EPS. Strain YAS34 was also found to lessen the effect of water stress on sunflower growth, which is particularly essential in tropical zones, where prolong dry period spells occur. Amellal *et al.*, (1998) reported the effect of EPS secretion by *Pantoea agglomerans* strain NAS206 on rhizosphere soil physical properties. Inoculation of wheat seedlings with strain NAS206 facilitated increased colonization at rhizoplane and in root-adhering soil (RAS). Inoculated strain NAS206 improved RAS macroporosity (pore diameter, 10-30 mm) compared to the noninoculated control, particularly when the soil was nearly water saturated.

Cho *et al.* (2008) reported that treatment of *Arabidopsis thaliana* by *Pseudomonas chlororaphis* O6 induced tolerance in host plants against biotic and abiotic stresses due to the production of a volatile metabolite, 2R, 3R-butanediol. Studies with *Arabidopsis* mutant lines indicated that the role of salicylic acid (SA), ethylene and jasmonic acid-signalling pathways in inducing drought tolerance.

Induction of drought tolerance in cucumber plants by a consortium of three PGPR Strains was reported by Wang *et al.* (2012). A consortium comprising three PGPR strains *i.e* *Bacillus cereus* AR156, *Bacillus subtilis* SM21 and *Serratia* sp. XY21, named BBS proved to a promising biocontrol agent in mitigating drought stress in cucumber plants. BBS treated plants showed increased leaf proline content and root growth by 3.45-fold and 50%, respectively; and also maintained the leaf chlorophyll content in cucumber plants under drought stress. Besides, in relation to the control, the BBS treatment significantly enhanced the superoxide dismutase (SOD) activity and mitigated the drought-triggered down-regulation of the expression of the genes *cAPX*, *rbcL*, and *rbcS* encoding cytosolic ascorbate peroxidase, and ribulose-1, 5-bisphosphate carboxy/oxygenase (Rubisco) large and small subunits, respectively, in cucumber leaves. These results revealed that the consortium induced systemic tolerance to drought stress in cucumber plants, by protecting plant cells, maintaining photosynthetic efficiency and root vigor and increasing antioxidase activities.

The effect of inoculation of two bacterial endophytes *Burkholderia phytofirmans* strain PsJN and *Enterobacter* sp. FD17 on growth, water status and photosynthetic activity of two maize cultivars under drought stress conditions was

investigated by Naveed *et al.* (2014). The results revealed that bacterial inoculation significantly minimized the drought stress-imposed effects in the treated plants by increasing, root biomass, shoot biomass, leaf area, chlorophyll content, photosynthesis, and photochemical efficiency of PSII. Marulanda *et al.* (2007) studied the effect of four arbuscular mycorrhizal (AM) fungal isolates (two autochthonous presumably drought-tolerant *Glomus* sp and two allochthonous presumably drought-sensitive strains) on Lavender plants (*Lavandula spica*) growing under drought conditions. It was found that the autochthonous strains increased plant biomass concurrently with an increase in N, K and water content and steep decline in the antioxidant compounds which are thought to play an important role in plant protection and metabolic functioning under drought conditions.

Lim and Kim, (2013) studied the effect of *B. licheniformis* K11 on pepper plant in mitigating drought stress. *B. licheniformis* K11 was found to reduce ethylene concentration of pepper plants by cleaving ACC under drought stress and hence increased plant growth.

2.11 Studies on gene expression in chickpea

Upasani *et al.* (2016) studied gene expression in against Fusarium wilt caused by *Fusarium oxysporum* f.sp. *ciceri* (*Foc*). They transformed *Foc* race 2 using green fluorescent protein (GFP) gene and used it to characterize pathogen progression and colonization in wilt-susceptible (JG62) and wilt-resistant (Digvijay) chickpea cultivars using confocal microscopy. They also employed quantitative PCR (qPCR) to estimate the pathogen load and progression across various tissues of both the chickpea cultivars during the course of the disease. Additionally, the expression of several candidate pathogen virulence genes was analyzed using quantitative reverse transcriptase PCR (qRT-PCR), which showed their characteristic expression in wilt-susceptible and resistant chickpea cultivars. They suggested that the pathogen colonizes the susceptible cultivar defeating its defense; however, albeit its entry in the resistant plant, further proliferation is severely restricted providing an evidence of efficient defense mechanism in the resistant chickpea cultivar.

Cho and Muehlbauer (2004) elucidated defense responses in chickpea (*Cicer arietinum* L.) against two fungal pathogens, *Ascochyta rabiei* (Pass.) Labr. causing

ascochyta blight and *Fusarium oxysporum* f. sp. *ciceri* (*Foc*) causing fusarium wilt. They investigated the expression patterns of defense related genes in chickpea after pathogen inoculation and exogenous treatments with systemic signals such as SA and Me-JA. Two blight differential germplasm lines, FLIP84-92C(2) (blight resistant and SA- and Me-JA-sensitive) and PI359075(1) (blight susceptible and SA- and Me-JA-insensitive) showed significant differential expression patterns of the defense-related genes after *A. rabiei* inoculation and exogenous treatment with SA and Me-JA. However, blight resistance in the recombinant inbred lines generated from the cross of the two germplasm lines did not co-segregate with the expression of the genes induced either by pathogen inoculation or by signal chemicals. Fusarium wilt resistance in chickpea also did not require induction of the defense-related genes after *Fusarium* infection. These results revealed that systemic regulation of the defense-related genes at transcription level which is reportedly associated with disease resistance in other model plant species such as *Arabidopsis* might not confer resistance in chickpea against two necrotrophic fungal pathogens, *A. rabiei* (Pass.) Labr and *Foc*.

Gurjar *et al.* (2012) studied gene expression analysis of chickpea roots during *Fusarium oxysporum* f. sp. *ciceri* (*Foc*) infection using various approaches. cDNAs derived from total mRNA during infection process of susceptible (JG62) and resistant (Digvijay) cultivars, were amplified using random oligonucleotides. Sequence characterization of differentially expressed transcripts revealed their homology with many plant genes essential for various metabolic functions including defense. Further, expression patterns of specific candidate gene transcripts were analyzed in the *Foc* inoculated and uninoculated resistant and susceptible chickpea cultivars, on day 6 of infection. Semiquantitative RT-PCR analysis of defense related genes was performed using gene specific oligonucleotides in resistant and susceptible chickpea cultivars. The expression of fungal pathogenesis related genes and their race specific response was determined throughout the course of chickpea-*Foc* interaction. Temporal expression and race specific response of plant defense related and fungal virulence genes were studied in the resistant and

susceptible cultivars of chickpea inoculated with three races of *Foc* highlighting the host-pathogen interactions.

Kavousi *et al.* (2009) studied expression analysis of genes induced in general defense response which provided clues to elucidate major defense mechanisms against pathogen infection in chickpea plants. The role of key phenylpropanoid pathway enzymes response to *A. rabiei* in chickpea was studied using a reverse transcription and semi-quantitative polymerase chain reaction (SQ-PCR). Transcript accumulation of four genes encoding phenylalanine ammonia-lyase (PAL), chalcon synthase (CHS), isoflavone reductase (IFR) and Flavanone 3-Hydroxylase (F3H) induced in response to race 3 of *A. rabiei* was compared in resistant and susceptible genotypes. Results obtained in this study showed that in resistant genotype all 4 phenylpropanoid pathway genes: PAL, CHS, IFR and F3H were rapidly up regulated 6 h after inoculation with race 3 of *A. rabiei*. However, transcripts of PAL and IFR genes were rapidly accumulated in both resistant and susceptible cultivars. Therefore, induction of key enzymes of phenylpropanoid pathway appeared to be an important defense mechanism of chickpea plants against *A. rabiei*.

Garcia-Limones *et al.* (2009) compared the activity levels of oxidative stress-related enzymes in the root apoplast during the interaction of WR315 (resistant) and JG62 (susceptible) chickpeas with the highly virulent race 5 of *Fusarium oxysporum* f. sp. *ciceri*. The interactions were compared at three specific stages during disease development in JG62: (i) before symptom development (10 days after inoculation); (ii) at the time of appearance of the first disease symptoms (15-17 days after inoculation) and (iii) when all plants had developed disease symptoms (20-22 days after inoculation). Diamine oxidase (DAO), ascorbate peroxidase (APX), glutathione reductase (GR), guaiacol-dependent peroxidase and superoxide dismutase (SOD), but not catalase (CAT), were found in the apoplast of chickpea roots. In terms of APX activity, infection by the pathogen caused a different response in the incompatible compared to the compatible plant. In the case of GR, SOD and DAO activities, the pathogen caused the same response, but it developed earlier (i.e. GR and SOD) or to higher levels (i.e. DAO) in the incompatible interaction. Expression of *apx*, *cat*, *sod*, *lipoxygenase* (*lox*) and *actin* genes was also analysed in infected roots. Infection

by *F. oxysporum* f. sp. *ciceri* race 5 only caused a significant change in the root expression of *lox* and *actin* genes. This up-regulation was earlier (*lox*) or higher (*actin*) in the incompatible than in the compatible interaction. Thus, changes in oxidative metabolism differ in compatible and incompatible interactions in *Fusarium* wilt of chickpea.

Arfaoui *et al.* (2007) investigated differential expression of phenylalanine ammonia-lyase (PAL), chalcone synthase (CHS) and isoflavone reductase (IFR) genes involved in phenylpropanoids metabolism using Northern blot analyses in chickpea seedlings bacterized with *Rhizobium* isolates (PchDMS and Pch43) and further challenged with *Fusarium oxysporum* f. sp. *ciceri* (Foc) race 0. Gene activation patterns in the moderately resistant accession INRAT87/1 were compared with those exhibited by the susceptible accession ILC482 at various time intervals after inoculation with Foc, to determine whether differences in levels or timing of transcript accumulation could be correlated with differences in the susceptibility of chickpea accessions to Foc. Results showed that the gene activation was higher in the moderately resistant accession INRAT87/1 than in the susceptible ILC482. Pre-treatment of chickpea seedlings with *Rhizobium* isolates before inoculation with the pathogen enhanced the accumulation of the three genes' mRNA transcripts. In parallel, changes in the soluble phenolic pool produced through pathways involving these enzymes were analysed in chickpea roots. A strong accumulation of these compounds was revealed at 72 hpi in both accessions. After that time, these high levels of phenolic compounds were maintained until the end of the experiment in the moderately resistant accession, while they have significantly declined in the susceptible accession. HPLC analyses revealed a very high accumulation of the constitutive isoflavones, formononetin and biochanin A and their glycoside conjugates in chickpea roots inoculated with *Rhizobium* isolates and/or challenged with Foc, as compared to the controls. Their results suggest that the increased accumulation of phenolic compounds, observed in chickpea seedlings inoculated with Foc, can be attributed to increased expression of genes in the phenylpropanoid pathway and that such gene expression is enhanced by pre-treatment with *Rhizobium* isolates.

Zarandi *et al.* (2011) studied the role of chitinases in the interaction between chickpea and pathogens. The expression of two classes of chitinase, I and III, was monitored in infected and non-infected plants with different disease resistance (resistant MCC10 and sensitive MCC724 genotypes). The expression levels of these genes in chickpea were studied at different times after inoculation with *F. oxysporum* f. sp. *ciceri* isolate 85. Semi-quantitative reverse transcription PCR was used to study the differential induction of the chitinase genes. Expression levels of chitinase I in non-infected and infected MCC724 plants were higher than those in MCC10. Since chitinase III may play an effective role in plant defense, its expression levels were quantified by reverse transcription PCR. Chitinase III was induced in MCC724 within 6 h and in MCC10 in 48 h after inoculation. However, the expression levels of chitinase III in MCC10 non-infected plants were significantly higher than in MCC724. The results showed that chitinase III can be effective to enhance resistance to *F. oxysporum* f. sp. *ciceri* isolate 85.

2.12 Management of dry root rot

Chickpea (*Cicer arietinum* L.) is the oldest and third most important pulse crop in world. The production of chickpea in the Indian continent and other countries in Asia is severely affected by many plant pathogenic fungi, bacteria, virus, and nematodes which cause the diseases such as Fusarium wilt, dry root rot, Colletotrichum blight, collar rot, bacterial blight, filiform virus and dirty root nematode. Among the several constraints affecting the productivity of chickpea, 10-35 per cent loss in yields are due to wilt and dry root rot diseases (Mahendra aPal, 1998). Thus management of the pathogen is very important because once it does get established in the field, can cause severe crop losses.

Several scientists have attempted to manage the dry root rot pathogen. However, the disease can be managed to a considerable extent by use of fungicides, host plant resistance, biological means combined with other management strategies (Majumdar *et al.*, 1996; Cardoso *et al.*, 1997; Mandhare and Suryawanshi, 2009).

Development of host plant resistance is an important method in combating pathogen attack. However, specific resistance to DRR in chickpea has been found

ineffective so far, as none of the lines showed consistent resistant reactions to the disease. Many researchers attempted for screening various germplasm and breeding lines (Nene *et al.*, 1981; Gurha *et al.*, 2003; Ashraf *et al.*, 2005). However, only four genotypes (GCP- 101, GBM-2, GBM-6, and ICCV-10) were found to be disease tolerant. Moreover, few resistant sources for DRR have also been reported by Gangwar *et al.*, (2002); Prajapati *et al.*, (2003); Pande *et al.*, (2006); Gupta *et al.*, (2012b); and Khan *et al.*, (2013). Few resistant sources of DRR/wilt complex were reported by Baker and Ahmed (1991) from Bangladesh. Some of the germplasm lines such as ICCV 05530, ICCV 05529, ICCV 08305, and ICCV 05532 have displayed a moderate level of resistance to DRR.

Earlier reports reveal that biocontrol agents like *Pseudomonas fluorescens* effectively colonize and reduce the germination of sclerotia of *M. phaseolina* and are considered to be a potent tool agent against root rot fungus (Srivastava *et al.*, 2001). Other bioagents such as *T. viride*, *T. harzianum*, *B. subtilis* and certain botanicals have also been found to have a profound effect in management of the pathogen in chickpea or other crops (Singh *et al.*, 1998; Ahamad and Srivastava, 2000; Ray and Mukerjee, 2002; Naik *et al.*, 2009).

Haram *et al.* (1996) evaluated that *T. harzianum* is an efficient biological agent that is commercially produced to prevent development of several soil seed pathogenic fungi like *R. bataticola*, *Fusarium solani* and *F. oxysporum*. Different mechanisms have been suggested as being responsible for their biocontrol activity, which include competition for space and nutrients, secretion of chitinolytic enzymes mycoparasitism and production of inhibitory compounds.

Parakhia and Vaishnow, (1986) reported that when chickpea seeds were pretreated with *T. harzianum* before sowing in pot inoculated with *R. bataticola* infection reached 18%. When the antagonist was applied as a soil drench, disease levels were 28% and incorporation of wheat husk bran cultures gave levels of 14% compared to 70% in the untreated controls. Kumar and Khare, (1990) investigated the antagonistic relationship of soybean with *R. bataticola* and *Sclerotium rolfsi*. It was inferred that the population increased and decreased

due to the antagonistic activity of *T. harzianum* and *B. subtilis*. Similarly, Singh and Mehrotra (1980) made another attempt in controlling dry root rot by coating seeds with *Bacillus* and *Streptomyces* spp. It was found that dry root rot disease can be reduced significantly by coating chickpea seeds with certain isolates of *Bacillus* and *Streptomyces* spp.

A significant control of the root disease of gram associated with *R. bataticola* was achieved by organic amendment of soil with wheat straw, maize straw, and sorghum straw whereas amending the same soil with legumes *i.e.* mung, urd and *Trifolium* straws was found to be ineffective and increase the disease incidence (Singh D, 1976).

Singh *et al.* (1992) reported that integrated application of various chemicals resulted in better control of DRR in chickpea compared to single applications. Higher yield was obtained upon pretreatment of seeds with carbendazim + Thiram, followed by 2 sprays of Endosulfan or monocrotophos. However, Singh *et al.* (2003) reported the efficacy of various biocontrol agents such as *Trichoderma harzianum*, *T. viride*, *T. hamatum*, *Gliocladium virens*, *Pseudomonas fluorescens* and *Bacillus subtilis* in controlling *R. bataticola* both *in vivo* and *in vitro*. Amongst all the bioagents tested, *T. harzianum* recorded highest control of the pathogen both in lab and field conditions.

Chickpea seeds on treatment with carbendazim, thiophanate methyl and vitavax reduced the incidence and severity of DRR of chickpea significantly over untreated check (Taya *et al.*, 1990; Bhardwaj 1995; Singh & Sindhan 1998; Rathore & Rathore 1999; Sharma & Gupta 2004). The combined use of host resistance with fungicide treatment resulted in better seedling emergence and delayed the onset of root rots. Gurha *et al.* (2003) reported that treating seeds with captan or thiram is also helpful in reducing the disease. In addition to it, seed treatment with bavistin and thiram reduced the DRR incidence in central and southern parts of India (Ghosh *et al.*, 2013).

Goyal and Mehrotra (1981) found Benlate, Bavistin, Thiophanate methyl and Dyrene to be promising among the nine tested fungicides against *R. bataticola* in Chickpea *in vitro*. Benomyl and Bavistin were reported to be most effective in

inhibiting fungal growth *in vitro*. In another *in vitro* study, Sarwar and Raju (1985) found that the antibiotic Topsin M-70 was highly effective at a concentration of 1000 ppm against *R. bataticola* among the eight tested fungicides. Taya *et al.* (1990), stated that the best control of *R. bataticola* was demonstrated with single application of carbendazim or in combination with thiram as seed treatment, presowing, soil- drench and seed treatment + drenching after sowing treated seed of chickpea.

Peshrey *et al.* (1992) reported that Thiram, Campton, PCNB, Mancozeb Iprodione, Carboxine, Carbendazim, Thiabendazole (all 0.2%) and tridemorph (0.07%) effectively controlled growth on sclerotial germination of *R. bataticola in vitro*. Carbendazim and thiobendazole were most effective and inhibited growth at the concentration of 0.006%.

Singh *et al.* (1982) reported that 3 chickpea cultivars did not be related to their resistance to *R. bataticola* carbohydrate content was higher in the susceptible cultivar than in the 2 resistant ones. Singh and Mehrotra (1982) reported the cultivars BG-203, G-543, and Hare Chhole to show resistance against *R. bataticola* when grown in infested soil. Singh and Mehrotra, (1980) investigated biological control of *R. bataticola* in chickpea and reported 4 bacteria and 6 actinomycetes isolates which proved to be antagonistic in culture with the exception of streptomyces isolates, all these reduced the disease symptoms and increased plant growth and dry matter when tested further by coating chickpea seed sown in field soil inoculated with the pathogen.

Ved Ratan *et al.* (2010) reported that change in sowing date could act as an efficient and economic approach towards management of dry root rot and wilt diseases of chickpea. Vijay-Mohan *et al.* (2006) reported that DRR could be managed by using carbendazim (0.2%) and etaconazole (0.1%) as seed treatment, soil drenching and seed treatment plus soil drenching.

The understanding of chickpea, *R. bataticola* and moisture deficit has provided noteworthy information and provided a way to deal with the emerging disease. Climate also plays a significant role in dry root rot emergence as high temperature and moisture stress conditions aggravate the incidence of DRR.

Thorough knowledge of genetics, behavior of host and pathogen in the process of disease development and host-pathogen interaction will be effective for developing suitable and effective disease management modules.



CHAPTER III

MATERIALS AND METHODS

MATERIALS AND METHODS

The present investigation entitled “**Physiological and molecular characterization of drought tolerant biocontrol agents for management of dry root rot of Chickpea (*Cicer arietinum* L.) caused by *Rhizoctonia bataticola***” was undertaken during the years 2013 to 2015. The details of the materials and methods used during the course of investigation have been described.

3.1 Experimental site

The *in vitro* experiment was conducted in the Biocontrol and Plant Health Clinic Laboratory of Department of Mycology and Plant Pathology, Institute of Agricultural Sciences, Banaras Hindu University, Varanasi, India. The *in vivo* experiment was carried out in the polyhouse and field of the same department, where chickpea crop was raised in pots (15 x 10 cm) and field and all physical precautions were kept in view in order to protect the crop from the external damage. The site of the experiment was unaltered during experimentation period. The university is located in the south- eastern part of Varanasi city which is at 25°18' N latitude, 83°3' E longitude and at an altitude of 75.7 m above mean sea level.

3.2 Collection of diseased samples from different districts of Eastern Uttar Pradesh

3.2.1 Survey for incidence of dry root rot from different zones of Eastern Uttar Pradesh and collection of diseased samples

Random roving method of survey was carried out to record the severity of dry root rot in chickpea. The survey was conducted during *Rabi* 2012 - 13 in ten districts of Eastern Uttar Pradesh i.e Varanasi, Mirzapur, Azamgarh, Faizabad, Sultanpur, Mau, Deoria, Basti, Jaunpur and Gorakhpur (Figure 3.1). The observations on stage of crop and disease severity were recorded in the rating scale of (1-9). Chickpea plants

showing typical root rot symptoms were collected separately in paper bags and brought to the laboratory for isolation of associated pathogen and further investigations. The diseased samples were collected from areas having maximum water deficit incidence in the locality as the pathogen aggravates in soil moisture deficit conditions.

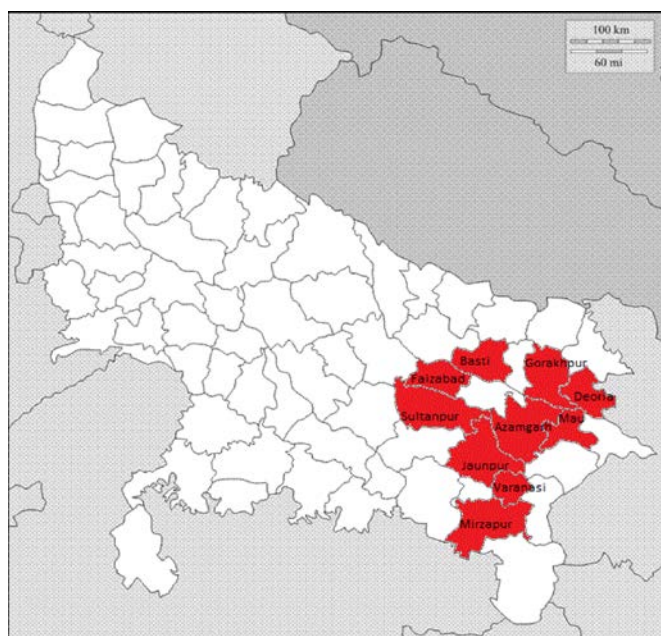


Figure 3.1 Survey of different districts of Eastern Uttar Pradesh for collection of *R. bataticola* isolates.

3.2.2 Isolation of the pathogen

The pathogen *Rhizoctonia bataticola* was isolated on PDA medium from infected root of chickpea plant. A small portion of diseased tissue along with a portion of adjacent healthy tissue were cut into small pieces (3 to 5 mm in length) and then surface sterilized with 0.1% HgCl_2 for 30 seconds. The pieces were then rinsed thrice with sterilized distilled water. Sterilized and rinsed pieces were inoculated aseptically on sterilized Petriplates containing PDA medium. The inoculated Petriplates were incubated at 20- 25 °C for five to six days. When the fungal colony developed a small cut was made on mycelium with cork borer and was transferred on another Petriplate

containing PDA medium to obtain pure culture. The mycelial bit was also transferred to fresh PDA slants in order to store it for future use.

3.1.3 Maintenance and storing of the pathogen

The pure culture of the pathogen *R. bataticola* was maintained on PDA slants throughout the period of investigation by periodic sub culturing on fresh media and stored in a refrigerator at 4 °C.

3.1.4 Pathogenecity test

The chickpea cultivar “**Radhey**” was used for testing the pathogenicity of the pathogen. Chickpea seeds were surface sterilized by using 1% sodium hypochlorite for 30 s and was rinsed twice with sterilized distilled water and then air-dried. Soil mixture containing sandy loam soil, vermicompost, and farmyard manure (2:1:1) was autoclaved for 30 min at 15 lbs pressure for three consecutive days. Half amount of soil was also mixed with crushed mycelial powder of *R. bataticola*. The seeds were sown in 15 × 10 cm² pots under greenhouse conditions. Untreated seeds sown in pathogen infected and pathogen uninfected soil served as positive and negative controls, respectively. The studies on pathogenicity test was done 60 DAS as the symptoms are not produced at the initial stages of crop growth..

3.3 Isolation and characterization of agriculturally important micro-organisms (AIMs) from Eastern Uttar Pradesh and Bundelkhand region

Soil samples from rhizosphere were collected from six districts of Eastern Uttar Pradesh and Bundelkhand region of Uttar Pradesh. Rhizospheric soil samples were carefully collected in poly bags from agricultural fields of six districts i.e Varanasi, Allahabad, Jhansi, Lalitpur, Banda and Jalaun districts of Uttar Pradesh (Figure 3.2). 1gm of soil sample was weighed and mixed in a test tube containing 10 ml sterilized distilled water and mixed properly. The tube serves as 10⁻¹ dilution. After its proper sedimentation 1ml of suspension was taken and dispensed in another test tube containing 9 ml of sterilized distilled water. This was 10⁻² dilution. This process was repeated up to 10⁻⁷ dilutions. The process of serial dilution was done for about 50 soil samples.

Materials and Methods

About 50 µl of suspension was taken from 10^{-6} and 10^{-7} dilutions and poured on Nutrient agar medium (NA) plates. The suspensions were spread on the medium with the help of a spreader. The plates were then kept in BOD incubator for 1-2 days at 25 ± 2 °C. Different colonies of bacteria appear on the plates after 1 day. A fresh plate containing NA medium is taken and 10 grids were prepared on it. The plate serves as a master plate and contains different types of bacteria. The colonies which vary with one another were selected and transferred to the master plate and are numbered serially.

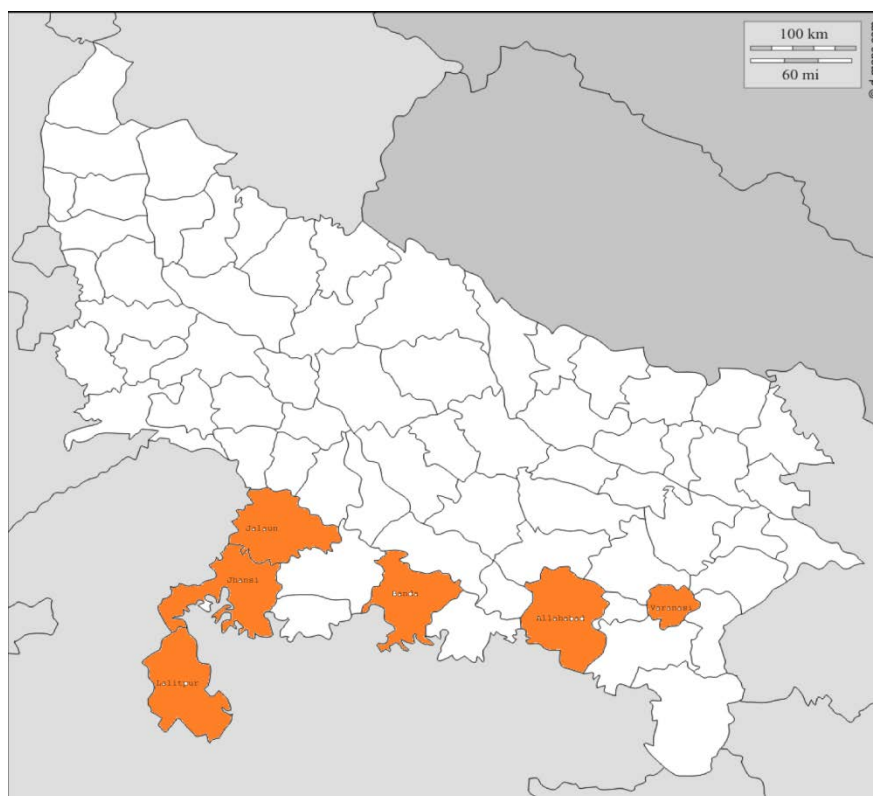


Figure 3.2 Isolation of AIMs from different districts of Eastern Uttar Pradesh and Bundelkhand Region

3.4 *In vitro* test for drought tolerance

60% NB+ PEG media have been used for testing drought tolerance of the bacterial isolates. Firstly, 120 glass vials were taken and 10 ml of nutrient broth media was transferred to each vial. A second set of 120 glass vials were taken and 10 ml of

60% NB+ PEG media was transferred into it. The vials were subjected to autoclave. The different bacteria were inoculated in the plain NB tubes and kept in shaker for a day. When sufficient growth of the bacteria was seen in the NB tubes then about 50 µl of the suspension was transferred to 60% NB+ PEG media containing tubes. The tubes were labelled properly. The tubes were placed in shaker and their growth was checked continuously for about 5 days. Those isolates which showed prominent growth up to 5 days were selected and their colony forming unit CFUml⁻¹ was measured.

3.5 Thermotolerance test

The selected drought tolerant bacteria were inoculated in plain NB tubes and were kept for about 5 days in different B.O.D whose temperature are maintained at a range of 40-55 °C. The growth potential of each bacterial isolate was checked per day by transferring 20 µl of the bacterial suspension to fresh NA plates for up to 5 days. Those isolates which showed prominent growth up to 5 days were selected for antagonistic test against the pathogen *R. bataticola*.

3.6 Antagonistic test of the isolates

The Petriplates containing PDA media were inoculated with the corresponding bacterial isolates at one edge as a straight line and with 5 mm mycelia disc of the pathogen on the opposite edge at a distance of 5 cm. Inoculated plates were incubated at 25 °C in BOD incubator

Observations were recorded up to 7 days of inoculation on area covered by tested bacterial isolates and the pathogen (*R. bataticola*). Inhibition of mycelial growth of pathogen fungi by each strain was recorded.

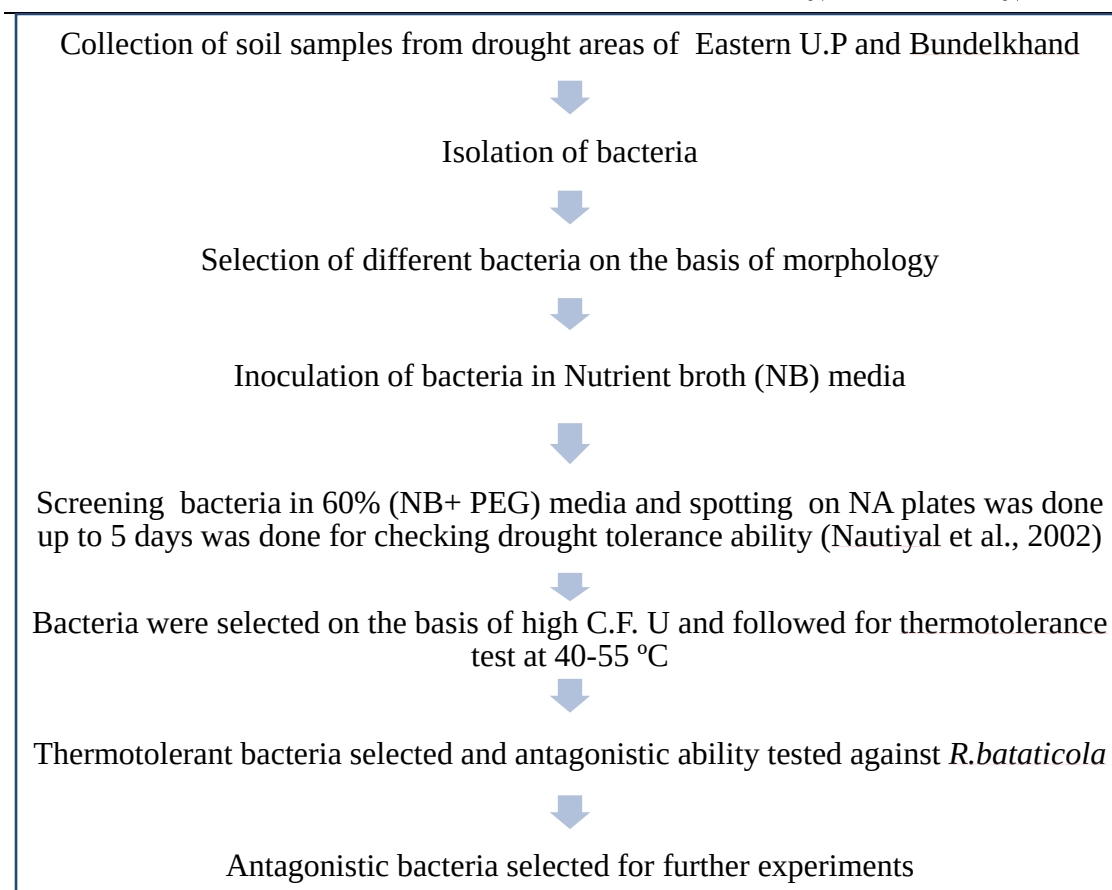


Figure 3.3 Flow chart representing different steps in isolation of drought tolerant bacteria antagonistic to *R. bataticola*

3.7 Biochemical test for identification of bacterial isolates

3.7.1. Pectolytic activity: For testing pectolytic activity, we used the medium as described by Hankin *et al.* (1971). The medium contains 500 ml of mineral solution, 1 gm yeast extract, 15 gm of agar, 5 gm of pectin (citrus or apple) and 500 ml of distilled water. The pH of medium was kept 7.0. After sterilization, the media was poured in sterilized Petriplates. The test bacteria were streaked at the centre of the plates. The plates were incubated at 28 ± 1 °C for 2 days. After 2 days when colony growths of isolates were seen, then the plates were flooded with 1 % aqueous solution of Cetyl trimethyl- ammonium bromide (CTAB). Transparent circle around the mycelium or streaked colonies seen was positive indication of pectinase activity.

3.7.2. Amylolytic activity: Amylolytic activity was detected by checking the ability of the organisms to degrade starch through production of amylolytic enzymes. To test the degradation of the amylase (starch) the medium was used which had Difco nutrient agar or PDA & 0.2 % soluble starch (Society of American Bacteriologists, 1951). The pH of medium was maintained at 7.0 and the bacterial isolates were grown in it. After 3-4 days of incubation, the plates were flooded with iodine solution. A yellow zone around the colony confirms the presence of amylase activity (Society of American Bacteriologists (1951)).

3.7.3. Proteolytic activity: For protease test, firstly media was prepared as described by Society of American Bacteriologists (1951) and poured in the sterilized Petri plates. The medium contained nutrient agar or PDA with 4% gelatin, (pH 6.0). 8% solution of gelatin in water was sterilized separately and added to the medium at the rate of 5 ml per 100 ml of medium. The test microbes were inoculated in the centre of the plates as 5 mm mycelial disc and incubated at 28 ± 1 °C for 2-3 days. The plates were then flooded with saturated solution of ammonium sulphate. Transparent circle around the mycelia or bacterial streak seen was positive indication of protease activity.

3.7.4. Lipolytic activity: For lipase test, firstly lipase media was prepared and was poured in sterilized Petri plates. The bacterial isolates were inoculated in the centre of the plates and were incubated at 28 ± 2 °C for 2-3 days. After that the plates were observed carefully to see as if there is any crystal formation which is a clear indication of lipase activity (Sierra, 1957).

3.7.5. Cellulolytic assay: Media prepared with cellulose as sole carbon source were used for detecting cellulolytic assay. After 3-5 days of incubation, plates were flooded with 1% Congo red solution for 10 minutes. Appearance of clear zones around the colonies confirmed the presence of cellulolytic activity (Lu *et al.*, 2004).

3.8) Preparation of PEG solution

For the preparation of 1 g/ml concentration of stock solution, weighed the PEG at the amount of 100 g and dissolved in 100 ml of double distilled water and stored in refrigerator at 4°C till use.

Table 3.1 List of different concentrations of PEG solution

Name of chemical	Concentrations used
Polyethylene glycol (PEG)	25 %
	30 %
	35 %
	40 %

The drought stress was induced by treating the seeds with PEG solution. PEG solution of four different concentrations as mentioned above were prepared. The treated seeds were then placed in moist blotting paper contained in a Petri plate.

3.8.1 Screening of PEG solutions

Screening of PEG solution was done on the basis of germination ability of the seeds. The concentration of PEG at which the seeds cease to germinate was taken as reference and the concentration just prior to the reference was selected for further experiment.

3.9 Experimental details

Chickpea seeds were surface sterilized by using 1% sodium hypochlorite for 30 s and was rinsed twice with sterilized distilled water and then air-dried. The seeds were treated with the prepared spore suspensions of BHU08 isolate as mentioned above for 1 h. Soil mixture containing sandy loam soil, vermin compost, and farmyard manure (2:1:1) was autoclaved for 30 min at 15 lbs pressure for three consecutive days. Half amount of soil was also mixed with mycelia powder of *R. bataticola*. The seeds were sown in $15 \times 10 \text{ cm}^2$ pots under greenhouse conditions. Untreated seeds sown in pathogen infected and pathogen uninfected soil served as positive and negative controls, respectively. Similarly, the treated and untreated seeds were sown in field also. Plants germinated from untreated seeds, which were pricked but not inoculated with pathogen served as control. Sampling for plant growth promotion was done after 30,60 and 90 days of sowing and growth parameters like per cent germination, plant height, shoot and root length, shoot and root dry weight, number of roots, pod number and number of grains per pod were measured. The experiment was conducted with three replications and the data were pooled for analysis.

Treatment No.	Treatment
T ₁	Control (C)
T ₂	Chickpea+ drought stress (D)
T ₃	Chickpea plant + drought stress + Biocontrol agent (D + B)
T ₄	Chickpea Plant + Pathogen (P)
T ₅	Chickpea Plant + Pathogen + Biocontrol agent (P + B)
T ₆	Chickpea Plant + Drought stress + Pathogen (D + P)
T ₇	Chickpea Plant + Drought stress + Pathogen + Biocontrol agent (D + P + B)
T ₈	Chickpea Plant + Biocontrol agent (B)

Table 3.2. Combination of treatments used for conducting experiment**3.10. Details of the layout plan**

1. Experimental design: Randomized completely block design
2. Number of treatment: 8
3. Number of replication: 3
4. Date of sowing:
 - First year(S₁): October 21, 2013
 - Second year (S₂): October 30, 2014
5. Total number of plots: 24 x 2 x 2
6. Spacing: Row to row: 30 cm
7. Spacing: Plot to Plot: 30 cm

3.11 Observations recorded

Random sampling technique was adopted for recording the observations of various morpho-physiological characters after 30, 60 and 90 days after sowing (DAS).

Three plants of each treatment from each replication were selected at random at the time of recording the data on various characters. Data of the plants were averaged replication wise and mean data was used for statistical analysis. Recommended package of practices were applied to raise a healthy crop.

3.11.1 Morphological parameters

1. Plant height (cm)

It was measured in centimeter from the ground level (base of the plant) to the tip of the main axis of the plant after stretching the main shoot of plant at the time span of 30 days after sowing with the help of meter scale.

2. Shoot length (cm)

It was measured in centimeter from the ground level (base of the plant) to the tip of the main axis of the plant after stretching the main shoot of plant at the time span of 30 days after sowing with the help of meter scale.

3. Number of roots

The numbers of roots were counted from one randomly selected plant per replication.

4. Length of roots (cm)

It was measured in centimeter from the ground level (base of the plant) to the tip of the main root of the plant at the time span of 30 days after sowing with the help of meter scale.

5. Number of pods/plant

The pods from one randomly selected plants were taken from each replication and their number were counted.

6. Number of seeds/pod

The seeds from ten pods from each replication was randomly collected and counted to get number of seeds/pod.

3.11.2 Physiological trait:

1. Relative water content (RWC) (%)

Relative water content was determined by the method described by Weatherley (1950). 100mg leaf material was taken and kept in double distilled water in a Petridish for two hours to make the leaf tissue turgid. The turgid weights of the leaf materials were taken after carefully soaking the tissues between the two filter papers. Subsequently this leaf material was kept in a butter paper bag and dried in oven at 65°C for 24 hours and their dry weights were recorded. The RWC was calculated by using the formula.

$$\text{RWC (\%)} = \frac{(\text{Fresh weight} - \text{Dry weight})}{(\text{Turgid weight} - \text{Dry weight})} \times 100$$

2. Determinations of dry weight (g)

After washing the plants in the tap water and softly wiped with using blotting paper, fresh weight was determined by using an electronic balance (Sartorius BT-224S) and the values were expressed in grams. After taking fresh weight, the plants were placed to 100°C pre-heated hot air oven for one hour. Then they were placed in an oven, maintained at 60±2°C for drying purpose. The weight was measured regularly till to get constant value, expressed in grams.

3.12 Biochemical parameters: Observations to be recorded at 24, 48, 72 and 96 hours after pathogen inoculation.

1. Proline content (Bates *et al.*, 1973)
2. Catalase (EC 1.11.1.6) (Aebi, 1984)

3. Superoxide dismutase (EC 1.15.1.1) (Fridovich *et al.*, 1974)
4. Phenylalanine ammonia lyase (PAL) (Brueske, 1980)
5. Total phenol content (TPC) (Zheng and Shetty, 2000)

3.12.1. Sample collection for biochemical analysis

From each treatment, randomly three plants were uprooted (one from each pot) carefully without causing any damage at 24 h intervals after the pathogen inoculation up to 96 h. Nodal leaves (3rd to 5th nodes) from the bottom were collected as samples. Collected leaves were washed in running tap water, dried with blotting paper and stored in a deep freezer (80 °C) until used for biochemical analysis.

3.12.2. Free Proline content

Proline content was measured by the method described by Bates *et al.* (1973). Leaf tissues (0.1 g) were homogenized with 5 ml of sulphosalicylic acid (3%) in a prechilled mortar and pestle. The homogenate was centrifuged at 10000 x g for 15 min. To 2 ml of the extract, 2 ml each of glacial acetic acid and ninhydrin (1.25 g ninhydrin was dissolved in 30 ml glacial acetic acid and 20 ml 6 mol l⁻¹ phosphoric acid and the content was incubated in a water bath (100 °C) for one hour. The reaction was terminated in an ice bath to stabilize the purple colour of the extract and was brought to room temperature. Toluene (4 ml) was added to each tube, and the content was mixed thoroughly for 15-20s. The absorbance for top purple aqueous layer was recorded at 520 nm against a reagent blank. The concentration of proline in samples was determined using the standard curve of L- proline.

3.12.3. Catalase assay

Catalase (CAT) (E.C. 1.11.1.6) activity was measured according to the method described by Aebi (1984). Leaf samples (0.1 g) were homogenized in 50 mM Tris–HCl buffer (pH 8.0) containing 0.5 mM EDTA, 2 % w/v polyvinylpyrrolidone, and 0.5 % (v/v) Triton X100 using a prechilled mortar and pestle. The homogenate was centrifuged at 15,000g for 10 min at 4 °C, and the supernatant was used for enzyme assay. The reaction mixture consisted of 300 mM phosphate buffer (pH 7.2) and 100 mM H₂O₂ in 1 ml enzyme extract. Activity was determined by recording O₂ released from enzymatic dissociation of H₂O₂ in darkness for 1 min. O₂ produced

by the enzymatic reaction was estimated by measuring the decrease in H_2O_2 absorption at 240 nm (extinction coefficient of H_2O_2 is $0.036 \text{ mM}^{-1} \text{ cm}^{-1}$) and enzyme activity was expressed as $\text{IM H}_2\text{O}_2$ oxidized $\text{min}^{-1} \text{ g}^{-1} \text{ FW}$.

3.12.4. Superoxide dismutase (SOD) assay

SOD (EC 1.15.1.1) activity was assayed following the method of Fridovich (1974) by measuring the ability of enzyme extract from samples to inhibit photochemical reduction of nitroblue tetrazolium (NBT) chloride. Fresh leaves (0.1 g) from each of the treatments were homogenized in 2.0 ml of extraction buffer (0.1 mol l^{-1} phosphate buffer containing 0.5 mmol l^{-1} EDTA at pH 7.5) in a prechilled mortar and pestle. The homogenate was centrifuged at $15\,000 \text{ g}$ for 20 min at 4°C . The reaction mixture contained 200 mmol l^{-1} methionine, 2.25 mmol l^{-1} NBT, 3 mmol l^{-1} EDTA, 100 mmol l^{-1} phosphate buffer (pH 7.8), 1.5 mol l^{-1} sodium carbonate and enzyme extract. The final volume was maintained to 3 ml. Reaction was started by adding 2 l mol l^{-1} riboflavin (0.4 ml), and the tubes were illuminated with two 15-W fluorescent lamps for 15 min. Reaction mixture without enzyme served as control. The reaction was terminated by putting the light off and keeping the tubes in dark until the absorbance was recorded at 560 nm. One unit of the SOD activity was defined as the amount of enzyme reducing the absorbance to 50% in comparison to control lacking enzyme.

3.12.5. Phenylalanine ammonia-lyase (PAL) assay

Leaf sample of 0.1 g from each of the treatments was homogenized in 2 ml of (0.1 mol l^{-1} sodium borate buffer (pH 7.0; 4°C) containing 1.4 m mol l^{-1} β -mercaptoethanol and centrifuged at $16\,000 \text{ g}$ at 4°C for 15 min. The supernatant was used as enzyme source. To the reaction mixture containing 0.2 ml of enzyme extract, 0.5 ml of 0.2 mol l^{-1} borate buffer (pH 8.7) and 1.3 ml of water were added. The reaction was initiated by the addition of 1 ml of 0.1 mol l^{-1} L-phenylalanine (pH-8.7) and incubated for 30 min at 32°C . The reaction was terminated by addition of 0.5 ml of trichloroacetic acid (TCA, 1 M). PAL (EC 4.1.3.5) activity was measured following the formation of trans-cinnamic acid at 290 nm as described by Brueske (1980) and was expressed in terms of $\mu\text{mol l}^{-1}$ TCA per g fresh weight (FW).

3.12.6. Total phenolic content (TPC) assay

The TPC was determined following the method of Zheng and Shetty (2000). Leaf tissue (0.1 g) was placed in 5 ml of 95% ethanol and kept at 0 °C for 48 h. The samples were homogenized individually and centrifuged at 13000g for 10 min. To 1 ml of the supernatant, 1 ml of 95% ethanol and 5 ml of sterile distilled water and 0.5ml of 50% Folin–Ciocalteu reagent were added, and the content was mixed thoroughly. After 5 min, 1 ml of 5% sodium carbonate was added, the reaction mixture was allowed to stand for 1 h and the absorbance of the colour developed was recorded at 725 nm. Standard curves were prepared for each assay using various concentrations of gallic acid (GA; Sigma-Aldrich-27645) in 95% ethanol. Absorbance values were converted to mg GA equivalents (GAE) per g FW.

3.13 DNA extraction

Pure cultures of the selected bacterial isolate BHU08 were maintained on NA slants and were revived on freshly poured NA plates for 1-2 days. A small loopfull of bacteria was aseptically inoculated in flask containing NB and was kept for incubation at 27 °C for 1-2 days without shaking. After three days, the genomic DNA was isolated using Hi PurA TM SP Bacterial DNA Mini Kit (Himedia Laboratories Pvt., Ltd., Mumbai, India). The quality of isolated genomic DNA was analyzed on 1.2% Agarose gel in 1× TAE buffer. Sharp and distinct bands indicated a good quality of isolated DNA.

For taxonomic identification of the selected isolate, the 16S rDNA amplification was done using universal bacterial primers forward (5' AGAGTTGATCYTCGCTC 3') and reverse (5' GYTACCTTGTTACGACT 3') obtained from Bangalore GeneI Pvt. Ltd., Bengaluru, Karnataka, India. A 25ml reaction mixture containing 50 ng of genomic DNA, 1 × PCR Buffer, 0.5 µL Taq DNA Polymerase, 1 pM concerned primer was considered for PCR amplification in thermocycler (Techne, UK). The amplification profile for the reactions was denaturation at 94 °C for 4 min followed by 35 cycles of 30 sec each at 94 °C and 55

°C followed by 60 s at 72 °C and final extension at 72 °C for 15 min (Mishra and Nautiyal, 2011). PCR products obtained were analyzed on 2% agarose gel in 1× TAE buffer containing ethidium bromide (0.5 mg/ml) by horizontal electrophoresis. Further, a phylogenetic tree was constructed using MEGA 6 software to authenticate the classification.

3.14 HPLC analysis

HPLC system (Shimadzu, Japan) equipped with two LC-10 pumps and a UV detector SPD-10A was used in present analysis. The filtered 20 µl sample was injected in a C-18 HPLC column (RP- Hydro, 4 X, 250 x 4.6 mm, Phenomenex, USA). Separation of compound was achieved with a two pump linear gradient for pump A (1% acetic acid in HPLC grade water) and pump B (HPLC grade Acetonitrile), with detection at 254 nm (Jain *et al.*, 2014). The program was initially started with 18% acetonitrile from pump B, changing to 32% in 15 and finally to 50% in 30 min which was followed by 25 min washing. Samples were identified by comparing retention time (Rt) and peak of the samples (µg/g FW) obtained with that of standard (Class VP series software, Shimadzu, Japan).

3.14.1 Sample preparation for HPLC analysis

Leaf samples (1g) were collected individually from each control and treatment sets. The extract was prepared by crushing in 10 ml of methanol: water (1:1) in a mortar and pestle. The extract was incubated overnight at room temperature and filtered through sterilized Whatman No. 1 filter paper. The filtrate was further fractionated thrice using ethyl acetate. The ethyl acetate fractions were collected and evaporated under reduced pressure in a rotatory evaporator. Dried residue was resuspended in 1 ml of HPLC grade methanol and filtered through a membrane filter (0.2 µm, Merk) before analysis.

3.15 Analysis of expression of different defense genes during pathogen challenge in chickpea:

3.15.1 RNA Isolation

Sample for RNA isolation were done from all the at different time intervals (0 hrs, 24 hrs, 48 hrs, 72 hrs and 96 hrs).

3.15.2 Materials required

Sure prepTM RNA purification kit (Invitrogen), Mortar and pestle, lyses solution, liquid Nitrogen, RNase-free micro centrifuge tube, ethanol (70%), RNA wash solution, RNA elution solution, Thermal cycler, formaldehyde agarose gel electrophoresis, ethidium bromide, RNA loading dye.

3.15.3 Procedure

One g leaf sample was used for RNA isolation using SureprepTM RNA purification kit. The leaf sample was ground in liquid nitrogen using pre cooled mortar and pestle to make fine powder. 700µl of lysis solution (Buffer) was added to the sample and further ground until the sample has been homogenized. The lysate was transferred into RNase-free micro centrifuge tube by using the pipette. The lysate was then centrifuged for 2 minute, 12500 rpm at room temperature for sedimentation of cell debris and then supernatant was transferred to another RNase free micro-centrifuge tube. Equal volume of 70% ethanol was added and then vortexed to mix. Assembled a spin column with one collection tube and clarified lysate (600 µl) with the ethanol added into the column and centrifuged for 2 min at 12500 rpm RNA wash solution (400 µl) was added to the column and centrifuged for 2 min at 12500 rpm. The flow through was discarded and the column was reassembled with the collection tube. Column was washed a second time by adding another 400 µl of RNA wash solution and centrifuged for two min in dry column. The collection tube was discarded with the flow through and the column was placed into fresh centrifuge tube. 50 µl of RNA elution solution was added and centrifuged for 1 minute at 12000 rpm. If the entire volume has not been eluted, the column was Spin at 12000 rpm

(23182.56g) and the purified RNA sample was stored at -80°C for future use. RNA quality was tested by OD260/OD280 ratio and gel electrophoresis.

3.15.4 Visualization of RNA

To assess integrity and purity of the RNA, sample were analyzed by formaldehyde agarose gel electrophoresis (1.3% agarose, 5% formaldehyde in 1X MOPS) buffer (20 mM 3-[n-morpholino] propanole-sulphonic acid, 5 mM sodium acetate, 10 mM EDTA). Five µg of total RNA was mixed with 3.5µl formaldehyde solution (40% w/v), 10 µl formamide, 2 µl of 10 × MOPS buffer and 1 µl of ethidium bromide (1mg/ml) in a total volume of 20 µl, then heated to 95°C for 5 min and 5µl RNA loading dye (8% ficoll, 0.02% bromophenol blue, 0.04% xylene cyanol FF) added prior to loading.

3.16 cDNA construction

3.16.1 Materials required

SuperScript™ III RT, 5X First-Strand Buffer (250 mM Tris-HCl, pH 8.3 at room temperature; 375 mM KCl; 15 mM MgCl₂), 0.1 M DTT, 20 mM Tris-HCl (pH 7.5), 100 mM NaCl, 0.1 mM EDTA, 1 mM DTT, 0.01% (v/v) NP-40, 50% (v/v), dNTP (10m mole), poly(A) oligo(dT) primer, H RNase, sterile water.

3.16.2 Procedure

1 µl Oligo (dT), 2 pmole gene-specific primer, 2 ng total RNA, 1µl dNTP Mix and 12 µl distilled water were added in a nuclease-free micro centrifuge tube. Mixture was heated at 65°C for 5 min and quickly chilled on ice and centrifuged for 1 minute at 12000 rpm. The supernatant was collected in fresh nuclease-free micro centrifuge tube. 4 µl 5X First-Strand Buffer and 0.1 M DTT 1 µl was added in the tube. Contents of the tube were mixed gently and incubate at 42°C for 2 min. 1 µl of SuperScript™ III RT was added and mix by pipetting gently up and down, then incubate at 50°C for 60 minute and heated at 70°C for 15 min for inactivation of reaction. RNase (Ribonuclease) was added in the tube to degrade the remaining RNA template in

cDNA. Finally incubate at 37⁰C for 20 minute. The cDNA can now be used in the further experiment and stored at -80°C.

Table 3.3. List of primers used for studying gene expression in chickpea against *R.bataticola*.

S. No	Defense Gene in Chickpea	Primer length	Primer sequence	Gene name
1	Phenylalanine ammonia lyase	20	AAGCTCACTTGCAGAGACAT	<i>PAL1</i>
		22	ACCTGATGCAGTGATTGGGTGC	
2	Isoflavone reductase	19	GCTGCTGCTAACCCTGAAA	<i>IFR</i>
		19	GTTCCACATCAGCCTCAGT	
3	Flavonone 3 - hydroxylase	20	GAGGCCAGCAACATGAAGAT	<i>F3H</i>
		20	CCAGAATGGCCTTCTAATCC	
4	Ubiquitin	21	CGACAACGTCAAGGCGAAGAT	<i>Ub1</i>
		20	CCAAAGCCACGGCACAAGTT	
5	<i>Gro ES2</i>	20	CTGAGGTGGAGTTCGATGGT	<i>Gro ES2</i>
		20	GCCATGACATCTGAGGCTCT	
6	Isoflavanone synthase 2	20	CAGGTGAAGGTGATGCAGGT	<i>HOS2</i>
		21	GCCCATGTACCATCACTAGCA	

3.17 Semi-quantitative RT- PCR for studying gene expression

Single strand cDNA for all the four treatments were PCR amplified using chickpea primers after equalization with internal control actin. The PCR conditions used for the amplifications were 95°C for 3 min, 94°C for 30 sec and annealing at 55°C for 30 sec and extension at 72°C for 30 sec for a total cycle of 28 and final extension at 72°C for 10 min. Amplified PCR products were checked on 1.5% agarose gel.

3.18 Isozyme analysis of the induced defense proteins

Sample preparation:

Plant tissue (0.1g) was crushed in 2 ml of sterilized phosphate buffer (0.1 M; pH 7.0). The homogenate was further centrifuged at 10,000 rpm for 20 min and the supernatant was collected for activity gel electrophoresis. The total protein content was estimated by Bradford method (Bradford, 1976). Volume of the macerate equivalent to about 50 µg of total protein was loaded onto the wells of native PAGE gels comprising of 5% stacking and 12% resolving gel. The samples were allowed to run at 200 V and 4° C for 45-50 min till the loading dye reached the bottom of gel.

Activity gel electrophoresis:

3.18.1 Peroxidase (PO)

For PO isozyme analysis, the incubating solution was prepared by dissolving 80 mg of benzidine in 1 ml of 96% ethanol (ice-cold), and further diluting it to 80 ml with distilled water. To the above solution, 800 µl of glacial acetic acid was added to obtain a clear solution and the solution was filtered. To the filtrate, 400 µl of H₂O₂ was added. The gels were incubated in the above prepared solution in dark for 30 min till attainment of blue bands. Acetic acid (7%) was used for fixing the bands (Csiszar *et al.*, 2012).

3.18.2 Polyphenol oxidase (PPO)

The gels were incubated for 30 min in dark in 0.1% p-phenylene diamine in 0.1 M phosphate buffer (pH 7.0) and incubated in the same buffer in the presence of 10 mM catechol with gentle shaking till the appearance of discrete bands. The gels were further rinsed in distilled water and photographed (Senthilraja *et al.*, 2013).

3.18.3 Superoxide dismutase (SOD)

The gels were soaked in 5 ml of riboflavin-nitroblue tetrazolium chloride (NBT) solution (1 mg riboflavin + 2.5 mg NBT in 10 ml distilled water) for 15 min in dark. The gels were then removed from riboflavin-NBT solution, rinsed in distilled water and soaked in 0.1% 3, 3' tetramethylene diamine (TEMED) solution in the dark for 15

min. The gels were further removed from the above solution, rinsed in distilled water and exposed to constant light till the appearance of colorless bands on a purple stained background (Jaiswal *et al.*, 2013).

3.19 Statistical analysis

The data in the tables were analysed by using one way ANOVA. The data are expressed as the mean of three independent replications $\pm$ standard errors. Means were compared by DMRT ($P \leq 0.05$), using SPSS version 16.0. Different letters superscript in the column data indicates significant difference between the variants across the treatments.



CHAPTER IV

EXPERIMENTAL FINDINGS

EXPERIMENTAL FINDINGS

The results of experiments conducted on different aspects of the concerned objectives of the thesis are presented in this chapter.

4.1 Collection of diseased samples from different districts of Eastern Uttar Pradesh, isolation and purification of *Rhizoctonia bataticola*, test of pathogenicity and study of per cent disease incidence of selected *R. bataticola* isolates on Chickpea plants in pots

4.1.1 Survey for root rot incidence at different zones of Eastern Uttar Pradesh and collection of diseased samples

A roving survey was conducted during January, 2013 in 10 districts of Eastern Uttar Pradesh to analyze the status of dry root rot incidence and to collect diseased samples infected by *R. bataticola* under field condition. The areas having maximum root rot incidence were selected and infected samples were collected from those areas. The data pertaining to observations on the crop stages of chickpea showing the typical root rot symptoms, the disease severity of the surveyed plots and the name of the isolates of *R. bataticola* were presented in Table 4.1. The disease severity of the infected plants was recorded on a rating scale of 1-9 (Table 4.2). The disease severity of plants varies with area and different crop stages like flowering, podding and post-podding stages. Forty samples were collected from different districts of Eastern Uttar Pradesh from which twenty different isolates of *R. bataticola* were isolated in laboratory in aseptic condition on potato dextrose agar (PDA) plates.

Table 4.1 Collection of diseased plant sample from different Chickpea growing districts of Eastern Uttar Pradesh

S. No.	Isolate name	Location	District	Stage	Disease Severity rating (1-9 rating scale)
1	Rb 1	Rajatalab	Varanasi	Flowering	7.2
2	Rb 2	BHU	Varanasi	Flowering	9.1
3	Rb 3	Ayodhyapur	Varanasi	Podding	5.5
4	Rb 4	Adalpura	Varanasi	Flowering	6.2
5	Rb 5	Chunar	Mirzapur	Podding	4.8
6	Rb 6	Tedia	Mirzapur	Podding	6.7
7	Rb 7	Nihalgarh	Faizabad	Flowering	3.6
8	Rb 8	Mewai	Faizabad	Podding	8.0
9	Rb 9	Dhanpatganj	Sultanpur	Podding	6.3
10	Rb 10	Lambhua	Sultanpur	Podding	5.8
11	Rb 11	Bhosi	Mau	Flowering	6.1
12	Rb 12	Tajaipur	Mau	Flowering	7.2
13	Rb 13	Bagapar	Deoria	Podding	4.8
14	Rb 14	Dalan	Deoria	Flowering	3.1
15	Rb 15	Kaptanganj	Basti	Flowering	5.4
16	Rb 16	Haria	Basti	Podding	8.9
17	Rb 17	Anaura	Azamgarh	Podding	8.1
18	Rb 18	Dharmapur	Jaunpur	Flowering	5.3
19	Rb 19	Sujanganj	Jaunpur	Podding	6.2
20	Rb 20	Bijauli	Gorakhpur	Flowering	7.1

Table 4.2 Rating scale used to record disease severity of dry root rot in Chickpea

Rating	Observation
1	No infection on roots
>1- ≤3	Very small lesions on roots
>3- ≤5	Lesions on root clear but small new roots free from infection
>5- ≤7	Lesions on roots many , new roots generally free from lesions
>7- <9	Roots infected and completely discoloured

4.1.2 Isolation, purification, maintenance of *R. bataticola* isolates

The isolates were purified by single sclerotial isolation technique and were identified as *R. bataticola* based on morphological and cultural characters using the descriptions given by C.M.I (1970). The isolates were designated serially from Rb 1 to Rb 20. The purified isolates were maintained in PDA slants and stored at 4°C for further use.

4.1.3 Test of pathogenicity of *R. bataticola* isolates

After isolation and purification of *R. bataticola* from the collected samples, they were subjected to the pathogenicity tests on susceptible genotype of chickpea cultivar ‘Radhey’ through soil inoculation method. Out of 20 isolates of *R. bataticola* tested for pathogenicity, 10 isolates showed typical root rot symptoms like drying and blackening of the roots. The tap root of the plants got almost rottened with less number of lateral and fine roots. Thus, 10 isolates showed positive result for Koch’s postulate while the remaining 10 isolates failed to prove Koch’s postulate indicating their non-pathogenicity to chickpea. Those 10 isolates which showed positive result for Koch’s postulate were

selected as representative samples for further study. The selected isolates were Rb 1, Rb 2, Rb 6 Rb 8, Rb 9, Rb 12, Rb 16, Rb 17, Rb 19 and Rb 20. The colony characteristics of these isolates were studied .These selected isolates were further studied for Per cent disease incidence (PDI) through ‘soil inoculation’ method. The results of pathogenicity are presented in Table 4.3.

Table 4.3 Test of pathogenicity of *R. bataticola* isolates collected from different districts of Eastern Uttar Pradesh

S. No.	Isolate name	Root rot symptoms	Koch postulates
1	Rb 1	+ve	+ve
2	Rb 2	+ve	+ve
3	Rb 3	-ve	-ve
4	Rb 4	-ve	-ve
5	Rb 5	-ve	-ve
6	Rb 6	+ve	+ve
7	Rb 7	-ve	-ve
8	Rb 8	+ve	+ve
9	Rb 9	+ve	+ve
10	Rb 10	-ve	-ve
11	Rb 11	-ve	-ve
12	Rb 12	+ve	+ve
13	Rb 13	-ve	-ve
14	Rb 14	-ve	-ve
15	Rb 15	-ve	-ve
16	Rb 16	+ve	+ve
17	Rb 17	+ve	+ve
18	Rb 18	-ve	-ve
19	Rb 19	+ve	+ve

20	Rb 20	+ve	+ve
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4.1.4 Studies on cultural variability in the selected *R. bataticola* isolates

Variability in the cultural and morphological characters of ten isolates of *R. bataticola* was studied by growing on PDA medium. The variations were recorded on the basis of colony diameter, colony colour, colony texture and presence or absence of aerial mycelium (Table 4.5).

Colony diameter. The data on colony diameter presented in the table 4.5 revealed that there was a significant difference in the colony growth recorded among the ten isolates of *R. bataticola* which varied from 39.2 to 73.0 mm. Isolate Rb 1, Rb 2, Rb 8, Rb 16 and Rb 17 showed significantly highest colony growth ≥ 60 mm while the least colony diameter was observed with the isolate Rb 19 (39.2 mm).

Colony colour. Based on visual observation on colony colour, the isolates were divided into three groups. Black colour colony was observed in 5 isolates i.e Rb 1, Rb 6, Rb 12, Rb 16 and Rb 20 while Rb 2 and Rb 17 showed black with grey mycelium. Remaining 3 isolates i.e Rb 4, Rb 5 and Rb 9 showed grey type of colony.

Colony texture. The isolates were categorized into three groups on the basis of colony texture. 5 isolates produced appressed colony while 2 isolates had fluffy texture. Rest 3 isolates had produced velvety growth. The different types of colony texture are depicted in Photo 4.1. The microscopic view of microsclerotia of Rb 2 is observed at 10 X and 40 X (Photo 4.2).

Aerial mycelium. Out of the 10 isolates, only 2 i.e Rb 2 and Rb 5 produced aerial mycelium while in majority of the isolates aerial mycelium was absent.

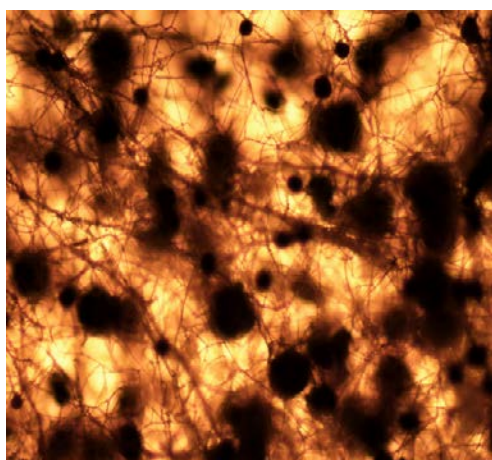


Appressed

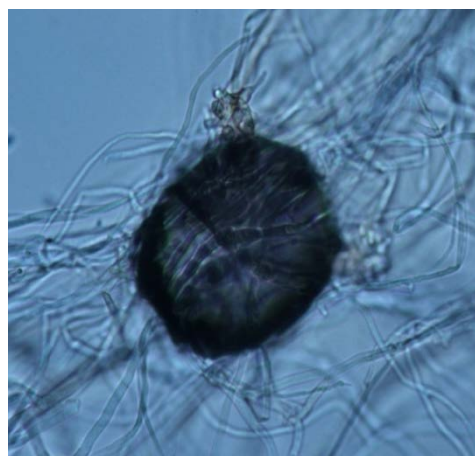
Fluffy

Velvety

Photo 4.1. Variability in colony texture of selected *R. bataticola* isolates



(a)



(b)

Photo 4.2. Microscopic view of microsclerotia of *R. bataticola* at **(a)** 10 X **(b)** 40 X

Table 4.4 Variability in cultural characteristics of selected isolates of *R. bataticola*

S. No.	Isolate name	Radial growth (mm)	Colony texture	Pigmentation	Aerial mycelium
1	Rb1	61.0	Appressed	Black	Absent
2	Rb 2	73.0	Fluffy	Black with grey	Present
3	Rb_6	57.7	Appressed	Black	Absent
4	Rb_8	63.4	Velvetty	Grey	Absent
5	Rb 9	43.0	Velvetty	Grey	Present
6	Rb 12	57.2	Appressed	Black	Absent
7	Rb 16	69.4	Appressed	Black	Absent
8	Rb 17	67.3	Fluffy	Black with grey	Absent
9	Rb 19	39.2	Velvetty	Grey	Present
10	Rb 20	45.8	Appressed	Black	Absent

4.1.5 Study of the effect of selected isolates of *R. bataticola* on chickpea plants in pots

The studies of percent disease incidence (PDI) of 10 selected isolates of *R. bataticola* were studied by soil inoculation methods in pots under greenhouse conditions. The data were recorded from 30 DAI to 120 DAI (Table 4.5). It is evident from Table 4.4, none of the isolate showed PDI up to 30 DAI while four isolates i.e Rb2, Rb8, Rb16 and Rb17 recorded PDI of 21.52%, 17.43%, 15.84% and 19.41% respectively. All the ten isolates recorded different levels of PDI at 90 DAI. Maximum PDI i.e 36.42% was recorded in treatment with Rb2 followed by 29.67% PDI by Rb8 and similarly the values of PDI decreased with other isolates. Rb2 recorded the highest PDI amongst all isolate from 60-120 DAI while Rb 8 recorded the second highest position from 90-120 DAI. Four other isolates i.e Rb1, Rb12, Rb16 and Rb17 also recorded PDI more than 70.00% while the remaining ones i.e Rb 6, Rb9, Rb19 and Rb20 recorded PDI less than 70.00%.

From the observations of Table 4.4, Rb 2 isolated from Varanasi, was found to be the most aggressive amongst the ten isolates of *R. bataticola*, and thus was selected as test pathogen for carrying out further experiments.

Table 4.5 Effect of selected *R. bataticola* isolates on disease incidence of chickpea through soil inoculation method. Results are expressed as mean of triplicates $\pm$ S.D.

Name of the isolate	Per cent disease incidence (%)				
	Days after inoculation				
	30	60	90	105	120
Control	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0	0 $\pm$ 0
Rb1	0 $\pm$ 0	0 $\pm$ 0	25.63 $\pm$ 6.25	48.95 $\pm$ 8.33	74.35 $\pm$ 14.33
Rb 2	0 $\pm$ 0	21.52 $\pm$ 6.25	36.42 $\pm$ 7.45	53.85 $\pm$ 8.33	89.63 $\pm$ 14.33
Rb_6	0 $\pm$ 0	0 $\pm$ 0	22.75 $\pm$ 6.38	41.62 $\pm$ 10.25	65.85 $\pm$ 14.33
Rb_8	0 $\pm$ 0	17.43 $\pm$ 5.17	29.67 $\pm$ 6.25	49.82 $\pm$ 8.33	80.26 $\pm$ 10.25
Rb 9	0 $\pm$ 0	0 $\pm$ 0	22.40 $\pm$ 7.45	33.52 $\pm$ 10.25	61.70 $\pm$ 14.33
Rb 12	0 $\pm$ 0	0 $\pm$ 0	26.52 $\pm$ 6.38	40.87 $\pm$ 14.33	70.35 $\pm$ 10.25
Rb 16	0 $\pm$ 0	15.84 $\pm$ 6.38	27.31 $\pm$ 6.25	43.63 $\pm$ 14.33	74.30 $\pm$ 14.33
Rb 17	0 $\pm$ 0	19.41 $\pm$ 7.54	23.20 $\pm$ 6.38	45.36 $\pm$ 8.33	77.34 $\pm$ 14.33
Rb 19	0 $\pm$ 0	0 $\pm$ 0	21.64 $\pm$ 8.33	35.74 $\pm$ 6.38	65.42 $\pm$ 10.25
Rb 20	0 $\pm$ 0	0 $\pm$ 0	23.15 $\pm$ 6.25	38.56 $\pm$ 6.25	68.36 $\pm$ 14.33

4.2 Isolation and characterization of agriculturally important micro-organisms (AIMs) from Eastern Uttar Pradesh and Bundelkhand region, test for drought tolerance and thermotolerance ability.

A roving survey was conducted during June-July, 2012 in Varanasi, Allahabad, Banda, Jhansi, Lalitpur and Jalaun districts of Uttar Pradesh order to collect soil samples for isolating thermotolerant microorganisms. Soil samples were collected from various sites of these districts and the temperature of the region while taking the samples was recorded (Table 4.6).

Table 4.6 Isolation site of AIMs from extreme regions of Eastern Uttar Pradesh and Bundelkhand Region

S. No.	District	Latitude/ Longitude	Location	Temperature (°C)	Year of collection
1	Varanasi	25.2765° N/ 83.0335° E	Ramnagar	45.2	2012
2	Allahabad	25.4253° N/ 81.9126° E	Phulpur	43.6	2012
3	Jhansi	25.5739° N/ 79.2987° E	Garautha	46.4	2012
4	Lalitpur	24.5831° N/ 78.7278° E	Mahroni	44.5	2012
5	Banda	25.5503° N/ 80.7048° E	Baweru	47.4	2012
6	Jalaun	26.0296° N/ 79.6558° E	Pipraya	43.5	2012

4.2.2 Isolation, purification, maintenance of the bacteria

The bacteria were isolated from collected soil samples through serial dilution method. The bacteria having distinct colony characteristics were selected and grown in individual NA plates. 120 bacteria were isolated having different morphology and colony characters and are designated serially from BHU01 to BHU120. Purification of the

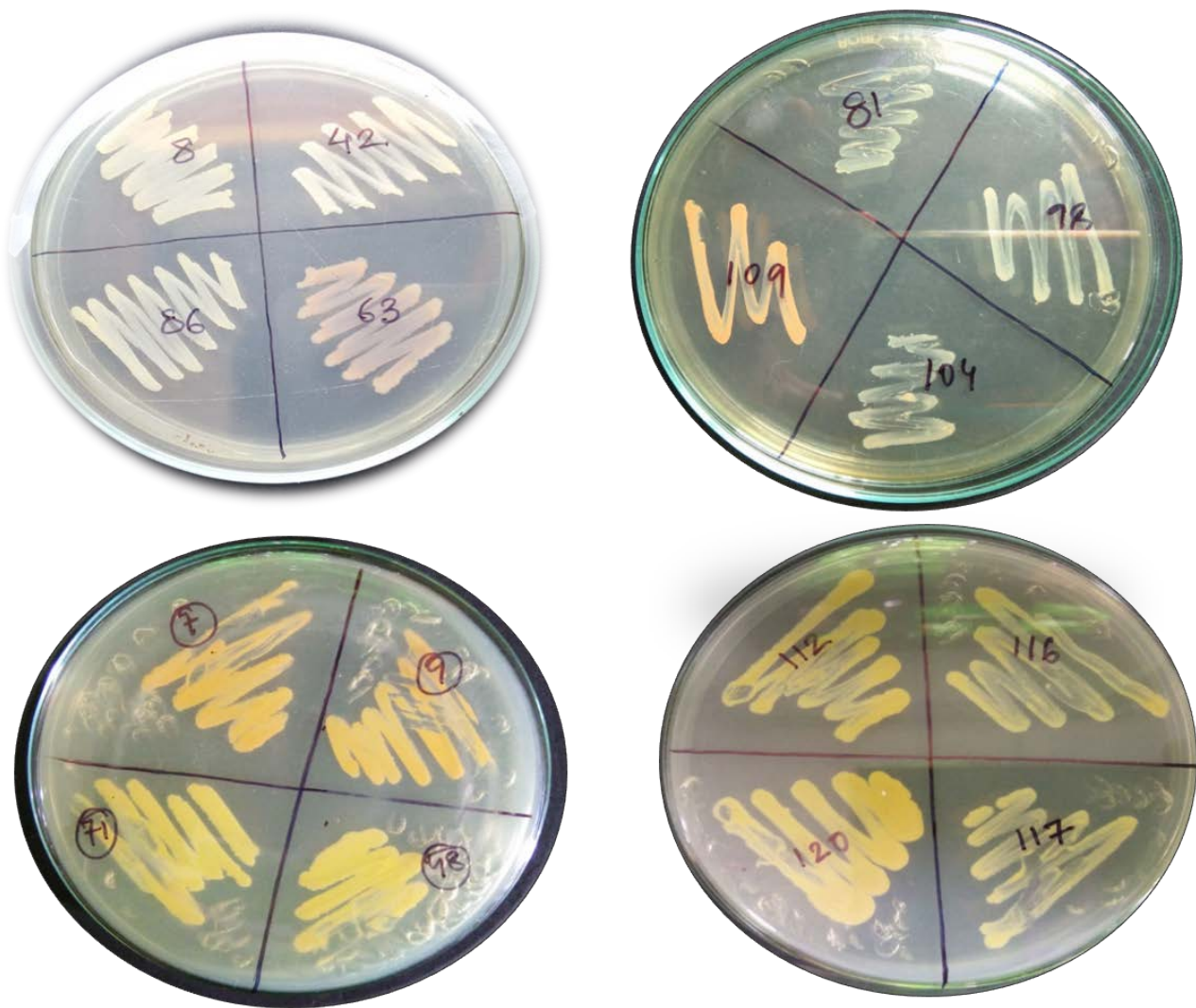


Photo 4.3. Selected samples of drought tolerant bacteria isolated from extreme soils of Eastern Uttar Pradesh and Bundelkhand region.

bacterial isolates was done in NA plates. The Pure cultures were maintained in NA slants at stored at 4 °C for future use. Selected samples of drought tolerant bacteria have been presented (Photo 4.3)

4.2.3 *In vitro* drought tolerance test of the isolates

The 120 bacteria were subjected to drought tolerance test and their growth up to 5 DAI in NB+PEG media was checked. Ten isolates i.e BHU7, BHU 8, BHU 9, BHU 42, BHU 63, BHU 71, BHU 81, BHU 86 and BHU98 were selected on the basis of their high C.F.U after 5 DAI compared to the rest and were subjected to thermo tolerance test. The C.F.U ml⁻¹ of the 10 isolates were calculated from 1 -5 DAI and data was recorded (Table 4.7).

Table 4.7 Survival of selected bacterial isolates at different time intervals

Name of the isolate	CFUml ⁻¹				
	Days after inoculation				
	1	2	3	4	5
BHU07	5.53 x 10 ⁶	4.26 x 10 ⁴	1.42 x 10 ⁵	3.46 x 10 ⁶	6.33 x 10 ⁴
BHU08	1.146 x 10 ⁵	6.82 x 10 ⁴	3.21 x 10 ⁴	7.33 x 10 ⁴	5.73 x 10 ⁴
BHU09	6.28 x 10 ⁴	3.41 x 10 ³	4.26 x 10 ³	1.70 x 10 ³	7.06 x 10 ²
BHU42	1.42 x 10 ⁹	4.53 x 10 ⁸	5.60 x 10 ⁸	1.24 x 10 ⁷	8.0 x 10 ⁸
BHU48	6.40 x 10 ⁶	5.73 x 10 ⁴	2.6 x 10 ³	1.78 x 10 ³	1.84 x 10 ³
BHU63	1.36 x 10 ⁹	1.06 x 10 ⁷	6.8 x 10 ⁶	5.62 x 10 ⁶	5.08 x 10 ⁴
BHU71	7.42 x 10 ⁵	2.2 x 10 ⁵	1.08 x 10 ⁵	9.86 x 10 ⁴	3.24 x 10 ³
BHU81	7.41 x 10 ⁶	5.21 x 10 ⁴	2.30 x 10 ⁴	1.58 x 10 ³	5.6 x 10 ²
BHU86	2.09 x 10 ⁹	1.74 x 10 ⁷	1.16 x 10 ⁷	1.6 x 10 ⁷	5.33 x 10 ⁵
BHU98	4.41 x 10 ⁶	8.10 x 10 ⁴	1.65 x 10 ³	1.18 x 10 ³	2.13 x 10 ³

4.2.4 *In vitro* thermotolerance test of the isolates

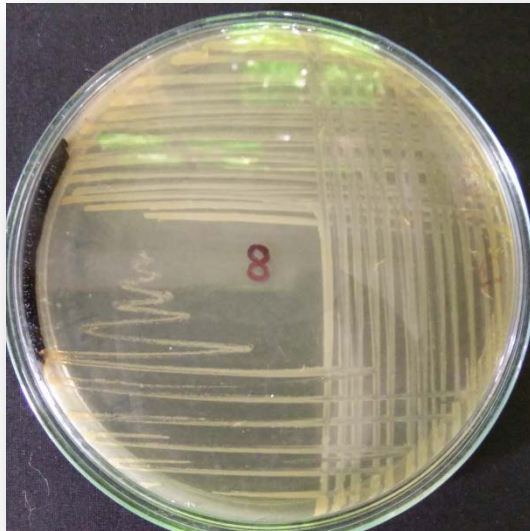


Photo 4.4. Plate of *Bacillus* sp. BHU 08



Photo 4.5. Dual culture test of *R. bataticola* with *Bacillus* sp. BHU 08

Ten isolates were selected for checking their thermotolerance ability. The selected isolates were inoculated in NB tubes and kept at different temperature range from 40⁰C - 55⁰C. The tubes were kept for about 5 days and their growth potential was checked each day by transferring 20 µl of the bacterial suspension to fresh NA plates. Isolate BHU 07 and BHU 81 showed growth up to 45 ⁰C. However, 4 isolates namely BHU 08, BHU 42, BHU63 and BHU86 showed growth at temperature range of 55⁰C (Table 4.8). The 4 above mentioned isolates were selected and subjected to antagonism test against *R.bataticola*.

Table 4.8 Thermotolerance test of the selected bacterial isolates

Isolates	40 °C	45 °C	50 °C	55 °C
BHU07	+	+	-	-
BHU08	+	+	+	+
BHU09	-	-	-	-
BHU42	+	+	+	+
BHU48	+	-	-	-
BHU63	+	+	+	+
BHU71	+	-	-	-
BHU81	+	+	-	-
BHU86	+	+	+	+
BHU98	+	-	-	-

4.3 Antagonism test against *R. bataticola*

Out of the 4 isolates, only BHU 08 showed antagonistic ability against *R.bataticola* in dual culture, while the remaining 3 isolates BHU 42, BHU 63 and BHU 86 failed to show antagonistic ability. So for carrying out further experiments BHU 08 was selected as it is

drought tolerant, thermotolerant and antagonistic to dry root rot pathogen *R.bataticola* (Photo 4.4 and 4.5)

4.4 Biochemical test conducted for identification of bacterial isolates

The 10 selected bacterial isolates i.e BHU 07, BHU 08, BHU 09, BHU 42, BHU 48, BHU 63, BHU 71, BHU 81, BHU 86 and BHU98 were grown in different media for their qualitative assay.

4.4.1 Pectolytic activity – The isolates BHU 08, BHU 42, BHU 81, BHU 86 and BHU 98 showed a transparent circle around the bacteria when the plates were flooded with 1 % aqueous solution of hexadecyl trimethyl-ammonium bromide (CTAB) which is an indicator of presence of pectolytic enzymes.

4.4.2 Lipolytic activity –The isolates BHU 08, BHU 09, BHU 42, BHU 63 and BHU86 showed crystal formation in the lipase media which is a positive indicator for lipase activity (Photo 4.6 a)

4.4.3 Protease activity –. The isolates BHU 08 and BHU71 showed transparent circle around the bacterial streak, when the plates were flooded with saturated solution of ammonium sulphate (Photo 4.6 b)

4.4.4 Amylase activity - The isolates BHU 08, BHU 09, BHU 42 and BHU 63 showed a yellow zone around the colony when the plates were flooded with iodine solution and hence were tested positive (Photo 4.6 c)

4.4.5 Cellulolytic activity – All the isolates showed negative results.

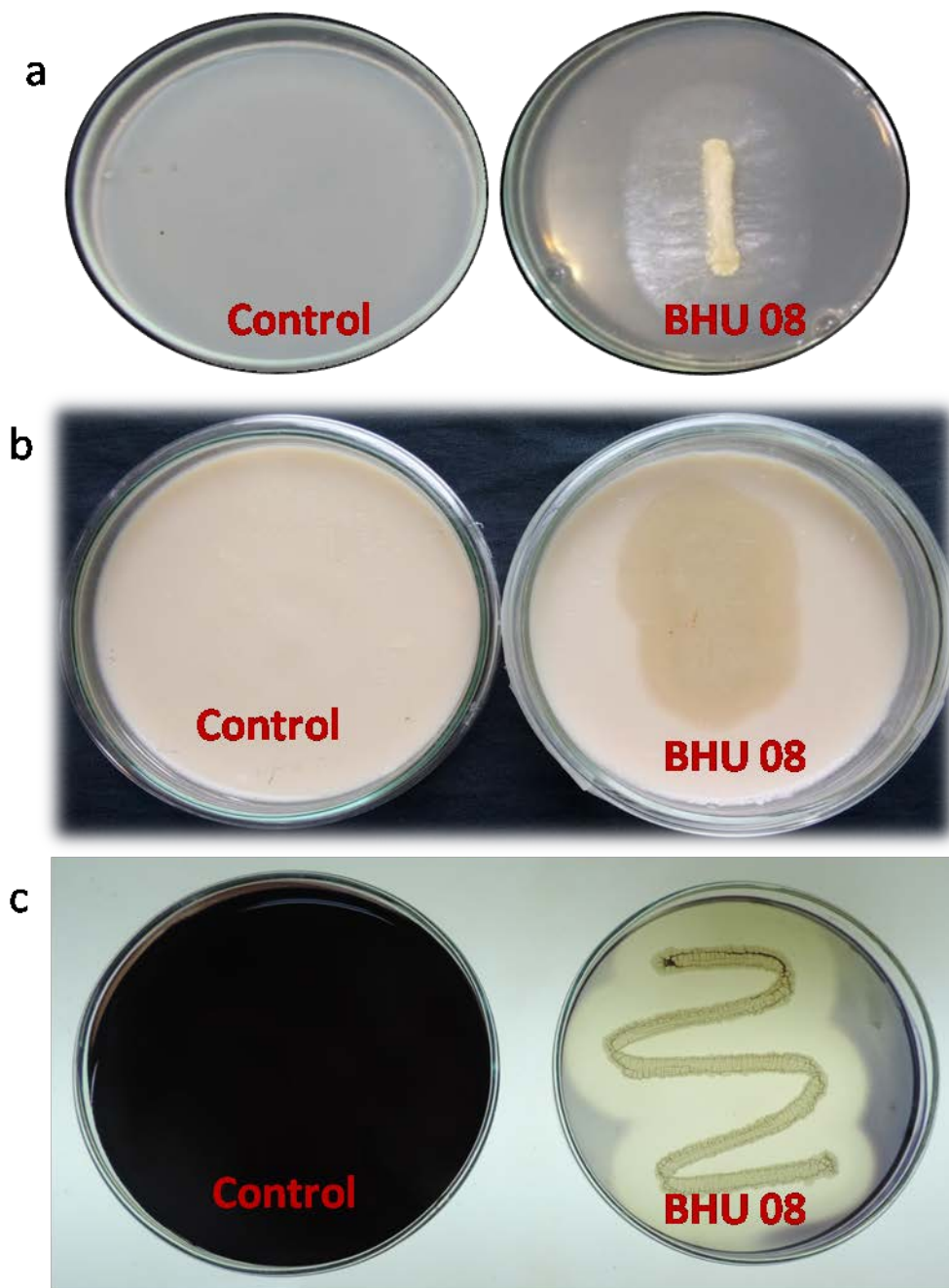


Photo 4.6. (a) Lipolytic (b) Proteolytic and (c) Amylolytic assay of Control vs *Bacillus* sp. BHU 08 *in vitro* respectively.

Table 4.9 Study of various enzymatic ability of the selected bacterial isolates

Isolates	Pectolytic activity	Amylolytic activity	Proteolytic activity	Lipolytic activity	Cellulolytic activity
BHU07	-	-	-	-	-
BHU08	++	++	++	++	-
BHU09	-	++	-	++	-
BHU42	++	++	-	++	-
BHU48	-	-	-	-	-
BHU63	+	++	-	++	-
BHU71	-	-	++	-	-
BHU81	++	-	-	-	-
BHU86	++	-	-	++	-
BHU98	++	-	-	-	-

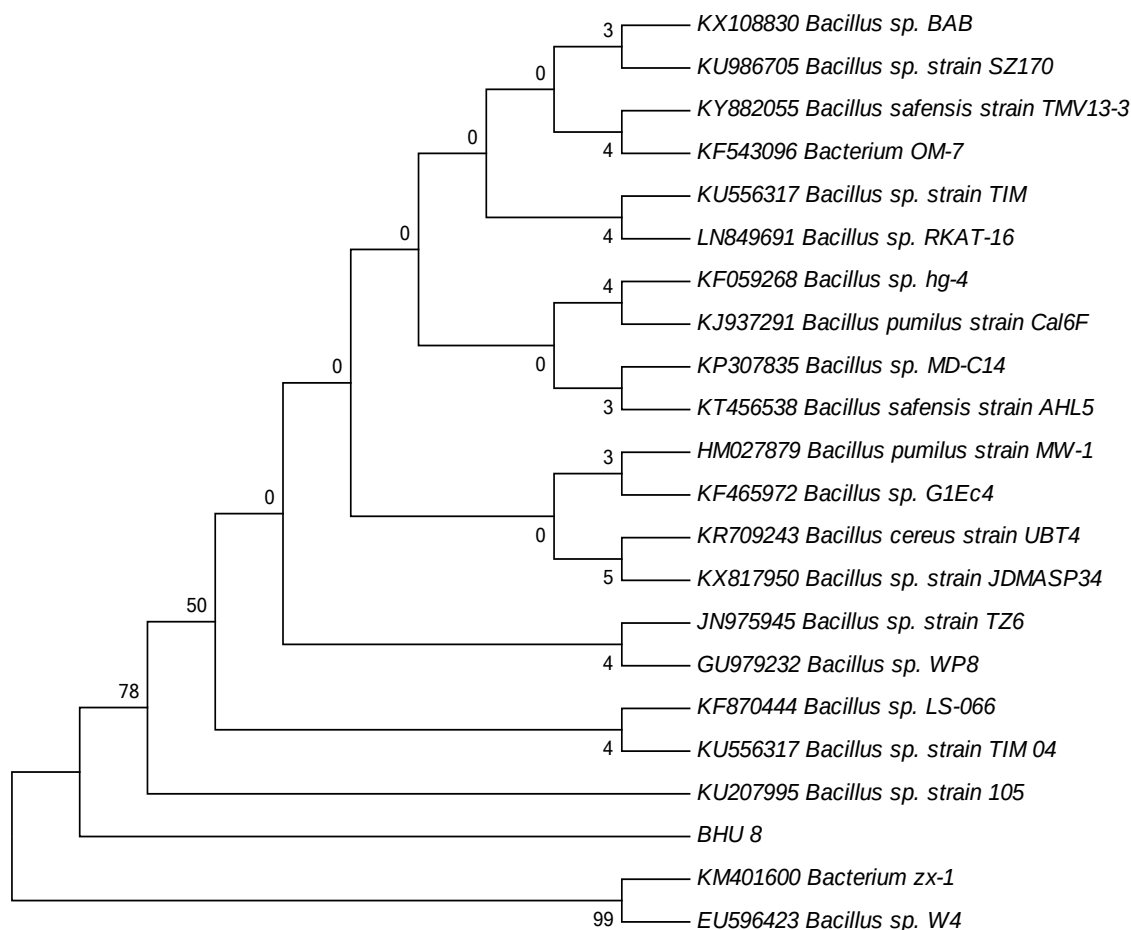


Figure 4.1 The phylogenetic tree of BHU 08 isolates constructed according to 16S rDNA sequence. Bootstrap values are based on 1,000 replications. Bar 0.005 substitutions per site

Molecular identification of bacterial isolate BHU 08 was performed by 16S. The ribosomal DNA of BHU 08 was amplified and the product (1.5 kb) was sequenced. Comparative analyses of the amplified 16S rRNA portion of the genome of the bacterial strains, using BLAST method, revealed that the tested bacterial isolate showed closest phylogenetic relationship with various *Bacillus* sp. with 98% homology (Fig. 4.3).

Table 4.10 Effect of various treatments on growth parameters of Chickpea crop (Year 2013-14). *Results are expressed as mean of triplicate $\pm$ SD. Different letters indicate significant difference among treatment results taken at same time interval according to Duncan's multiple range test at $p \leq 0.05$.

Days after sowing (DAS)	Growth parameters	Treatments								
		T1	T2	T3	T4	T5	T6	T7	T8	CD at 5 %
30	Plant height (cm)	28.2 $\pm$ 1.10 ^{ab}	28.5 $\pm$ 0.3 ^{ab}	28.2 $\pm$ 1.4 ^{ab}	26.1 $\pm$ 2.31 ^{bc}	28.6 $\pm$ 1.1 ^{ab}	25.6 $\pm$ 1.74 ^c	28.3 $\pm$ 0.96 ^{ab}	29.1 $\pm$ 0.88 ^a	2.25
	Shoot length (cm)	23.7 $\pm$ 1.81 ^{ab}	21.9 $\pm$ 1.68 ^{ab}	24.2 $\pm$ 2.68 ^{ab}	22.1 $\pm$ 2.05 ^{bc}	23.3 $\pm$ 0.55 ^{ab}	21.4 $\pm$ 0.66 ^c	24.0 $\pm$ 0.97 ^{ab}	24.5 $\pm$ 0.30 ^a	2.40
	Root length (cm)	8.9 $\pm$ 3.23 ^a	7.1 $\pm$ 0.8 ^a	9.1 $\pm$ 2.31 ^a	7.6 $\pm$ 1.26 ^a	8 $\pm$ 1.45 ^a	6.1 $\pm$ 0.45 ^a	9.7 $\pm$ 3.41 ^a	9.3 $\pm$ 0.20 ^a	3.32
	Number of roots	21 $\pm$ 2.0 ^a	18 $\pm$ 2.0 ^a	21 $\pm$ 6.5 ^a	15.6 $\pm$ 2.5 ^a	18.3 $\pm$ 1.15 ^a	15.6 $\pm$ 1.52 ^a	18 $\pm$ 3.0 ^a	20.6 $\pm$ 3.0 ^a	5.09
	Fresh weight (g)	2.6 $\pm$ 0.2 ^{ab}	1.63 $\pm$ 0.09 ^e	2.2 $\pm$ 0.6 ^{cd}	1.48 $\pm$ 0.08 ^e	2.41 $\pm$ 0.7 ^{bc}	0.86 $\pm$ 0.3 ^f	2.16 $\pm$ 0.41 ^d	2.66 $\pm$ 0.36 ^a	0.15
	Dry weight (g)	0.57 $\pm$ 0.03 ^a	0.43 $\pm$ 0.07 ^{ab}	0.41 $\pm$ 0.02 ^{ab}	0.38 $\pm$ 0.03 ^{ab}	0.51 $\pm$ 0.05 ^a	0.24 $\pm$ 0.01 ^b	0.47 $\pm$ 0.02 ^a	0.55 $\pm$ 0.13 ^a	0.78
	Relative water content (%)	67.20 $\pm$ 0.28 ^{ab}	59.71 $\pm$ 0.54 ^a	61.87 $\pm$ 0.37 ^{ab}	56.21 $\pm$ 0.46 ^{bc}	62.20 $\pm$ 1.25 ^{ab}	54.36 $\pm$ 0.86 ^{bc}	60.44 $\pm$ 0.58 ^{ab}	72.45 $\pm$ 1.10 ^{ab}	1.24
60	Plant height (cm)	41.8 $\pm$ 2.45 ^a	37.6 $\pm$ 3.75 ^a	39.8 $\pm$ 4.45 ^a	30.4 $\pm$ 2.45 ^b	41.6 $\pm$ 3.05 ^a	29.7 $\pm$ 3.45 ^b	40.1 $\pm$ 3.75 ^a	45.6 $\pm$ 3.15 ^a	5.57
	Shoot length (cm)	37.6 $\pm$ 1.15 ^a	34.8 $\pm$ 3.35 ^a	36.4 $\pm$ 3.95 ^a	26.3 $\pm$ 1.25 ^b	38.4 $\pm$ 3.12 ^a	24.2 $\pm$ 2.55 ^b	35.8 $\pm$ 3.15 ^a	38.8 $\pm$ 2.45 ^a	5.15
	Root length (cm)	8.6 $\pm$ 0.45 ^{abc}	7.3 $\pm$ 0.40 ^e	8.2 $\pm$ 0.35 ^{bcd}	7.2 $\pm$ 0.70 ^e	9.2 $\pm$ 0.55 ^a	7.4 $\pm$ 0.55 ^{de}	7.9 $\pm$ 0.30 ^{cde}	9.00 $\pm$ 0.45 ^{ab}	2.90
	Number of roots	38.6 $\pm$ 6.02 ^{bc}	32.6 $\pm$ 5.50 ^{cd}	35.0 $\pm$ 3.60 ^{bcd}	27.6 $\pm$ 1.52 ^d	43.6 $\pm$ 4.04 ^{ab}	28.0 $\pm$ 4.58 ^d	37.6 $\pm$ 4.04 ^{bc}	48.0 $\pm$ 4.58 ^a	8.09
	Fresh weight (g)	32.8 $\pm$ 1.35 ^a	22.5 $\pm$ 1.05 ^e	27.6 $\pm$ 1.75 ^c	20.4 $\pm$ 0.85 ^f	30.5 $\pm$ 0.95 ^b	18.5 $\pm$ 0.75 ^f	25.4 $\pm$ 1.35 ^d	34.7 $\pm$ 0.65 ^a	1.61
	Dry weight (g)	11.6 $\pm$ 1.15 ^b	8.6 $\pm$ 0.85 ^d	6.9 $\pm$ 0.26 ^e	7.2 $\pm$ 0.05 ^e	10.3 $\pm$ 0.55 ^c	5.8 $\pm$ 0.15 ^f	7.6 $\pm$ 0.26 ^{de}	14.5 $\pm$ 0.40 ^a	0.76
	Relative water content (%)	76.24 $\pm$ 0.74 ^{ab}	55.16 $\pm$ 0.38 ^{bc}	61.12 $\pm$ 0.46 ^{ab}	50.47 $\pm$ 0.75 ^b	73.18 $\pm$ 1.16 ^{bc}	42.54 $\pm$ 0.78 ^{ab}	58.15 $\pm$ 1.65 ^{bc}	80.66 $\pm$ 0.74 ^{ab}	1.16
	Plant height (cm)	54.6 $\pm$ 4.05 ^{ab}	47.1 $\pm$ 3.25 ^{bc}	50.2 $\pm$ 6.45 ^{ab}	42.1 $\pm$ 1.32 ^c	52.1 $\pm$ 2.45 ^{ab}	34.6 $\pm$ 4.85 ^d	49.4 $\pm$ 4.25 ^{abc}	56.8 $\pm$ 3.85 ^a	4.01
	Shoot length (cm)	46.7 $\pm$ 1.8 ^a	39.4 $\pm$ 3.85 ^c	44.3 $\pm$ 1.55 ^{abc}	34.2 $\pm$ 3.35 ^d	44.6 $\pm$ 3.15 ^{ab}	28.2 $\pm$ 2.45 ^e	41.2 $\pm$ 1.65 ^{bc}	49.4 $\pm$ 3.25 ^a	4.82

90	Root length (cm)	9.76 ± 1.84 ^a	7.4 ± 0.7 ^a	7.7 ± 0.55 ^a	7.5 ± 0.65 ^b	8.76 ± 0.85 ^a	6.8 ± 0.45 ^b	7.56 ± 0.65 ^a	8.43 ± 0.85 ^a	13.19
	Number of roots	42.7 ± 8.08 ^a	36.4 ± 4.0 ^{bc}	38.7 ± 7.0 ^{bc}	22.6 ± 6.42 ^{bc}	39.4 ± 8.08 ^{ab}	16.25 ± 4.0 ^c	40.6 ± 7.0 ^{bc}	48.6 ± 6.42 ^{bc}	1.70
	Fresh weight (g)	42.31 ± 1.59 ^a	28.60 ± 0.75 ^f	32.36 ± 0.55 ^d	29.81 ± 1.55 ^e	41.43 ± 1.45 ^b	21.60 ± 2.15 ^f	28.25 ± 0.45 ^e	47.26 ± 0.95 ^c	2.26
	Dry weight (g)	20.3 ± 0.39 ^a	10.5 ± 0.74 ^f	13.9 ± 0.58 ^d	11.9 ± 0.63 ^e	18.76 ± 1.28 ^b	9.19 ± 1.12 ^f	13.7 ± 1.15 ^d	24.16 ± 1.18 ^c	0.58
	Relative water content (%)	71.42 ± 0.29 ^{bc}	58.16 ± 0.35 ^a	64.85 ± 0.95 ^{bc}	52.74 ± 1.05 ^{bc}	69.48 ± 0.63 ^{ac}	42.15 ± 0.75 ^{bc}	60.37 ± 1.25 ^{ac}	76.54 ± 0.58 ^{bc}	1.58

4.5 Study of effect of various treatments on growth parameters of the crop

The chickpea plants were undergone different treatments as mentioned in Table 3.2. The morpho-physiological parameters were checked at 30, 60 and 90 DAS and observations were recorded. A comparison of various morphological parameters between control and treated plants were carried out (Photo 4.7-4.10)

Morphological parameters taken at 30, 60 and 90 DAS (Table 4.10)

4.5.1 Plant height

Plant height was recorded at 30, 60 and 90 DAS. Plant height was increased with increasing age of the plants. The sole BHU 08 treated (T8) plant showed maximum height in comparison to control and stressed plants in each treatment levels. It was clear from the data that the maximum height was recorded in T8 treatment level which was followed by control (T1) at 60 and 90 DAS. While comparing the stress regimes, T6 showed minimum plant height in all the time durations. While comparing to the individual stress provided to crop, T4 treatment showed less plant height compared to T2. The values of plant height did not significantly varied at 30 DAS as the pathogen *R. bataticola* shows symptoms only after 50 DAS. However the data recorded at 60 DAS and 90 DAS reflects a significant difference in plant height of stress challenged (T2, T4, and T6) and BHU 08 treated plants (T3, T5, T7 and T8) (Figure 4.8)

4.5.2 Shoot length

Shoot length was recorded at 30, 60 and 90 DAS. Shoot length was increased with increasing age of the plants. Maximum shoot length was attained in T8 at all time durations. At 30 DAS T3 recorded the second highest value. While comparing the stress regimes T6 showed minimum shoot length at in all the time durations. On comparison of the individual stress provided to crop T4 showed less plant height compared to T2. The data recorded at 60 DAS and 90 DAS reflects a significant difference in shoot length of stress challenged (T2, T4, and T6) and BHU 08 treated plants (T3, T5, T7 and T8).

4.5.3. Root length

Root length was recorded at 30, 60 and 90 DAS. At 30 DAS T7 recorded the highest value, while at 60 DAS T5 was maximum. At 90 DAS T1 recorded highest root length. While comparing the stress regimes T6 showed minimum root length at in 30 DAS and 90 DAS, while in 60 DAS T4 recorded minimum root length. The data recorded at 60 DAS and 90 DAS reflects a significant difference in root length of stress challenged (T2, T4, and T6) and BHU 08 treated plants (T3, T5, T7 and T8).

4.5.4 Number of roots

Number of roots was recorded at 30, 60 and 90 DAS. Maximum number of roots was attained in T8 at 60 DAS and 90 DAS. At 30 DAS T1 recorded the maximum number of roots value. While comparing the stress regimes T6 showed minimum number of roots at all the time durations. On comparison of the individual stress provided to crop T4 showed less number of roots at 30 DAS along with T6. The data recorded at 60 DAS and 90 DAS reflects a significant difference in number of roots of stress challenged (T2, T4, and T6) and BHU 08 treated plants (T3, T5, T7 and T8).

4.5.5 Fresh weight

Fresh weight was recorded at 30, 60 and 90 DAS. Maximum fresh weight was attained in T8 at all time regimes. While comparing the stress regimes T6 showed minimum number of roots at all the time durations. The data recorded at 60 DAS and 90 DAS reflects a significant difference in fresh weight of stress challenged (T2, T4, and T6) and BHU 08 treated plants (T3, T5, T7 and T8).

4.5.6 Dry weight

Dry weight was recorded at 30, 60 and 90 DAS. Maximum dry weight was attained in T8 at 60 DAS and 90 DAS, while at 30 DAS T1 showed maximum dry weight. While comparing the stress regimes T6 showed minimum number of roots at 30 DAS and 90 DAS, while at 60 DAS T1 exhibited minimum dry weight. The data recorded at 60 DAS and 90 DAS reflects a significant difference in dry weight of stress challenged (T2, T4, and T6) and BHU 08 treated plants (T3, T5, T7 and T8).

4.5.7 Relative water content

Relative water content was recorded at 30, 60 and 90 DAS. Maximum relative water content was attained in T8 at all-time regimes. While comparing the stress regimes T6 showed minimum number of roots at 30 DAS and 90 DAS, while at 60 DAS T4 exhibited minimum relative water content. The data recorded at 60 DAS and 90 DAS reflects a significant difference in relative water content of stress challenged (T2, T4, and T6) and BHU 08 treated plants (T3, T5, T7 and T8).



Control



Plant + Biocontrol agent



Plant + drought



Plant + Pathogen



Plant + drought + Pathogen

Photo 4.7. Comparison of various treatments with control in field

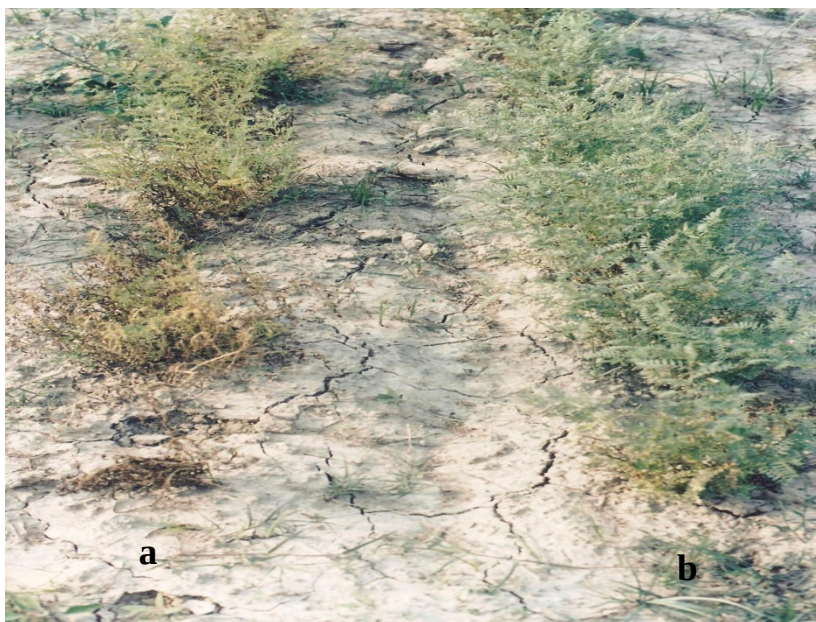


Photo 4.8. Comparison of (a) root rot infected chickpea plants with (b) *Bacillus* sp. BHU 08 treated plants

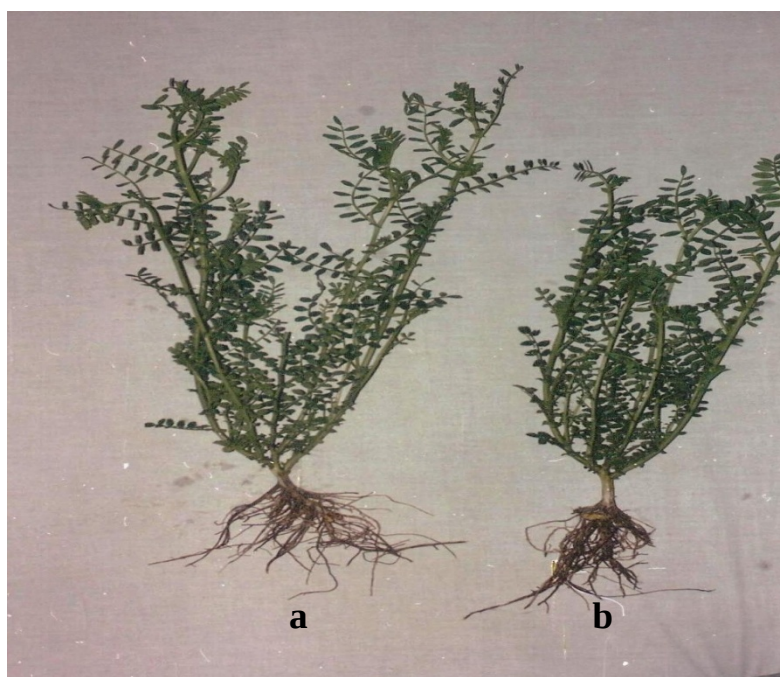


Photo 4.9. Comparison of (a) *Bacillus* sp. BHU 08 treated plants with (b) control

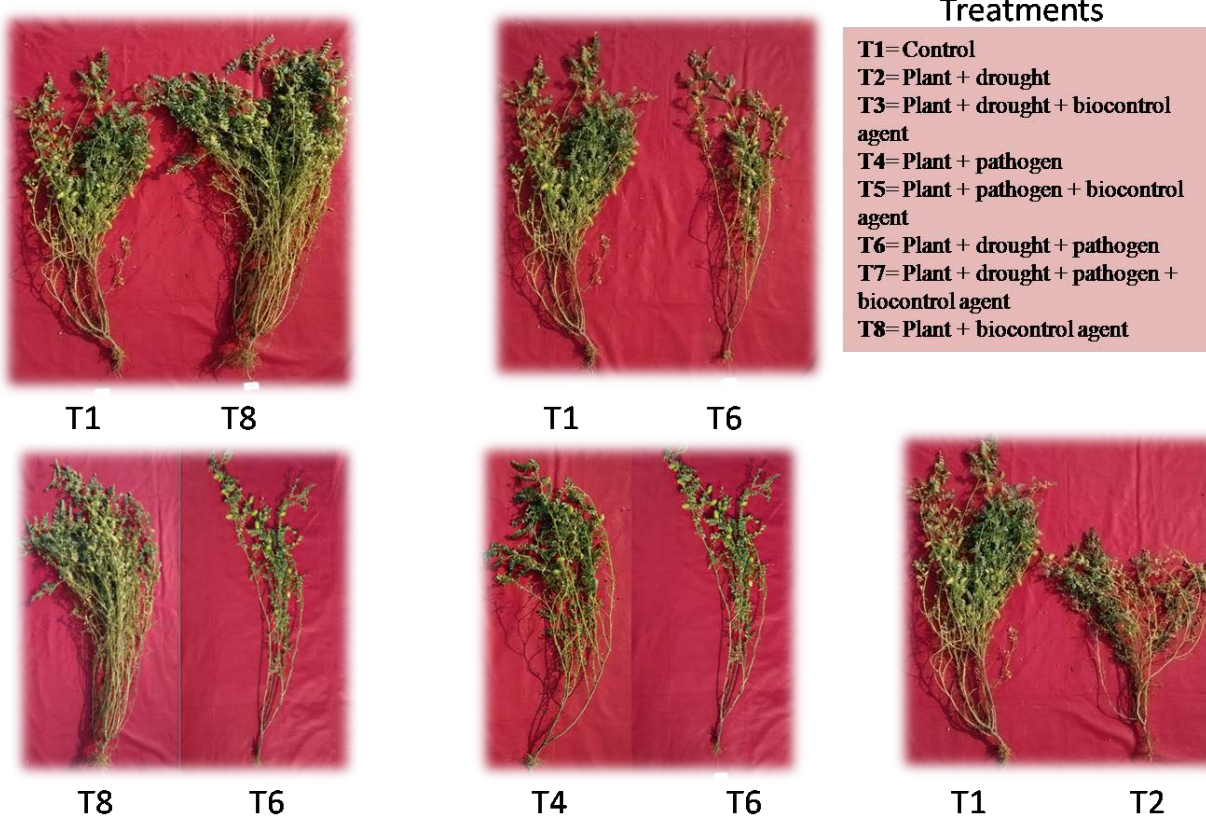


Photo 4.10. Comparison of various treatments with control of plants harvested at 90 DAS

4.6 Biochemical estimation

4.6.1 Phenylalanine ammonia lyase (PAL)

PAL levels increased significantly in all treatments up to 48 h, followed by a decline in its activity. T7 showed maximum accumulation of PAL at 48 h, followed by T3 at 24 h. At 48 h, T7 exhibited 1.95 folds increase compared to control (T1). The treatments in which only stress was applied i.e T2, T4 and T6 recorded 1.32, 1.29 and 1.51 folds increase compared to control. T3 and T5 in which both stress and biocontrol agent i.e BHU 08 was applied showed an increment of 1.69 and 1.48 folds as compared to control. Similarly, the plants treated with only biocontrol agent (T8) showed 1.52 folds increase in PAL values with reference to control.

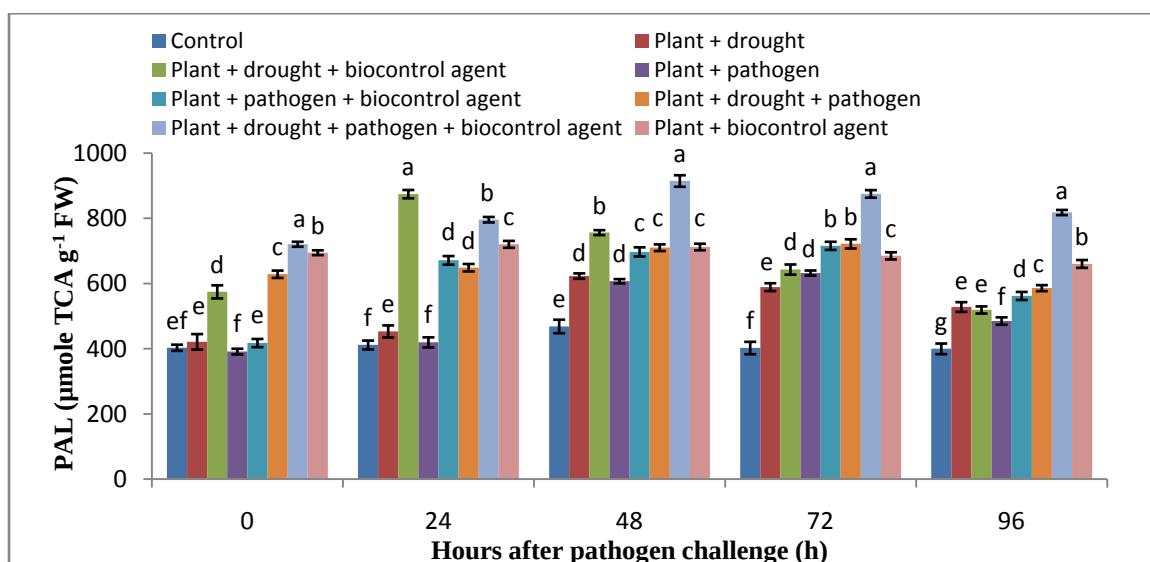


Figure 4.2 PAL activity at different time intervals in chickpea raised from seeds treated with *Bacillus* sp. BHU 08, challenged with *R. bataticola* and drought individually and in combination. Results are expressed as means of three replicates and vertical bars indicate standard deviation of the mean. Different letters indicate significant difference among treatment results taken at same time interval according to Duncan's multiple range test at $p \leq 0.05$

4.6.2 Total Phenol content (TPC)

The TPC followed a similar trend that of PAL with maximum increment at 48 h in T7 followed by a sharp decline in its activity. The amount of TPC content shows significant variation among different treatments. The highest phenolic content was observed at 48 h in T7. The total phenolic content was higher by 0.89, 1.08, 0.86, 1.1, 1.02, 1.39 and 1.01 folds in T2, T3, T4, T5, T6, T7, and T8 respectively, as compare to control (T1) at 48 h.

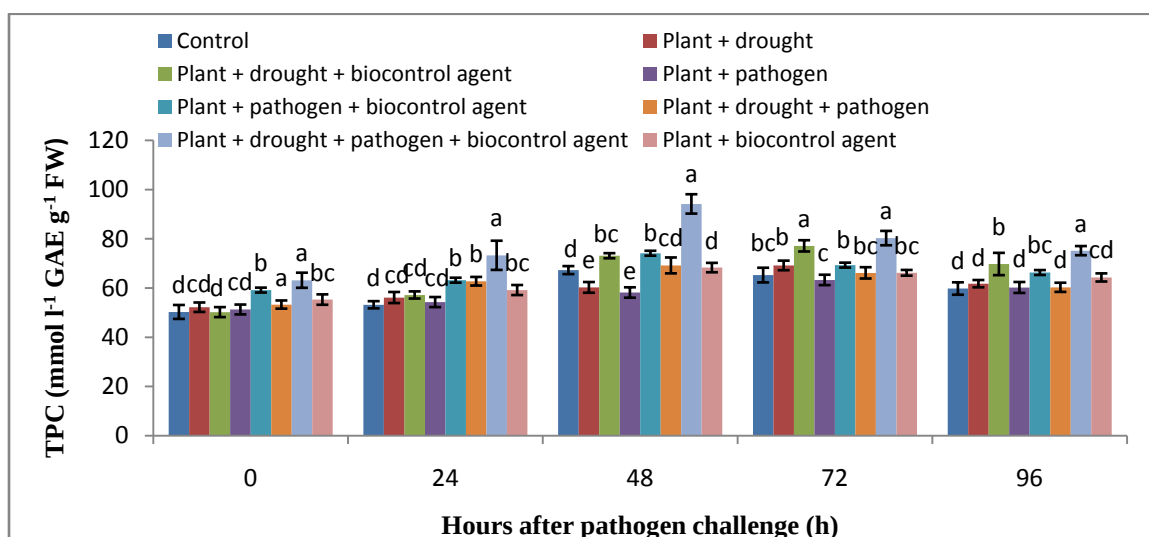


Figure 4.3 TPC activity at different time intervals in chickpea raised from seeds treated with *Bacillus* sp. BHU 08, challenged with *R. bataticola* and drought individually and in combination. Results are expressed as means of three replicates and vertical bars indicate standard deviation of the mean. Different letters indicate significant difference among treatment results taken at same time interval according to Duncan's multiple range test at $p \leq 0.05$

4.6.3. Superoxide dismutase (SOD)

SOD activities increased consistently and attained their maximum levels at 72 h in all treatments and thereafter its value declined. The highest SOD activities were observed

at 72 h in T7 followed by T6. The SOD values were 1.11, 1.48, 1.00, 1.70, 1.89, 2.17 and 1.83 folds in T2, T3, T4, T5, T6, T7, and T8 respectively, as compare to control (T1) at 72 h.

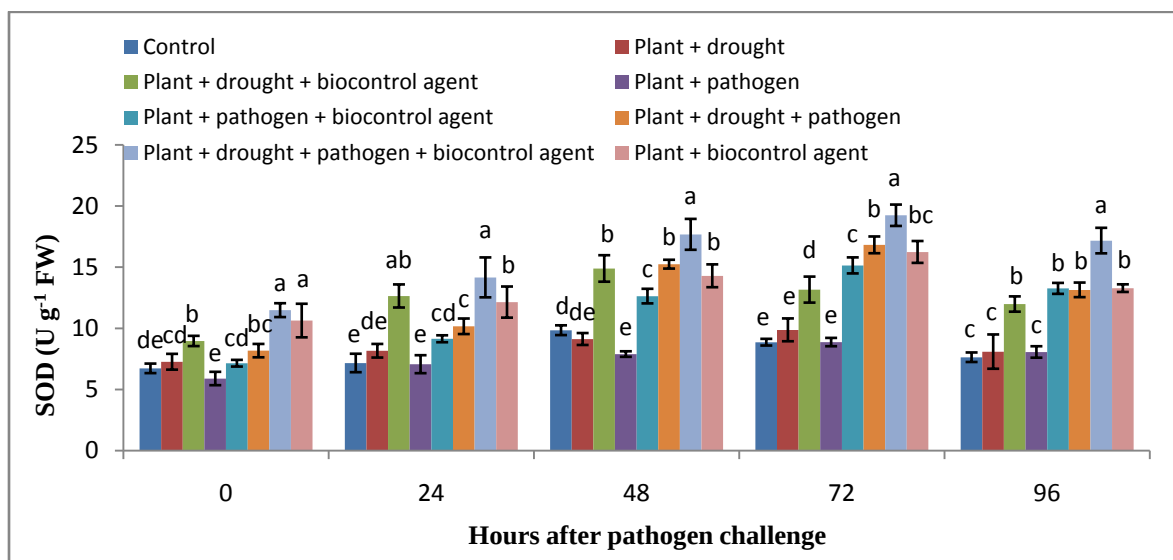


Figure 4.4 SOD activity at different time intervals in chickpea raised from seeds treated with *Bacillus* sp. BHU 08, challenged with *R. bataticola* and drought individually and in combination. Results are expressed as means of three replicates and vertical bars indicate standard deviation of the mean. Different letters indicate significant difference among treatment results taken at same time interval according to Duncan's multiple range test at $p \leq 0.05$

4.6.4. Proline

Proline accumulation followed a trend similar to that of SOD and increased significantly in all the treatments up to 72 h, thereby declining up to 96 h. Proline accumulation increased consistently and attained their maximum levels at 72 h in T7 followed by T8 and T3. Proline accumulation increased by 1.13, 1.55, 1.44, 1.54, 1.36, 1.74 and 1.55 folds in T2, T3, T4, T5, T6, T7, and T8 respectively, as compare to control at 72 h.

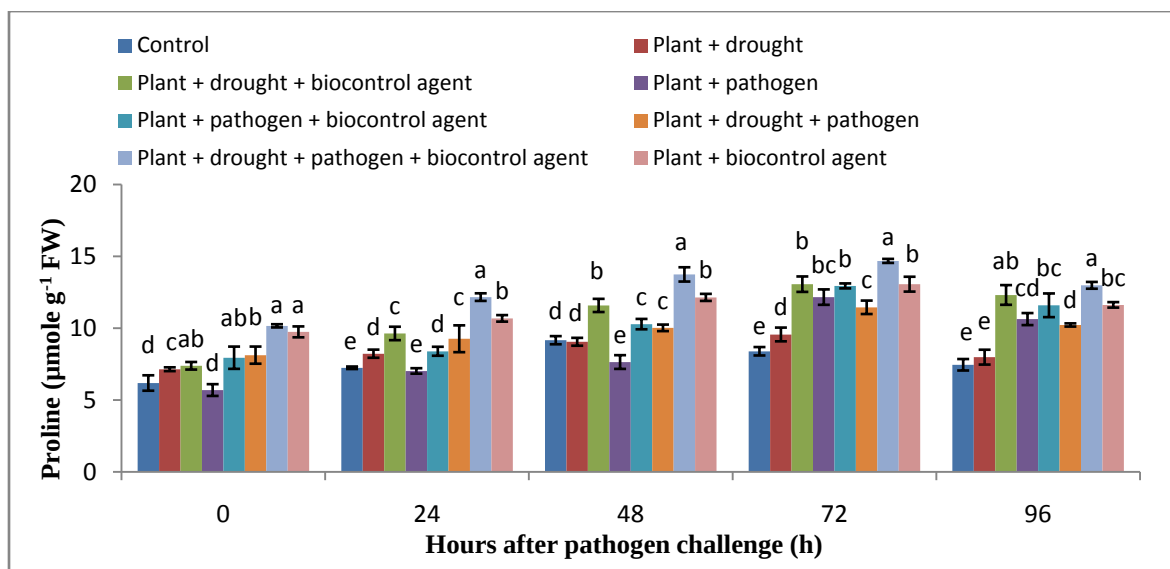


Figure 4.5 Proline activity at different time intervals in Chickpea raised from seeds treated with *Bacillus* sp. BHU 08, challenged with *R. bataticola* and drought individually and in combination. Results are expressed as means of three replicates and vertical bars indicate standard deviation of the mean. Different letters indicate significant difference among treatment results taken at same time interval according to Duncan's multiple range test at $p \leq 0.05$

4.6.5. Catalase

Catalase activity has increased from 24 h and reached its maximum level at 48 h, thereafter it started declining. The Catalase activity was found to be highest at 48 h in T7. On comparing the other treatments it can be seen that catalase activity increased by 1.23, 1.60, 1.10, 1.36, 0.93, 1.80 and 0.80 folds in T2, T3, T4, T5, T6, T7, and T8 respectively, as compared to control (T1) at 48 h.

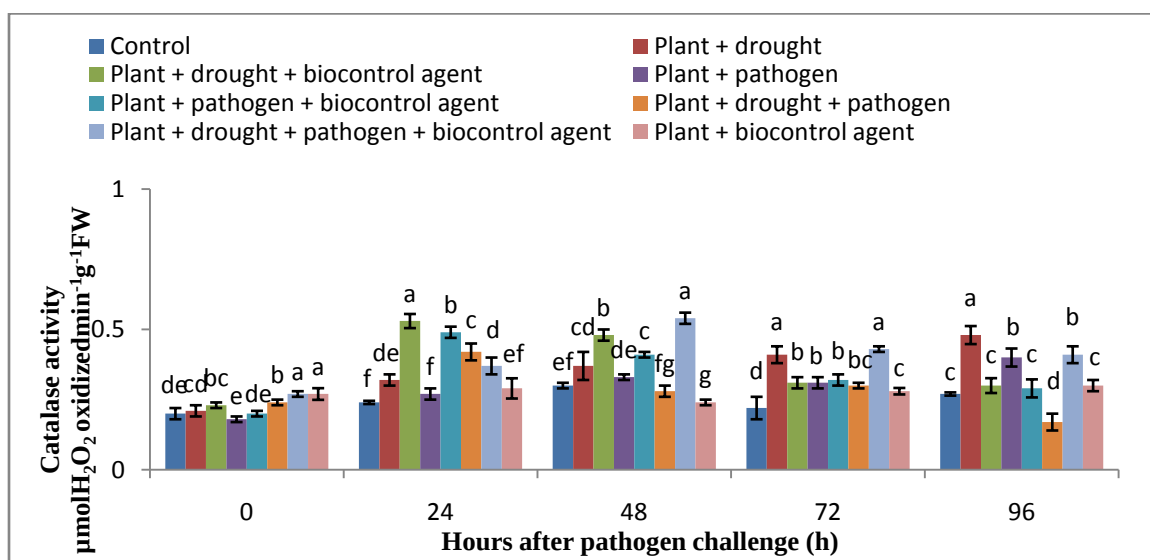


Figure 4.6 Catalase activity at different time intervals in Chickpea raised from seeds treated with *Bacillus* sp. BHU 08, challenged with *R. bataticola* and drought individually and in combination. Results are expressed as means of three replicates and vertical bars indicate standard deviation of the mean. Different letters indicate significant difference among treatment results taken at same time interval according to Duncan's multiple range test at $p \leq 0.05$

4.7 HPLC analysis

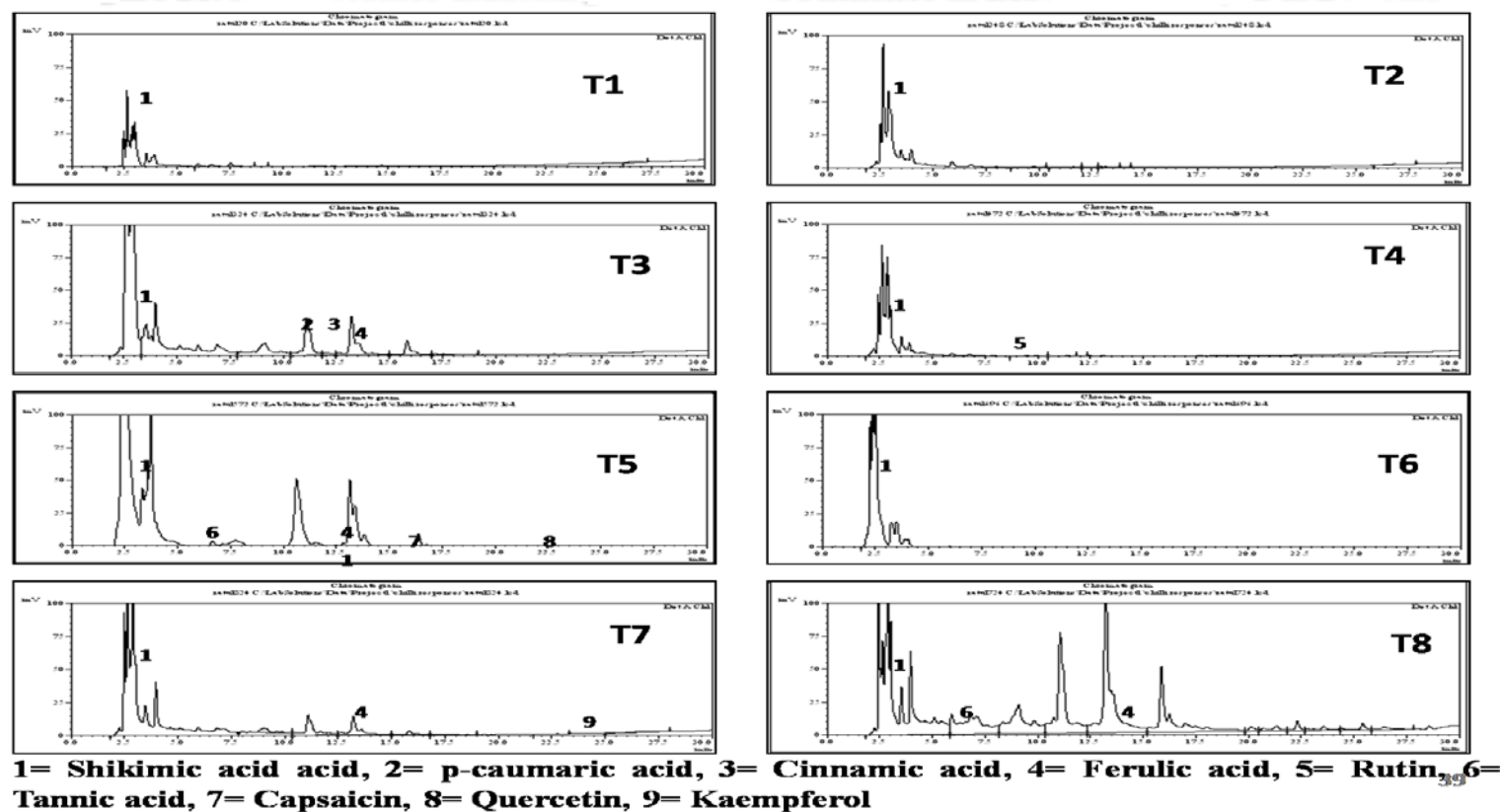


Figure 4.7 HPLC Profiles Exhibiting Induction of Phenolics in Leaves of Chickpea at 24 hours

Qualitative and quantitative analysis of phenolics

In order to gain insight regarding the interaction of various stress parameters with the biocontrol agent (*Bacillus* sp. BHU 08), qualitative and quantitative enhancement of phenolic profiles were studied from the leaf samples of combined treatments of *R. bataticola* and selected bioagents (*Bacillus* sp. BHU 08) through HPLC at 24 hrs after pathogen challenge and the results were presented in Fig. 4.4. Nine different phenolic compounds were detected in chickpea leaves, however the systemic accumulation of shikimic acid was observed in all the treatments. The control plants (C), recorded the lowest production of shikimic acid i.e 3920 µg/ g FW whereas in other treatments there is a significant increase in level of shikimic acid compared to C. The highest amount of shikimic acid was recorded in T5 (P + B) i.e 30120 µg/ g FW. The treatments in which drought and pathogen stress was applied i.e T2 (D), T4 (P) and T6 (D + P) recorded shikimic acid content of 7620, 7140 and 10300 µg/ g FW respectively. Similarly in T3 (D + B), T7 (D + P + B) and T8 (B), shikimic acid values were 14740, 14632 and 14286 µg/ g FW which is more than the individual stress challenged plants.

p- coumaric acid and cinnamic acid was present in T3 (D + B) only with a value of 136 µg/ g FW. The level of ferulic acid was found in a range of 44 – 204 µg/ g FW. T8 (B) recorded highest amount of ferulic acid i.e 204 µg/ g FW followed by T5, T3 and T7 which recorded 116, 54 and 44 µg/ g FW. However in treatments T1, T2, T4 and T6 it was found to be absent. Tannic acid was present in treatments T5 and T8 with values of 18 and 304 µg/ g FW respectively. Rutin was present in only one treatment (T4) with value of 1.028 µg/ g FW. Capsaicin, quercetin and kaempferol also followed the pattern of rutin and were present in single treatment only i.e T5 and T7. Capsaicin and quercetin are present in T5 and recorded a value of 370 and 0.194 µg/ g FW. Kaempferol was present in treatment T7 and its concentration was 2.2 µg/ g FW.

4.9 Study of gene expression in Chickpea

The semi quantitative expression studies for the genes selected from the literature corresponding to the defense response activation in Chickpea depicts considerable variation in their expression. All the studied genes i.e *PAL1*, *IFR*, *F3H*, *Ub1*, *Gro ES2*

and *HOS2* were up regulated in the treatment P+B, compared to the other treatments. This indicates that in combination of *R. bataticola* and *Bacillus* sp. BHU 08, the defensive genes in the plant are up regulated in order to activate the induced systemic resistance (ISR) of the plant. The control exhibited lowest expression of all the genes. The expression profiles of BHU 08 treated plants exhibited almost similar expression profiles depicting the acceptability of the beneficial microbes by the plants without treating them as harmful agents. On *R. bataticola* challenge, significant fold increase was recorded in the expression profile of *F3H* and *HOS2* while the remaining genes have a similar level (Figure 4.5).

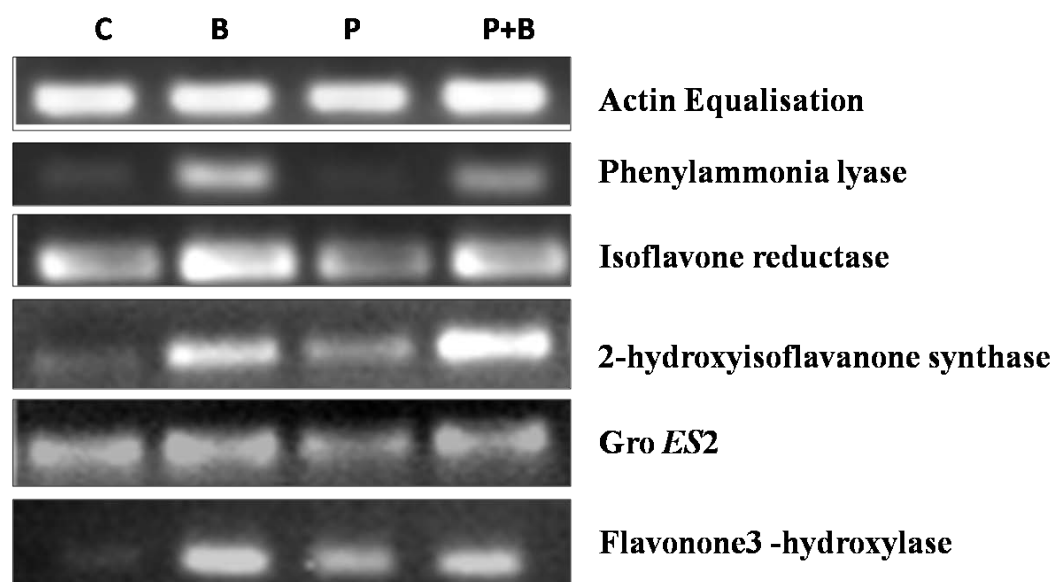


Figure 4.8 Relative expression of various genes using semiquantitative RT-PCR in Chickpea leaves at 48 hpi under the influence of different treatments. Where C = Control, B = BHU 08, P = *R. bataticola*, P+B = *R. bataticola* + BHU 08

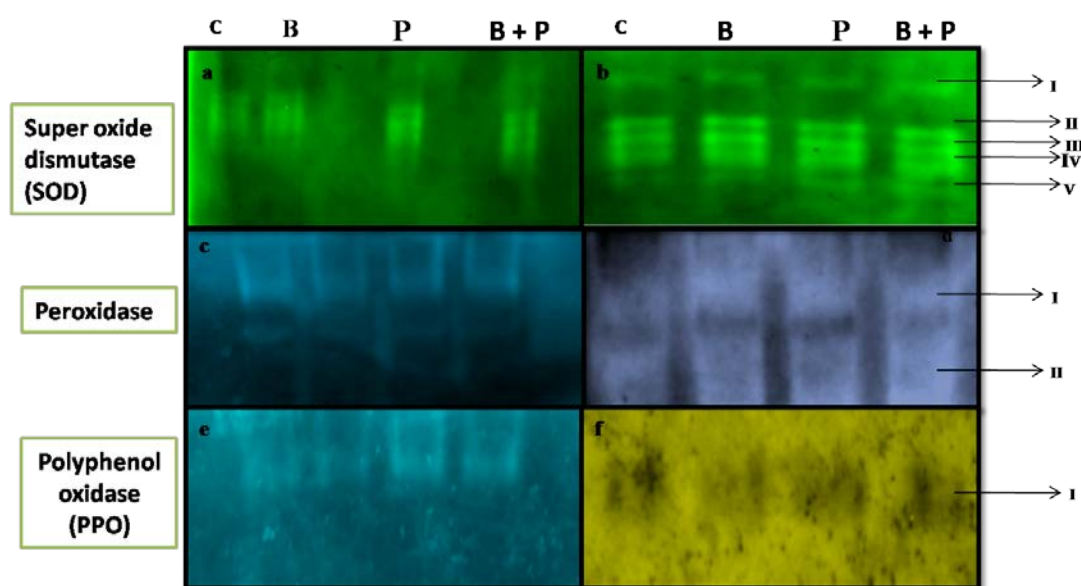


Figure 4.9 Variations in isozyme profiling of various treatments i.e Control (C) , BHU 08 (B), *R. bataticola* (P) and *R.bataticola* + BHU 08 (B + P) at 48 h (a,c,e) and 72h (b,d,f) hours post infection with *R. bataticola* in chickpea The SOD (a,b) showed 5 isozymes (I,II,III,IV,V) with increased intensity at 72h in comparison to 48h. The PO (c,d) showed two isoforms (I,II) with enhanced activity at 72 h as compared to 48h. Finally, the PPO produced a single isoform

Isozyme analysis of the induced defense proteins

The native PAGE analysis of SOD isoforms revealed five principal bands with band II, III and IV exhibiting enhancement in activity with time(Figure). The treatment (P + B) exhibited highest enhancement in SOD activity at 72 h post infection followed by P, B and C, thereby depicting maximum production of H_2O_2 at 72 h. Similarly, PO isoforms depicted two significant isozyme bands (I, II) in unprimed control plants post 48 h of pathogen challenge. The plants treated with *R.bataticola* and BHU 08 i.e (P + B), exhibited a highest intensity of activity particularly in both the isoforms, with the followed by P, B and C. The pattern of expression of PO was similar to that of SOD; however difference lies in no of band formation. Further, the level of PO activity increased at 72 h of pathogen stress, thereby suggesting the augmentation in H_2O_2 quenching activity and a subsequent polymerization of the soluble phenolics through the

phenylpropanoid pathway. The native PAGE analysis of PPO bands, however, revealed single isoforms which showed an enhancement in its activity with time.



CHAPTER V

DISCUSSION

DISCUSSION

Fungi are responsible for many devastating plant diseases and are a major threat for agriculture with regard to both yield and quality. Dry root rot (DRR) of chickpea caused by *Rhizoctonia bataticola*, is one of serious disease emerging as a potential threat to chickpea production (Pande *et al.*, 2010; Sharma *et al.*, 2010; Ghosh *et al.*, 2013). Savary *et al.* (2011) described DRR as an acute-emerging disease that occurs irregularly, both temporally and spatially, and may cause enormous disruptions in system performances with expanding ranges to new areas. The disease usually appears around flowering and podding time of the crop in the form of scattered dried plants. The disease may also appear at seedling stage; however, the old plants are more susceptible towards infection. The incidence of disease is usually seen when day temperature exceeds 30°C. Microsclerotia serve as the primary source of inoculum and have the ability to persist in the soil for around 15 years (Short *et al.*, 1980). They are generally found in clusters on the soil surface and have been reported up to the depth of 0-20 cm in soil (Alabouvette, 1990; Campbell and Van der Gaag, 1993) and are well adapted to survive under adverse environmental conditions (Short *et al.*, 1980). Several scientists have been attempting to manage the dry root rot pathogen. However, the disease can be managed to a considerable extent by use of fungicides, host plant resistance, biological means combined with other management strategies (Majumdar *et al.*, 1996; Cardoso *et al.*, 1997; Mandhare and Suryawanshi, 2009).

5.1 Collection, identification and purification of the pathogen causing dry root rot of chickpea from different districts of Eastern Uttar Pradesh.

A roving survey was conducted during January, 2013 in 10 districts of Eastern Uttar Pradesh to analyze the status of dry root rot incidence and to collect diseased samples infected by *R. bataticola* under field condition. The areas having maximum root rot incidence were selected and infected samples were collected from those areas on the basis of disease severity rating scale of 1-9. Symptoms on affected plants were observed as bronzing of the leaves on one or more of

the lower branches, leaves became yellow to brown in plants showing advanced disease symptoms. The affected branches and leaf stalks were stiff, turned upwards and the leaflets stand more or less vertically and were shed prematurely. The terminal part of the tap root and lateral roots became brown to black and shriveled. The tap root was devoid of any lateral roots. Dastur (1935) reported similar type of symptoms of dry root rot as yellowing of leaves which later turn brown in colour. The affected branches and leaflets were stiff and turned upwards, stand vertically and shed prematurely. Nene *et al.* (1991) observed infection in seedling stage. When the plants were uprooted, the lower portion of tap root remained inside the soil and devoid of most of lateral roots. The mycelium was initially white in colour which was later converted to dark brown to black in colour. Production of aerial mycelium was also observed in some isolates. The vegetative mycelium was characterized by the formation of barrel shaped cells and the formation of septum near the origin of branch of the mycelium. Branching occurred mostly at right angle to parent hyphae, but branching at acute angles was also observed. The above observations were in accordance with the descriptions given by Short and Willie (1978) as branching occurs at right angle to parent hyphae but branching at acute angles was also common.

Twenty different isolates of *R. bataticola* were isolated from different areas which vary considerably in their cultural and morphological patterns like colony diameter, colony colour, colony texture, pigmentation and presence or absence of aerial mycelium. Results of the pathogenicity test revealed that out of 20 isolates, only 10 of them were found to produce dry root rot symptom in chickpea cultivar 'Radhey'. The results found were in accordance to isolation of *R. bataticola* carried out by various researchers in which every isolated failed to prove the pathogenicity test (Purkyastha *et al.*, 2006, Ahmed and Ahmed, 1969; Dhingra and Sinclair, 1973; Raut and Ingle, 1989). Similar studies on cultural and morphological characters of *R. bataticola* were conducted by Sharma *et al.* (2012) and Gupta *et al.* (2012). Sharma *et al.* (2012) reported the variation among 94 isolates of *R. bataticola* collected from chickpea and concluded that the light black (52.12%) colony colour was more predominant. Similarly, Agakhani and Dubey (2009) isolated 23 isolates of *R. bataticola* collected from 10 different major chickpea growing states of India having high variability in their morphological and cultural characters along with pathogenicity/ virulence.

5.2 Isolation and characterization of agriculturally important micro-organisms (AIMs) from Eastern Uttar Pradesh and Bundelkhand region.

The survey was conducted during June-July, 2012 in six districts of eastern Uttar Pradesh and Bundelkhand region for isolating AIMs having drought tolerant potential. The main objective was to use these microbes for management of drought tolerance in plants. This would support to the research work carried out by various scientists in which different types of AIMs were used in mitigating drought stress in different plants (Figueiredo *et al.*, 2008; Kohler *et al.*, 2008; Arshad *et al.*, 2008; Amellal *et al.*, 1998; Marulanda *et al.*, 2007; Sandhya *et al.*, 2009a; 2009b; Creus *et al.*, 2004; Alami *et al.*, 2000). Some of the AIMs which induce abiotic stress tolerance are *Azospirillum* sp., *Pseudomonas syringae*, *P. fluorescens*, *Bacillus* sp., *Bacillus polymyxa*, and *Pseudomonas alacaligenes*. Other than bacteria, salinity and drought-tolerant isolates of *Trichoderma harzianum* and the effect of other strains of *Trichoderma* in amelioration of such abiotic stresses have also been reported. Arbuscular mycorrhizal fungi (*Glomus mosseae*, *G. etunicatum*, *G. intraradices*, *G. fasciculatum*, *G. macrocarpum*, *G. coronatum* etc.) help in alleviating abiotic stresses in different crops. Different types of bacteria were isolated and were categorized on the basis of their colony characters. The drought tolerance was checked by inoculating the bacteria in 60% (NB + PEG) and spotting in NA plates for 5 consecutive days (Rehman and Nautiyal, 2002). *In vitro* test for drought tolerance revealed that out of 120 different bacteria, only 30 of them possess drought tolerance ability. The 10 isolates on the basis of high cfu ml⁻¹ were selected for checking their thermotolerance ability and 4 of them were found to grow at 55 °C. The isolates were selected and subjected to antagonism test against *R. bataticola*. Out of the 4 isolates, only BHU08 showed antagonistic ability against the dry root rot pathogen in dual culture and was selected for further experiments. The antagonism test of drought tolerant isolates was a novel experiment and has not been conducted by any researcher.

5.3 Study on the impact of individual and combined application of AIMs on chickpea growth and to elucidate their mode of action against *R. bataticola*

The experiments on growth promotion and disease management on chickpea plants were conducted and it was found that there is a significant increase in the morpho-physiological

parameters of the crop. Also the isolated bacteria BHU 08, showed the capability of imparting resistance to dry root rot as there are significant increase in all the growth parameters in the treated plants compared to the single pathogen challenged plants. Results also revealed that in the treatment where both drought and pathogen stress were applied (T6), showed the least value of all the studied traits. However T3, T5 and T7 showed better results compared to their stress counterparts i.e T2, T4 and T6. The highest growth promotion was seen in T8 where only BHU 08 was present. The results of growth promotion in plants by PGPR against drought have been found to support the results of numerous researchers. Under drought stress, co-inoculation of bean (*Phaseolus vulgaris* L.) with *Rhizobium tropici* and two strains of *P. polymyxa* resulted in augmented plant height; shoot dry weight and nodule number (Figueiredo *et al.*, 2008). Interestingly, the effect on IST and increased nodule number was greater when a mix of two strains of *P. polymyxa* was applied than when one strain was applied, suggesting some synergistic effects from the use of strain mixtures. Similarly, the ACC deaminase activity of *Achromobacter piechaudi* was shown to confer tolerance to water deficit in tomato and pepper, resulting in significant increases in fresh and dry weights (Mayak *et al.*, 2004b). Similarly, osmotic stressed plants inoculated with PGPR stains such as *B. subtilis* GB03 and *Pseudomonas* spp. significantly enhanced leaf RWC and dry weight of the plants than those in plants without inoculation (Sandhya *et al.*, 2010). Seed bacterization of maize with strains of *Proteus penneri* (Pp1), *Pseudomonas aeruginosa* (Pa2) and *Alcaligenes faecalis* (AF3), significantly improved soil moisture contents, plant biomass, root and shoot length and relative water content of treated plants with respect to control (Naseem and Bano, 2014). Similarly, inoculation with drought-tolerant *Bacillus* spp. strains resulted in high dry biomass, aggregate stability (AGS), mean weight diameter (MWD), relative water content and leaf water potential of maize plants under stressed conditions (Sandhya *et al.*, 2009).

5.4 Study on the resistance response of chickpea under the pathogen challenged condition at physiological and biochemical level.

5.4.1 Biochemical parameters

The resistance response was evaluated by studying the various biochemical parameters of plant pathogen interaction. All the major enzymes which contribute inducing systemic resistance were

studied and their up regulation /down regulation has been checked. Application of *Bacillus* sp. BHU 08 induced accumulation of high level of phenol, proline, activities of PAL, SOD in chickpea plants when compared with control. Induction of systemic resistance has been established as a new mechanism by which plants defend themselves against pathogen attack (Van Loon *et al.*, 1998). Various reports confirm the induction of systemic resistance by biocontrol agents against fungal pathogens (Yedida *et al.*, 1999; Meena *et al.*, 2000). Phenylalanine ammonia-lyase (PAL) plays an important role in the biosynthesis of various defense chemicals in phenylpropanoid metabolism. Activity of PAL could be induced in plant-pathogen interactions and fungal elicitor treatment. De Mayer *et al.* (1999) reported that rhizosphere colonization by *P. aeruginosa* 7NSK2 activated PAL in the roots and increased the salicylic acid levels in leaves. In the present study, activity of PAL was increased up to 48 h and then declined thereafter. The maximum level of PAL was recorded in T7 which indicate that in association of both biotic and abiotic stress along with the pathogen resulted in more production of PAL in order to extent the defense mechanism to its zenith. The time required to activate the defense mechanisms is important for the suppression of the invading pathogen. Earlier and higher levels of expression of defense enzymes and accumulation of inhibitory compounds at the infection site certainly prevent the fungal mycelial colonization. The activity of PAL was maintained at a higher level throughout the experimental period in all treated plants.

The phenolic compounds as constituents of lignin may contribute to enhance the mechanical strength of the host cell wall and may also inhibit fungal growth as they are fungitoxic in nature. The induction of phenols might be due to the activation of the shikimic acid pathway, through which the aromatic amino acids, phenylalanine and tyrosine are formed and channeled for the synthesis of phenolics. M'Piga *et al.* (1997) found that the hyphae of the pathogen surrounded by phenolic substances exhibited considerable morphological changes including cytoplasmic disorganization and loss of protoplasmic content. In the present study maximum accumulation of phenolics was observed in chickpea treated with biocontrol agents, in T7. Singh *et al.* (2013a) also reported higher accumulation of phenols in chickpea at 48 h, with maximum accumulation in the consortium of *Trichoderma*, *Pseudomonas* and *Rhizobium* sp. against *Sclerotium rolfsii*. Similar results were reported for black gram against dry root rot and

groundnut against *C. personata* (Meena *et al.*, 2000). Singh *et al.* (1998) found that seed and soil application of *T. viride* enhanced the phenolics concentration in chickpea plant that led to induced resistance to *Macrophomina phaseolina*.

Superoxide dismutase (SOD) catalyzes dismutation of superoxide into oxygen and hydrogen peroxide. As such, they are an important antioxidant defense in nearly all cells exposed to oxygen. In higher plants, SOD act as antioxidants and protect cellular components from being oxidized by reactive oxygen species (ROS). At 72 h, SOD activity in the T7 treatment was found to be highest than those in the other challenged control and unchallenged healthy control. The enzymes, SOD and PO work together with other enzymes of the ascorbate–glutathione cycle to promote scavenging of free radicals (Hernandez *et al.*, 2001). Analysis of plants after pathogen infection indicated that pea seeds treated with the three-microbe consortium followed by two-microbe consortium of *B. subtilis* and *P. aeruginosa* exhibited maximum activities of PO and SOD compared with the plants treated with single BCAs and untreated control (Jain *et al.*, 2011; 2014).

Similar results were also obtained in a previous study, where microbial mixture of *Bacillus* strains IN937a and IN937b was found to induce maximum SOD and PO activity compared with untreated control (Jetyanon, 2007). Several individual PGPR strains have been demonstrated to induce generation of antioxidants including SOD and PO in the host plants upon pathogen attack as an ISR response that provides protection against diverse pathogens (Silva *et al.*, 2004; Singhai *et al.*, 2011; Singh *et al.*, 2013b). Some amount of natural SOD activity in unchallenged healthy control was consistently presented throughout the assays. Increase in SOD activity elicited by the treatments could explain the observed disease suppression in all the tested pathosystems.

Proline is a class of imino acid which protects membranes and proteins against the adverse effects of high concentrations of inorganic ions and temperature extremes. Proline acts

as an osmoregulant and helps to maintain the water potential of plant to extract water from soil (Hanson *et al.*, 1979). Proline accumulation under drought condition is a good indicator of drought resistance capacity of plants (Singh and Singh, 1983). These are hydrophilic highly soluble molecules, able to produce a salvation surface, that capture water molecules to be later available during water limitation; these are non toxic molecules at high concentration, thus not interfering with cellular metabolism (Yancey *et al.*, 1982; Bartels and Sunkar 2005). Proline is also known to act as an antioxidant agent apart from its well-known role in osmoprotection (Schobert and Tschesche, 1978). Proline accumulation is a common metabolic response of higher plants to water deficits, and salinity stress. In the present study, at 72 h, proline activity in the T7 was found to be maximum than those in the other challenged control and unchallenged control. The results found were in accordance with Jain *et al.* (2011) where a microbial consortium of *T. harzianum* and *P. aeruginosa* and *B. subtilis* lead to higher accumulation of proline compared to infected and healthy control.

Catalase has one of the highest turnover rates for all enzymes with the potential to directly dismutate H_2O_2 into H_2O and O_2 , and is indispensable for ROS detoxification in peroxisomes during stress condition (Sairam and Srivastava 2001). As in the present study, it is already found that SOD activity also increased, and SOD detoxifies superoxide anion free radicals (O_2^-) by forming H_2O_2 , and then the H_2O_2 can be eliminated by CAT (Hasheminasab *et al.*, 2012). It was notable that highly induced SOD activity is well correlated with the enhanced activity of catalase (Bin *et al.*, 2010). In the present study, catalase activity has increased from 24 h and reached its maximum level at 48 h, thereafter it started declining. The catalase activity was found to be highest at 48 h in T7.

5.4.2 HPLC analysis

The elevated levels of accumulated biochemical enzymes involved in phenylpropanoid and shikimic acid pathways supported the observed level of phenols in the treated chickpea plants under drought and *R. bataticola* challenge when observed in HPLC profiling. Elevated levels of shikimic acid was found in the *Bacillus* sp. BHU 08 indicated the positive role of microbes in enhancing the defense network of plants by modulating the shikimic acid pathway

(Lattanzio *et al.*, 2006). Further production of p-caumaric acid, cinnamic acid and ferulic acid was seen in T3, where chickpea plant was challenged by drought in presence of *Bacillus* sp. BHU 08. Role of ferulic acid has been advocated as a precursor for phenylpropanoid derivatives like flavonoids which possess antimicrobial properties and fall in category of phytoalexins (Dixon *et al.*, 2002; Facchini *et al.*, 2002). In our present study maximum amount of ferulic acid was formed in plant treated individually with *Bacillus* sp. BHU 08. The specific formation of Capsaicin, kaempferol and quercetin in stress induced plants along with BHU 08 indicate the role of flavonols in contributing resistance to the pathogen which have been described previously (Chan *et al.*, 2007).

5.5 Adaptation of proteomic and genomic approach to understand the resistance response induced by AIMs in chickpea against *Rhizoctonia bataticola*.

5.5.1 Study of gene expression in Chickpea against *R. bataticola*

The semi quantitative expression studies for the genes selected from the literature corresponding to the defense response activation in chickpea depicts considerable variation in their expression. Although these genes are already present in chickpea plant, the defense imparting ability is determined by the level of expression of these genes. All the studied genes i.e. *PAL1*, *IFR*, *F3H*, *Ubl1*, *Gro ES2* and *HOS2* were up regulated in the treatment P+B, compared to the other treatments. This indicates that in combination of *R. bataticola* and *Bacillus* sp. BHU 08, the defensive genes in the plant are up regulated in order to activate the induced systemic resistance (ISR) of the plant. The up regulation of these genes has a direct co relation with increased defense of the plant against *R. bataticola* pathogen. The results obtained were in accordance with experiments of Kavousi *et al.* (2009) in which expression analysis of genes induced in general defense response in chickpea plants were studied against *Ascochyta rabiei*. Transcript accumulation of four genes encoding phenylalanine ammonia-lyase (PAL), chalcon synthase (CHS), isoflavone reductase (IFR) and Flavanone 3-Hydroxylase (F3H) induced in response to race 3 of *A. rabiei* was compared in both resistant and susceptible genotypes. The results revealed that in resistant genotype all the 4 phenylpropanoid pathway genes: PAL, CHS, IFR and F3H were rapidly up regulated 6 h after inoculation with

race 3 of *A. rabiei*. Thus, induction of key enzymes of phenylpropanoid pathway seems to be an important defense mechanism of chickpea plants against *A. rabiei*. Similarly, Gurjaret *et al.* (2012) studied the response of plant defense genes specific to *Foc* races. All the genes namely *GroES2*, *CHS*, *60srp* and *IFR*, showed higher expression in resistant chickpea cultivar “Digvijay” in comparison with susceptible cultivar JG62, when exposed to race 1 of *Foc*. The results clearly suggested that, for establishment of resistance against *Foc* race 1 response of all the three above mentioned genes was essential.

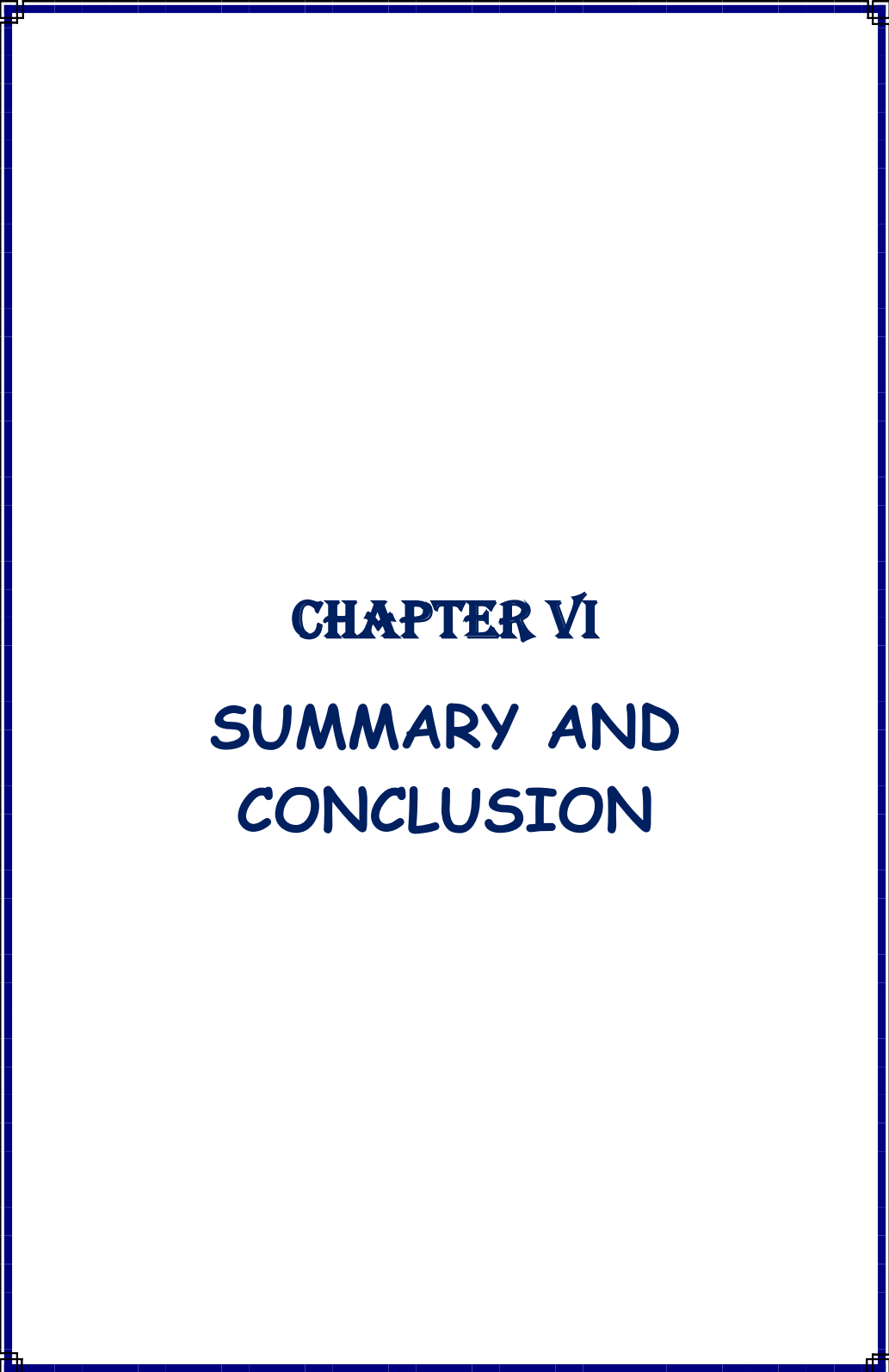
5.5.2 Response of antioxidant isozymes to biotic stress

Phenolic acid secretion in plants is a mode of defense strategy wherein the induced phenolics are also secreted in the form of root exudates so as to attract the beneficial microbes and repel the harmful ones. However, if the concentrations of pre-existing antifungal phenolics are not sufficient to stop the development of the infectious process, plant cells usually respond by enhancing the level of pre-existing antifungal phenols at the infection site, after an elicited activity of the key enzymes (PAL and chalcone synthase) of the biosynthetic pathway (Singh *et al.*, 2012; Jain *et al.*, 2015). The increased level of phenolics provides an adequate substrate to oxidative reactions catalyzed by PPO and/or PO that, after consuming oxygen and producing fungitoxic quinones, create unfavourable conditions for further development of pathogens. Though many phenolic compounds present in plant tissues may not have any anti-microbial activity, there are various reports where the oxidized products of pre-existing phenolics have antimicrobial activity, which often results in inhibition of the cell wall degradation by extracellular enzymes, released by the pathogens (Jain *et al.*, 2013).

Production of phenolics in plants results from induction of defense response to any form of biotic/abiotic stress in the hosts (Mandal *et al.*, 2010). Phenolic acids possibly are the primary compounds most noted for their ability to bind to protein *in vitro*, forming soluble and insoluble complexes (Jain *et al.*, 2015). To gain an idea regarding the probable correlation between phenolic induction owing to biotic stress and the resulting physiological and molecular changes triggered by *R. bataticola* infection and bacterial colonization, the phenolic profiling of chickpea leaves was conducted in this work. In this study, the probable resistance alterations

in host morphology owing to pathogen challenge has been looked upon. Further, the *in planta* changes in phenolic profiles linked to the expression of defense genes has also been discussed. The possible reasons for host resistance have been studied at the phenolic and molecular level. However, a deep study is still needed to assess the various factors responsible for host defense both on the microbial as well as plant level.





CHAPTER VI

SUMMARY AND CONCLUSION

SUMMARY AND CONCLUSION

The present investigation was undertaken to meet the following objectives:

1. Collection, identification and purification of the pathogen causing dry root rot of chickpea from different districts of Eastern Uttar Pradesh.
2. Isolation and characterization of agriculturally important micro-organisms (AIMs) from Eastern Uttar Pradesh and Bundelkhand region.
3. To study the impact of individual and combined application of AIMs on chickpea growth and to elucidate their mode of action against *Rhizoctonia bataticola*.
4. To study the resistance response of chickpea under the pathogen challenged condition at physiological and biochemical level.
5. Adaptation of proteomic and genomic approach to understand the resistance response induced by AIMs in chickpea against *Rhizoctonia bataticola*.

Experiments were conducted during the cropping seasons 2013-14 and 2014-2015 at polyhouse and field of Department of Mycology and Plant Pathology, Institute of Agricultural Sciences, Banaras Hindu University, Varanasi.

1. Collection, identification and purification of the pathogen causing dry root rot of chickpea from different districts of Eastern Uttar Pradesh.

A roving survey was conducted during January, 2013 in 10 districts of Eastern Uttar Pradesh to analyze the status of dry root rot incidence and to collect diseased samples infected by *R. bataticola* under field condition. The areas having maximum root rot incidence were selected and infected samples were collected from those areas on the basis of disease severity rating scale of 1-9. 20 different isolates of *R. bataticola* were isolated and were subjected to pathogenicity test. Only 10 of them were found to prove pathogenicity on chickpea cultivar “Radhey”. The percent disease incidence of the 10 isolates was checked

and it was found that at 120 DAI, *R. bataticola* isolate Rb2 showed maximum PDI of 89.63% amongst all isolate and thus was selected for further experiments.

2. Isolation and characterization of agriculturally important micro-organisms (AIMs) from Eastern Uttar Pradesh and Bundelkhand region

A roving survey was conducted during June-July, 2012 in Varanasi, Allahabad, Banda, Jhansi, Lalitpur and Jalaun districts of Uttar Pradesh order to collect soil samples for isolating thermotolerant microorganisms. Soil samples were collected from various sites of these districts and the temperature of the region while taking the samples was recorded. Different types of bacteria were isolated and were categorized on the basis of their colony characters. The drought tolerance was checked by inoculating the bacteria in 60% (NB + PEG) and spotting in NA plates for 5 consecutive days. *In vitro* test for drought tolerance revealed that out of 120 different bacteria, only 30 of them possess drought tolerance ability. The 10 isolates on the basis of high cfu ml⁻¹ were selected for checking their thermotolerance ability and 4 of them were found to grow at 55 °C. The 4 isolates were selected and subjected to antagonism test against *R. bataticola*. Out of the 4 isolates, only BHU08 showed antagonistic ability against the dry root rot pathogen in dual culture and was selected for further experiments. The sequencing of BHU 08 was performed and its phylogenetic tree was constructed. BLAST results revealed that the bacteria showed 98% similarity with different *Bacillus* sp. and thus considered to come under the genus *Bacillus*.

3. To study the impact of individual and combined application of AIMs on chickpea growth and to elucidate their mode of action against *Rhizoctonia bataticola*.

The growth promotion and disease suppression effect of *Bacillus* sp. BHU 08 was performed in both polyhouse and field. 8 treatments were performed to assess the effect of *Bacillus* sp. BHU 08 in imparting both drought and pathogen tolerance. Morpho-physiological parameters were recorded 30, 60 and 90 DAS. During the initial period, much variation in the growth parameters is not seen as the pathogen remains dormant at this stage. But, the data at later results revealed significant variation in the parameters amongst the treatments. T8 (B) was found to achieve highest values of morphological parameters. However, increase in fresh weight, dry

weight and RWC was also found in T8. On the basis of increment in these parameters compared to control and stress challenged plants, it was estimated that *Bacillus* sp. BHU 08 has a great potential in mitigating drought and dry root rot incidence in chickpea.

4. To study the resistance response of chickpea under the pathogen challenged condition at physiological and biochemical level.

Application of *Bacillus* sp. BHU 08 induced accumulation of high level of phenol, proline, activities of PAL, SOD and catalase in chickpea plants when compared with control. The treatment T7 recorded highest value of all these enzymes at different hours after pathogen challenge varying from 48-72 h and then declined. Increased levels of these plant defense enzymes upon pathogen attack suggest that the ISR and SAR mechanisms of the plants get initiated and thus it will trigger a defense response in the host. These antioxidant enzymes quench the toxic reactive oxygen species and in turn protect plants from oxidative damage. PAL activity precedes rapid recognition of a microbial invader which potentiates the accumulation of disease resistance factors including phenolics, phytoalexins and lignin. An increase in lignification is often observed in response to attack by pathogen and is believed to represent one of the chief mechanisms adopted to block pathogen invasion due to its highly non-degradable and antimicrobial nature.

5. Adaptation of proteomic and genomic approach to understand the resistance response induced by AIMs in chickpea against *Rhizoctonia bataticola*.

The different genes i.e. *PAL1*, *IFR*, *F3H*, *Ub1*, *Gro ES2* and *HOS2* which are known to impart defense against *Fusarium oxysporum* f.sp. *ciceri* and *Ascochyta rabei* were tested for their potential in imparting defense to chickpea against *R. bataticola*. The experiments revealed that all the above mentioned genes were up regulated in the treatment P+B, compared to the other treatments. This indicates that in combination of *R. Bataticola* and *Bacillus* sp. BHU 08, the defensive genes in the plant are up regulated in order to activate the induced systemic resistance (ISR) of the plant. The up regulation of these genes has a direct co relation with increased defense of the plant against *R. bataticola* pathogen. The study through native PAGE revealed that all the three studied plant defense enzymes i.e. SOD, PO and PPO have increased levels in the treatment P+B, at 72 h compared to the other treatments. The increased level of phenolics provides an adequate substrate to oxidative reactions catalyzed by PPO and/or PO that, after consuming

oxygen and producing fungitoxic quinones, create unfavourable conditions for further development of pathogens.

Conclusions:

Following conclusions has been drawn from the present investigation

- Novel strain of *Bacillus* sp. BHU 08 has been reported as drought tolerant strain to be further used as thermotolerant strain for managing dry root rot of chickpea.
- Chickpea plants treated with *Bacillus* sp. BHU 08 showed significant augmentation in levels of antioxidant enzymes and phenylpropanoid pathway.
- It positively modulates the oxidative and defense network in the host.
- *Bacillus* sp. BHU 08 has a significant effect on the physiological parameters of chickpea plant.
- The levels of defence gene in chickpea against *R. bataticola* were upregulated on treatment with *Bacillus* sp. BHU 08.
- Based on the above observations, *Bacillus* sp. BHU 08 proved to be a promising biocontrol agent in management of dry root rot of chickpea under field conditions.



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